

SINGULAR POINTS OF RANK-ONE DEL PEZZO SURFACES IN CHARACTERISTIC THREE

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ABSTRACT. We prove that a normal projective klt del Pezzo surface of Picard number one over an algebraically closed field of characteristic three has at most seven singular points. An example of Bernasconi shows that this bound is sharp. The proof combines the hunt with rulings of small exceptional degree, local adjunction, and birational comparisons with open del Pezzo pairs and canonical del Pezzo surfaces carrying anticanonical curves of arithmetic genus one. The main result of this paper was obtained using the Danus system.

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1. INTRODUCTION

A klt del Pezzo surface is a normal projective surface X with Kawamata log terminal singularities and ample anticanonical divisor $-K_X$. We write $\rho(X)$ for its Picard number and $\text{Sing}(X)$ for its singular locus. Bounding the number of singular points measures how much singular geometry can occur on a surface with a positive anticanonical divisor. Even when $\rho(X) = 1$, the answer depends on the characteristic of the ground field.

Over the complex numbers, Belousov proves that a klt del Pezzo surface of Picard number one has at most four singular points [Bel09]. Liu–Xie obtain the bound $2\rho(X) + 2$ for complex klt del Pezzo surfaces of arbitrary Picard number [LX25]. In characteristic three, Bernasconi constructs a rank-one klt del Pezzo surface with seven singular points [Ber21, Proposition 3.2 and Remark 3.4]. Thus the complex rank-one bound does not extend to characteristic three.

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The hunt of Keel–McKernan [KM97] replaces a rank-one surface by a $\mathbb{P}^1$ -fibration (called a *net*) or a new rank-one surface carrying a boundary curve. It is central to Lacini’s classification in characteristics different from two and three [Lac24]. Nagaoka’s review [Nag25a] separates the hunt statements and local arguments that remain valid in small characteristic from the hypotheses of that classification. We also use Nagaoka’s tiger reduction [Nag25b, Theorem 5.3]. Palka–Pelka provide complementary classifications by height and elliptic-boundary descendants [PP24a, PP25], together with a structure theorem for open del Pezzo surfaces [PP24b]. Applying these results in characteristic three requires treating inseparable multisections and the exceptional open-pair configurations. It also requires comparing each resulting surface with the original one: a bound for the number of singular points on an endpoint alone does not give the required bound.

We show that seven is an upper bound in characteristic three. In the following statement, k denotes the ground field.

Theorem 1.1. *Let k be an algebraically closed field of characteristic 3, and let X be a normal projective klt del Pezzo surface over k with $\rho(X) = 1$. Then X has at most seven singular points.*

The Picard-number-one hypothesis is part of the assertion. The theorem counts singular points, without restricting their local indices or the number of curves in their resolutions. Bernasconi’s example cited above shows that the bound is sharp.

The proof keeps track of the reduced exceptional divisor D on the minimal resolution $Y \rightarrow X$. Its connected components count the singular points of X . There are three main comparisons. First, a $\mathbb{P}^1$ -fibration on Y whose general fibre has intersection number at most four with D yields the seven-point bound. The degree-four argument includes an inseparable trisection together with a section; we treat it by fibre geometry, separately from the separable ramification argument. Second, local adjunction bounds the number of surface singularities on curves contracted during the hunt. Birational counting identities then transfer endpoint bounds back to X . Third, when a smooth first boundary is contracted in the second hunt, the new boundary is a rational cuspidal curve of arithmetic genus one. On a projective partial resolution, we retain the end component of the resolution chain that meets this curve. An adjoint contraction then produces a canonical del Pezzo surface whose singularities can be counted. These comparisons allow the low-height and open-pair results to be used with their actual hypotheses. In particular, a numerically trivial reduced adjoint divisor is treated by a separate pair hunt, and the exceptional characteristic-three open-pair alternatives are checked on a common resolution with the chosen boundary distinguished from the marked cuspidal curve.

Section 2 fixes notation. Section 3 proves the ruling bounds, and Sections 4 and 5 establish the local and birational comparisons and the open-pair inputs. Section 6 records the hunt and its choices. Sections 7 and 8 treat surfaces with a tiger on their minimal resolution (Definition 6.1). For surfaces without such a tiger, Section 9 treats the first net and singular first boundary; Section 10 treats the second net when the first boundary is smooth. The remaining birational outputs are treated in Sections 11 and 12, according to whether the old boundary is contracted or both boundaries survive. Section 13 assembles the proof.

Remark 1.2 (Relation with the work of Palka–Pelka). Theorem 1.1 agrees with the announced completion of the classification of Palka–Pelka. Using the height and graph notation of Section 2 and the descendant notion of Definition 3.1, their public papers classify the cases of height at most two and those with an elliptic-boundary descendant [PP24a, Theorems A and E], as well as the cases of height three without such a descendant [PP25, Theorem A]. In [PP24a, Remark 1.7], they give the characteristic-three type $4A_2 + [5] + [3] + [2]$ and announce that forthcoming work will prove its singular-point count maximal. The height-three seven-point type $3A_2 + 4[3]$ appears in [PP25, Table 5, row 6.7(1)].

In private communication, Karol Palka informed the author that their classification is complete. According to this communication, surfaces in characteristic three with no elliptic-boundary descendant and height at least four have at most six singular points, and the complete

classification identifies $4A_2 + [5] + [3] + [2]$ and $3A_2 + 4[3]$ as the only singularity types attaining seven. Palka also described their method as emphasizing global geometry and uniqueness, whereas this paper follows the hunt method.

The announced complete classification would therefore also imply Theorem 1.1. The cited public papers do not supply the proof of the remaining cases. Neither the forthcoming work nor Palka's private communication is used in our argument.

Remark 1.3 (Relation with FrontierMath). Paolo Cascini contributed the question addressed by Theorem 1.1 and brought it to the author's attention. FrontierMath Open Problems posed it as a benchmark problem: the prompt asks for a characteristic-three rank-one klt del Pezzo surface with at least eight singular points and specifies a verifier for proposed positive constructions in Macaulay2 [Epo26, Prompt].

Theorem 1.1 gives a negative answer to this construction problem. Greg Burnham explained to the author that the FrontierMath verification mechanism was designed to check positive constructions rather than exclusion or nonexistence proofs; this theorem therefore falls outside its scope. In its update of July 31, 2026, Epoch AI states that the problem was removed from the benchmark because the verifier would not detect correct solutions with sufficiently high fidelity [Epo26, Update of July 31, 2026]. The proof is available for independent mathematical scrutiny.

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2. NOTATION AND TERMINOLOGY

Set-up 2.1. Unless a statement explicitly allows arbitrary characteristic, the ground field k is algebraically closed of characteristic 3. Surfaces are integral projective schemes of dimension two over k . A rank-one log del Pezzo surface is a normal surface S with Kawamata log terminal singularities, with $-K_S$ ample and with Picard number $\rho(S) = 1$. We write $\# \text{Sing}(S)$ for the number of its singular points.

A log pair (S, Δ) consists of a normal surface and an effective $\mathbb{Q}$ -divisor such that $K_S + \Delta$ is $\mathbb{Q}$ -Cartier. For a proper birational morphism $f : V \rightarrow S$, we use the coefficient convention

$$K_V + \Delta_V = f^*(K_S + \Delta), \quad e(F; S, \Delta) = \text{coeff}_F(\Delta_V).$$

Here canonical divisors are chosen compatibly, and F is a prime divisor on V . The log discrepancy is $1 - e(F; S, \Delta)$. We use the standard notions klt (Kawamata log terminal), plt (purely log terminal), and dlt (divisorially log terminal): klt requires all log discrepancies to be positive; plt allows coefficient-one boundary components and requires positive log discrepancies for exceptional divisors; dlt means that on a log resolution the exceptional log discrepancies are positive and the boundary has coefficients at most one. Intersections on normal surfaces are intersections of rational Cartier divisors, computed by pullback to a resolution. Anti-nef means that the negative of the divisor is nef; anti-ample means that its negative is ample.

For the minimal resolution $p : Y \rightarrow S$, put $D = \text{Exc}(p)_{\text{red}}$. A $\mathbb{P}^1$ -fibration is a morphism to a smooth projective curve with connected fibres and smooth geometric generic fibre isomorphic to $\mathbb{P}^1$. A curve on its source is horizontal if it dominates the base and vertical otherwise. The

height $\text{ht}(S)$ is the least value of $F \cdot D$ among such fibrations on Y , where F is a general fibre; the height is $+\infty$ when no such fibration exists. For a reduced boundary B on S , the height $\text{ht}(S, B)$ is defined on the minimal log resolution with its reduced total boundary, including all exceptional curves. Thus the two heights use different reduced divisors.

We always write $p_a(C)$ for arithmetic genus, and C^ν for the normalization of an integral curve. An ordinary node has two smooth transverse branches; an ordinary cusp has completed local ring $k[[u, v]]/(v^2 - u^3)$. A rational curve has normalization $\mathbb{P}^1$ and need not be smooth. The notation $[b_1, \dots, b_r]$ denotes a chain of smooth rational curves with self-intersections $-b_1, \dots, -b_r$ and adjacent intersections one. A sum of weighted graphs means their disjoint union. When the graph is the exceptional divisor of a resolution, its connected components count surface singular points; the number of vertices counts exceptional curves. We write A_r, D_r, E_r for Du Val graph types. For an irreducible curve C on S , its strict transform on a specified birational model V is denoted C_V .

3. RULED RESOLUTIONS AND ELLIPTIC BOUNDARIES

We use the height and exceptional-divisor notation of Section 2. We first count singular points in the low-height and elliptic-boundary classifications. We then study a ruling whose general fibre has intersection number four with the reduced exceptional divisor. The count in this case separates into a ramification argument and an argument for a purely inseparable trisection.

Definition 3.1. Let S be a normal surface, and let $p : Y \rightarrow S$ be its minimal resolution with $D = \text{Exc}(p)_{\text{red}}$. An *elliptic-boundary descendant* is a normal projective surface W with a birational morphism $\phi : Y \rightarrow W$ such that the reduced divisorial image $\phi_* D$ is an irreducible curve $C \subset W^{\text{reg}}$ with $p_a(C) = 1$.

The boundary here is the divisorial image of the exceptional divisor. Its arithmetic genus is one; the definition does not require it to be smooth.

Theorem 3.2 (Palka–Pelka). *Let S be a rank-one klt del Pezzo surface over an algebraically closed field of characteristic different from 2. If $\text{ht}(S) \leq 2$, then $\#\text{Sing}(S) \leq 4$.*

This is the singular-point bound in [PP24a, Theorem A and Remark 1.7]. In particular, it applies in characteristic three. The restriction on characteristic is part of the result.

Theorem 3.3 (Palka–Pelka). *Let S be a rank-one klt del Pezzo surface over an algebraically closed field. Assume that $\text{ht}(S) = 3$ and that S has no elliptic-boundary descendant. Then the singularity type of S is one of the types in [PP25, Tables 4–7], with the characteristic and parameter restrictions specified there. Each connected summand in those tables is the minimal-resolution graph of one singular point.*

This is [PP25, Theorem A(a)]. The characteristic restrictions in the tables distinguish the types that contribute to our count.

Proposition 3.4. *Let S be a rank-one klt del Pezzo surface over k . If $\text{ht}(S) = 3$ and S has no elliptic-boundary descendant, then $\#\text{Sing}(S) \leq 7$. If equality holds, its singularity type is $3A_2 + 4[3]$.*

The equality type has three two-curve chains and four single exceptional curves, giving seven connected components of the exceptional divisor, hence seven singular points.

Proof. By Theorem 3.3, it suffices to count the connected summands in the applicable classification rows. We first remove the rows excluded by characteristic and then count the remaining ones.

Step 1. In this step, we count the applicable rows occurring in several characteristics. The rows of [PP25, Table 4] valid in every characteristic have two summands. Among its rows valid in characteristic different from two, the rows numbered 5.15(1)–(6) have two summands and

5.15(7)–(16) have three. The final row requires characteristic different from both two and three, so it does not occur here. Tables 6 and 7 of the same classification occur only in characteristic two.

Step 2. In this step, we count the characteristic-three rows and identify the equality type. The numbers of connected summands in the eighteen rows of [PP25, Table 5] are

row	1	2	3	4	5	6	7	8	9
number	7	6	6	6	5	5	5	6	6
row	10	11	12	13	14	15	16	17	18
number	6	5	5	5	4	4	4	5	5

Thus every applicable row has at most seven summands. The unique row with seven is

$$[2, 2] + [2, 2] + [2, 2] + [3] + [3] + [3] + [3] = 3A_2 + 4[3].$$

Each connected summand represents one singular point, proving both assertions. $\square$

Theorem 3.5 (Palka–Pelka). *Let S be a rank-one klt del Pezzo surface over k , and let (W, C) be an elliptic-boundary descendant. For a normal surface Z , write $\Gamma(Z)$ for the weighted dual graph of the exceptional divisor of its minimal resolution. Let s be the number of vertices of $\Gamma(W)$. Then the following hold:*

- (i) *The curve C is singular, W is a canonical rank-one del Pezzo surface, and $C \in |-K_W|$. If $W \simeq \mathbb{P}^2$, then $\text{ht}(S) \leq 2$.*
- (ii) *If $\text{ht}(S) \geq 3$, then $s \geq 6$ and*

$$\Gamma(S) = \Gamma(W) \sqcup \mathcal{T}$$

for a graph $\mathcal{T}$ among the types in [PP24a, Lemma 6.2]. The type of W and the nodal or cuspidal type of C occur in the corresponding characteristic column of [PP24a, Table 15]. If C is cuspidal, then $\Gamma(W) \notin \{E_6, E_7, E_8\}$.

This combines [PP24a, Lemma 1.12, Theorem E(b),(c), and Notation 6.1]. The decomposition separates the singularities of the canonical descendant from those arising over its boundary. The integer s counts exceptional curves on its resolution.

Theorem 3.6. *Let S be a rank-one klt del Pezzo surface over k with an elliptic-boundary descendant. Then $\#\text{Sing}(S) \leq 7$.*

The bound consists of at most four singular points on the descendant and at most three contributed by the boundary construction.

Proof. If $\text{ht}(S) \leq 2$, Theorem 3.2 gives the stronger bound $\#\text{Sing}(S) \leq 4$. Assume $\text{ht}(S) \geq 3$, and choose a descendant (W, C) . We apply Theorem 3.5 and count the connected components in its graph decomposition.

Step 1. In this step, we bound the number of singular points of the canonical descendant W . The cited theorem says that C is singular, that W is a canonical rank-one del Pezzo surface, and that $C \in |-K_W|$. Also $W \not\simeq \mathbb{P}^2$, since the plane case has height at most two. Use the notation s , $\Gamma(W)$ and $\mathcal{T}$ from that theorem. Then $s \geq 6$ and

$$\Gamma(S) = \Gamma(W) \sqcup \mathcal{T}.$$

In the characteristic-three column of [PP24a, Table 15], retain the rows with $s \geq 6$ and a rational elliptic boundary. Their connected-component counts are as follows:

number	singularity types of W
1	$A_7, E_6, E_7, E_8, D_8, A_8$
2	$A_1 + A_5, A_1 + D_6, A_1 + E_7, 2A_4,$ $A_2 + A_5, A_2 + E_6, A_3 + D_5, A_1 + A_7$
3	$3A_2, A_1 + 2A_3$
4	$4A_2$

In particular, $\# \text{Sing}(W) \leq 4$. The rows $3A_1 + D_4$, $2A_1 + D_6$, $A_1 + A_2 + A_5$, $2D_4$ and $2A_1 + 2A_3$ have no such boundary in characteristic three. The rows $7A_1$, $4A_1 + D_4$ and $8A_1$ occur only in characteristic two and do not enter this count.

Step 2. In this step, we count the components contributed over the boundary singularity. For the graphs in [PP24a, Lemma 6.2], cases (1) and (2) have at most two connected components. In the cuspidal case, graph (3) is $[s - 3] + [3] + [2]$, with three connected components. Graphs (4)–(9) each have two. Graph (10) has two, except that its alternative with parameter two is one connected fork. Consequently $\#\pi_0(\mathcal{T}) \leq 3$. The disjoint graph decomposition now gives

$$\# \text{Sing}(S) = \# \text{Sing}(W) + \#\pi_0(\mathcal{T}) \leq 4 + 3 = 7.$$

□

3.1. Counting components in the fibres. We use the following properties of ruled minimal resolutions. They are [PP24a, Section 2B, Lemmas 2.3, 2.4, 2.7(a),(c),(d), and 2.8(b),(c)], specialized to klt surfaces.

Theorem 3.7 (Palka–Peřka). *Let S be a rank-one klt del Pezzo surface over k , let $p : Y \rightarrow S$ be its minimal resolution, and put $D = \text{Exc}(p)_{\text{red}}$. Let n_D be the number of irreducible components of D , and define the coefficients γ_E by*

$$p^*K_S = K_Y + \sum_{E \subseteq D} \gamma_E E.$$

Then D is a negative definite simple normal crossings divisor whose components are smooth rational curves and whose dual graph is a forest. Its connected components correspond to the singular points of S , $\rho(Y) = 1 + n_D$, and $0 \leq \gamma_E < 1$.

Suppose in addition that $f : Y \rightarrow B$ is a $\mathbb{P}^1$ -fibration over a smooth projective curve. Let F be its general fibre, let D_{hor} be the horizontal part of D , and for each degenerate fibre G let $r(G)$ be the number of components of G_{red} . Then:

(i) *Every G_{red} is a simple normal crossings rational tree. Its (-1) -curves have valency at most two, and a (-1) -curve of multiplicity one in G is a leaf. A component of G is a (-1) -curve if and only if it is not contained in D .*

(ii) *The Picard number satisfies*

$$\rho(Y) = 2 + \sum_G (r(G) - 1).$$

(iii) *If a component R of a degenerate fibre G satisfies $R \cdot D_{\text{hor}} = F \cdot D_{\text{hor}}$, then $R \subseteq D$ and G_{red} has exactly one component outside D .*

The Picard formula and the structure of each fibre control the number of vertical exceptional components. The coefficients γ_E will later bound the number of blowups on an inseparable trisection.

Set-up 3.8. Let S be a rank-one klt del Pezzo surface over k , let $p : Y \rightarrow S$ be its minimal resolution, and put $D = \text{Exc}(p)_{\text{red}}$. Fix a $\mathbb{P}^1$ -fibration $f : Y \rightarrow \mathbb{P}^1$ with general fibre F . Use the notation n_D , $r(G)$ and γ_E of Theorem 3.7, and write

$$D_{\text{hor}} = H_1 + \cdots + H_h, \quad d_i = F \cdot H_i,$$

and let c_{hor} be the number of connected components of D that contain a horizontal curve. For a degenerate fibre G , let D_G be the reduced sum of the components of D in G , and let $\sigma(G)$ be the number of components of G_{red} outside D . When $D_G \neq 0$, write $s(G)$ for the number of connected components of D_G , $t(G)$ for the number of those components meeting D_{hor} , and $\ell(G)$ for the number of leaves of G_{red} outside D . Define

$$v(G) = s(G) - t(G), \quad q(G) = 2 - \ell(G) - t(G).$$

Finally, let e_0 count the degenerate fibres with $D_G = 0$.

Thus $v(G)$ counts connected components of D contained entirely in G .

Lemma 3.9. *Assume Setup 3.8, and use the fibre notation above. Then*

$$(3.1) \quad \#\pi_0(D) = c_{\text{hor}} + h - 1 - e_0 + \sum_{D_G \neq 0} q(G).$$

The formula counts connected exceptional divisors. The lengths of their chains enter only through the Picard-number equality and disappear from the final expression.

Proof. We apply Theorem 3.7 to the given resolution and fibration. We first describe the components outside D , then use the Picard number to sum the counts over the fibres.

Step 1. In this step, we determine when components outside D can be adjacent in a degenerate fibre. No whole fibre is supported in D : its square is zero, whereas every nonzero divisor supported in D has negative square. In particular, $\sigma(G) \geq 1$.

Suppose two components L, M outside D are adjacent in G_{red} . They are (-1) -curves. Let their multiplicities in G be λ and μ . The identities $G \cdot L = G \cdot M = 0$ give

$$\lambda = \sum_{A \text{ adjacent to } L} \text{mult}_A(G) \geq \mu, \quad \mu = \sum_{A \text{ adjacent to } M} \text{mult}_A(G) \geq \lambda.$$

Both inequalities are equalities, so neither curve has another neighbour. Since the fibre is connected, $G_{\text{red}} = L + M$. Thus, when $D_G \neq 0$, no two components outside D are adjacent. When $D_G = 0$, the same argument shows that $r(G) = \sigma(G) = 2$. Here a degenerate fibre must have at least two components, since its components outside D are (-1) -curves and a single component would have square zero.

Step 2. In this step, we compute $v(G)$ by counting edges in the contracted fiber tree. For $D_G \neq 0$, contract each connected component of D_G to a vertex in the dual tree of G_{red} . The resulting bipartite tree has $s(G) + \sigma(G)$ vertices. Of its $\sigma(G)$ vertices outside D , exactly $\ell(G)$ have degree one and all others have degree two. Counting edges gives

$$s(G) + \sigma(G) - 1 = \ell(G) + 2(\sigma(G) - \ell(G)).$$

It follows that

$$(3.2) \quad v(G) = \sigma(G) - 1 + q(G).$$

Step 3. In this step, we use the Picard-number equalities to sum the component counts over the fibres. Every vertical component of D lies in a degenerate fibre, since a nondegenerate fibre cannot be supported in D . Consequently

$$\sum_G (r(G) - 1) = n_D - h + \sum_G (\sigma(G) - 1).$$

Substitution into the two Picard-number equalities yields

$$\sum_G (\sigma(G) - 1) = h - 1, \quad \sum_{D_G \neq 0} (\sigma(G) - 1) = h - 1 - e_0.$$

A connected component of D either contains a horizontal curve or is one of the $v(G)$ vertical components. Hence $\#\pi_0(D) = c_{\text{hor}} + \sum_{D_G \neq 0} v(G)$. Now sum (3.2) to obtain (3.1). $\square$

Put

$$Q_+ = \sum_{\substack{D_G \neq 0 \\ q(G) > 0}} q(G).$$

The component formula implies

$$(3.3) \quad \#\pi_0(D) \leq c_{\text{hor}} + h - 1 - e_0 + Q_+.$$

There are only three positive possibilities:

$$(3.4) \quad \frac{(\ell(G), t(G))}{q(G)} \mid \begin{array}{ccc} (0, 0) & (1, 0) & (0, 1) \\ 2 & 1 & 1. \end{array}$$

3.2. Separable horizontal curves.

Lemma 3.10. *Assume Setup 3.8, $F \cdot D = 4$, and that every map $H_i \rightarrow \mathbb{P}^1$ is separable. Then $\#\pi_0(D) \leq 7$.*

Ramification accounts for every positive fibre contribution except one possible fibre joining four sections. That exception itself reduces c_{hor} .

Proof. We bound the positive terms in (3.3) by ramification, and then handle the possible fibre at which that bound fails.

Step 1. In this step, we bound the total ramification of the horizontal curves. For $b \in \mathbb{P}^1$, let

$$R_b = 4 - \sum_{i=1}^h \#(H_i \cap f^{-1}(b)).$$

The points in this sum are counted on each H_i separately. If e_x is the local degree at such a point, then $R_b = \sum_x (e_x - 1)$. Each H_i is a smooth rational curve, and its different exponent at x is at least $e_x - 1$. Riemann–Hurwitz gives

$$(3.5) \quad \sum_b R_b \leq \sum_{i=1}^h (2d_i - 2) = 8 - 2h.$$

This inequality allows wild ramification.

Step 2. In this step, we compare positive fiber contributions with ramification and identify the possible exception. Let $G = f^{-1}(b)$ have $q(G) > 0$. If $(\ell(G), t(G)) = (0, 0)$, every horizontal point lies on an internal component outside D . Such a component has multiplicity at least two: a multiplicity-one (-1) -curve would be a leaf. Each point therefore has local degree at least two, so there are at most two horizontal points over b . Thus $R_b \geq 2 = q(G)$.

Suppose $(\ell(G), t(G)) = (1, 0)$ and $R_b = 0$. Every horizontal point has local degree one. It is a smooth point of a multiplicity-one component outside D , hence lies on the unique leaf L outside D . Therefore $L \cdot D_{\text{hor}} = 4 = F \cdot D_{\text{hor}}$, contrary to Theorem 3.7(iii). It follows that $R_b \geq 1 = q(G)$.

Finally suppose $(\ell(G), t(G)) = (0, 1)$ and $R_b = 0$. No horizontal point can lie on a component outside D , since such a point would require a multiplicity-one leaf. All horizontal points consequently lie on the unique connected component V of D_G meeting D_{hor} . If some $d_i \geq 2$, the curve H_i meets V at at least two distinct points. This gives a cycle in the dual graph of $H_i + V$, contrary to the forest property. Thus $h = 4$ and all four horizontal curves are sections.

Step 3. In this step, we treat the four-section exception and conclude the component bound. There is at most one fibre of this last kind. Two such fibres would give two different paths between the same pair of horizontal sections inside the forest D . If there is one, its connected component V joins all four sections, so $c_{\text{hor}} = 1$. Equation (3.5) gives $\sum_b R_b = 0$ in this case, and hence $Q_+ \leq 1$. We obtain

$$\#\pi_0(D) \leq 1 + 4 - 1 - e_0 + 1 \leq 5.$$

If there is no exception, $Q_+ \leq \sum_b R_b \leq 8 - 2h$, and

$$\#\pi_0(D) \leq c_{\text{hor}} + h - 1 - e_0 + 8 - 2h = 7 + c_{\text{hor}} - h - e_0 \leq 7.$$

These two alternatives prove the lemma. $\square$

3.3. A purely inseparable trisection. In characteristic three, an inseparable horizontal curve of degree at most four has degree three and purely inseparable degree three. Thus, when $F \cdot D = 4$, the only inseparable possibility is

$$D_{\text{hor}} = H + J, \quad F \cdot H = 3, \quad F \cdot J = 1,$$

where $H \rightarrow \mathbb{P}^1$ is purely inseparable and J is a section. We use this notation in the lemmas below.

Lemma 3.11. *Assume Setup 3.8 and $D_{\text{hor}} = H + J$ as above. Every fibre G with $q(G) > 0$ has $q(G) = 1$. For each such fibre one of the following holds:*

- (i) *Both H and J meet the same connected component of D_G . There is at most one fibre with this property.*
- (ii) *The unique point of H over the base point of G is a smooth point of a component outside D of multiplicity three in G , and H meets that component transversely.*

We call a positive-contribution fibre in (i) a *connector*. It joins the two horizontal curves inside D . All other positive contributions will require at least three blowups on H .

Proof. We classify the positive contributions and then bound the number of connectors.

Step 1. In this step, we determine the positions of H and J in a fiber with positive contribution. The section J meets a smooth point of a multiplicity-one component. If that component lies outside D , it is a leaf and $\ell(G) > 0$. If it lies in D_G , then $t(G) > 0$. Thus the case $(0, 0)$ in (3.4) cannot occur, and every positive value of $q(G)$ is one.

Suppose $(\ell(G), t(G)) = (1, 0)$. The section meets the unique leaf L outside D . The purely inseparable map on H has one geometric point over the base point of G . This point cannot be a fibre node: one branch in D_G would imply $t(G) > 0$, while two branches outside D would contradict the nonadjacency proved in Lemma 3.9. Hence H meets a smooth point of a component outside D . If m is its fibre multiplicity and I is the local intersection with H , then

$$mI = 3.$$

The choice $(m, I) = (1, 3)$ would put this point on L and give $L \cdot (H + J) = 4$, contradicting Theorem 3.7(iii). Thus $(m, I) = (3, 1)$, as in (ii).

Suppose $(\ell(G), t(G)) = (0, 1)$. The section meets the unique connected component V of D_G meeting D_{hor} . If H meets V , we have (i). Otherwise H again avoids fibre nodes and meets a component outside D , with $mI = 3$. The choice $m = 1$ would give a leaf outside D , contrary to $\ell(G) = 0$. This proves (ii).

Step 2. In this step, we prove that at most one connector occurs. Two connectors would give two distinct paths from H to J in the dual graph of D . That graph is a forest, so there is at most one. $\square$

Lemma 3.12. *Let Y be a smooth projective surface over k with a $\mathbb{P}^1$ -fibration $f : Y \rightarrow \mathbb{P}^1$. Let $H \subset Y$ be a smooth rational curve such that $f|_H$ is purely inseparable of degree three. Then there is a birational morphism $g : Y \rightarrow \mathbb{P}^1 \times \mathbb{P}^1$ over the base such that $Q = g(H)$ is smooth. If A is a section of square zero and L is a fibre on this product, then $Q \equiv 3A + L$ and $Q^2 = 6$.*

The morphism is dominated by the given surface Y . In the application where Y is the original minimal resolution, the blowups on Q can therefore be counted without replacing Y by a higher resolution.

Proof. We first make the image of H smooth by elementary transformations dominated by Y , then determine the resulting ruled surface.

Step 1. In this step, we make the image of H smooth by elementary transformations dominated by Y . Contract fibre components to obtain a relatively minimal ruled surface Z dominated by

Y , and let C be the image of H . Write $Z \simeq \mathbb{F}_n$ for $n \geq 0$, with a section A_n of square $-n$ and fibre class L_Z . For $C \equiv 3A_n + \beta L_Z$, adjunction gives

$$C^2 = -9n + 6\beta, \quad p_a(C) = 2\beta - 3n - 2, \quad C^2 = 3p_a(C) + 6.$$

Let z be a singular point of C . A point-blowup factorization of $Y \rightarrow Z$ must have a centre over z , since its strict transform H is smooth. The first such centre is z itself. Centres over other points can be moved past this first blowup, so Y dominates $\text{Bl}_z Z$. It therefore also dominates the elementary transformation obtained by contracting the strict transform of the fibre through z .

Put $m = \text{mult}_z(C)$. Since C has total intersection three with that fibre, $2 \leq m \leq 3$. After the elementary transformation the image C' satisfies

$$(C')^2 - C^2 = -m^2 + (3 - m)^2 = 9 - 6m, \quad p_a(C') - p_a(C) = 3 - 2m < 0.$$

The second equality uses the preceding adjunction identity on both ruled surfaces. An integral projective curve has nonnegative arithmetic genus. Repeating this operation therefore terminates with a smooth image Q on a ruled surface still dominated by Y .

Step 2. In this step, we identify the resulting ruled surface and compute the numerical class of Q . The curve Q is rational. Write the final ruled surface as $\mathbb{F}_{n_0}$, with $n_0 \geq 0$, a section A_{n_0} of square $-n_0$, and fibre class L_0 . Write $Q \equiv 3A_{n_0} + \beta_0 L_0$. Its genus gives

$$\beta_0 = \frac{3n_0 + 2}{2}, \quad Q^2 = 6, \quad Q \cdot A_{n_0} = \frac{2 - 3n_0}{2}.$$

The curves Q and A_{n_0} are distinct, so their intersection is nonnegative. Hence $n_0 = 0$ and $\beta_0 = 1$, proving the assertion. $\square$

Lemma 3.13. *Assume Setup 3.8 and $D_{\text{hor}} = H + J$ with H purely inseparable of degree three and J a section. Let g and Q be as in Lemma 3.12. In any point-blowup factorization of g , let N_H be the number of centres on the successive strict transforms of Q . Then $N_H \leq 11$.*

The bound uses the ampleness of $-K_S$ to restrict the square of the exceptional trisection on the minimal resolution.

Proof. We express N_H in terms of the square of H and then bound that square by canonical intersections.

Step 1. In this step, we compute the number of centers on the successive transforms of Q . Write $H^2 = -a$, where $a > 0$ by negative definiteness. The curve Q and all its strict transforms are smooth. Every centre on a strict transform lowers its square by one, and every other centre leaves its square unchanged. Thus

$$(3.6) \quad N_H = Q^2 - H^2 = 6 + a.$$

Step 2. In this step, we use the canonical pullback coefficients to prove $a \leq 5$. Intersect the pullback formula in Theorem 3.7 with H . Adjunction on the smooth rational curve H gives

$$0 = a - 2 - a\gamma_H + \sum_{\substack{E \subset D \\ E \neq H}} \gamma_E(E \cdot H), \quad \gamma_H \geq 1 - \frac{2}{a}.$$

All terms in the sum are nonnegative. On the general fibre, ampleness and the projection formula give

$$0 < p^*(-K_S) \cdot F = 2 - 3\gamma_H - \gamma_J.$$

Since $\gamma_J \geq 0$, these inequalities imply

$$3 \left(1 - \frac{2}{a}\right) \leq 3\gamma_H < 2.$$

Hence $a < 6$, so $a \leq 5$ and (3.6) gives $N_H \leq 11$. $\square$

Lemma 3.14. *Let $g : Y \rightarrow Z$ be a birational morphism of smooth projective surfaces over $\mathbb{P}^1$ over k , where $Z \rightarrow \mathbb{P}^1$ is a $\mathbb{P}^1$ -bundle. Let $Q \subset Z$ be smooth with purely inseparable map of degree three to the base, and let H be its strict transform. Fix a fibre G on Y . If H meets G at a smooth point of its reduced support, transversely to a component of multiplicity three, then every point-blowup factorization of g has at least three centres on successive transforms of Q over that base point.*

The first two blowups leave the trisection at a fibre node. The third separates it from the two branches and produces the required component of multiplicity three.

Proof. A purely inseparable degree-three map of smooth curves is the relative Frobenius followed by an isomorphism. Thus the map on completed local rings sends a base parameter to the cube of a parameter on Q . Choose formal coordinates (x, y) at the point of Q in question, with base parameter x , so that

$$Q : x = y^3.$$

Before a blowup, Q has contact order three with the multiplicity-one fibre $x = 0$. For the first blowup use $x = uv$, $y = u$. The strict transform of Q is $v = u^2$ and the fibre is $uv = 0$. Thus Q passes through the node of two multiplicity-one components.

For the second blowup, put $v = uw$. Then

$$x = u^2w, \quad Q : w = u.$$

The curve passes through the node of components of multiplicities two and one. For the third blowup put $w = uz$. Then

$$x = u^3z, \quad Q : z = 1.$$

Now Q meets a smooth point of the multiplicity-three component $u = 0$, transversely and away from its attaching points. A blowup whose centre is off the strict transform of Q is an isomorphism near that curve and does not change these local descriptions. Consequently the required final position cannot be reached with zero, one, or two centres on Q . $\square$

Lemma 3.15. *Assume Setup 3.8, $F \cdot D = 4$, and that some horizontal map is inseparable. Then $\#\pi_0(D) \leq 7$.*

The eleven centres allow at most three positive-contribution fibres other than a connector. There is at most one connector.

Proof. The inseparable degree is a power of three dividing the degree of the horizontal curve. Since the total horizontal degree is four, the horizontal part is $H + J$ as above. In particular, $h = 2$. Let M be the number of positive-contribution fibres, and let $j \leq 1$ be the number of connectors. By Lemma 3.11, all positive contributions are one, so $Q_+ = M$.

Choose the smooth ruled model from Lemma 3.12. Each of the $M - j$ fibres that is not a connector has the trisection in the position described in Lemma 3.14. Centres over distinct base points are distinct. Therefore

$$3(M - j) \leq N_H \leq 11$$

by Lemma 3.13. Since $M - j$ is an integer, $M - j \leq 3$, and hence $Q_+ = M \leq 4$. Finally (3.3), $c_{\text{hor}} \leq 2$, and $e_0 \geq 0$ give

$$\#\pi_0(D) \leq c_{\text{hor}} + 2 - 1 - e_0 + Q_+ \leq 2 + 1 + 4 = 7.$$

$\square$

Theorem 3.16. *Let S be a rank-one klt del Pezzo surface over k , let $p : Y \rightarrow S$ be its minimal resolution, and put $D = \text{Exc}(p)_{\text{red}}$. If Y admits a $\mathbb{P}^1$ -fibration $f : Y \rightarrow \mathbb{P}^1$ whose general fibre F satisfies $F \cdot D = 4$, then $\#\text{Sing}(S) \leq 7$.*

The hypothesis controls the total degree of the exceptional horizontal curves for this particular fibration; it does not require the fibration to realize the height.

Proof. The given resolution and fibration have the notation of Setup 3.8. By Theorem 3.7, $\# \text{Sing}(S) = \# \pi_0(D)$. If every horizontal map is separable, apply Lemma 3.10. Otherwise apply Lemma 3.15. The former lemma accounts for the ramification and the four-section exception; the latter accounts for the purely inseparable trisection. Both give $\# \pi_0(D) \leq 7$. $\square$

Corollary 3.17. *Let S be a rank-one klt del Pezzo surface over k , and let $p : Y \rightarrow S$ be its minimal resolution with $D = \text{Exc}(p)_{\text{red}}$. If there is a $\mathbb{P}^1$ -fibration on Y over a smooth projective curve whose general fibre F satisfies $F \cdot D \leq 4$, then $\# \text{Sing}(S) \leq 7$.*

This combines the exact degree-four argument with the lower-height classifications, and permits the fibration to have any smooth base.

Proof. The given fibration shows that $\text{ht}(S) \leq 4$. If $\text{ht}(S) \leq 2$, apply Theorem 3.2. If $\text{ht}(S) = 3$, apply Theorem 3.6 when an elliptic-boundary descendant exists, and Proposition 3.4 otherwise.

It remains to consider $\text{ht}(S) = 4$. The given fibration then has $F \cdot D = 4$, so D has a horizontal component. By Theorem 3.7, this component is a smooth rational curve. Its map to the smooth projective base shows that the base is rational: its function field is a subfield of $k(t)$, so Lüroth's theorem applies, also in the inseparable case. The base is therefore $\mathbb{P}^1$. Theorem 3.16 gives the required bound. $\square$

4. LOCAL ADJUNCTION AND BIRATIONAL COUNTS

We retain the intersection and discrepancy conventions of Section 2. We first compare resolutions and compute adjunction along a reduced dlt boundary. We then bound the singularities on a contracted curve and count the change in the singular locus under birational links. These arguments work over an algebraically closed field of arbitrary characteristic. The final subsection returns to k and constructs a partial resolution retaining one end of an exceptional chain.

4.1. Surface resolutions and local pairs. We recall the surface results used below, beginning with the construction and descent of morphisms between resolutions.

Theorem 4.1 (Resolution of indeterminacy). *Let B be a scheme, let V be a regular integral scheme of dimension two over B , and let R be proper over B . If a B -rational map $\varphi : V \dashrightarrow R$ is defined on a dense open subset U , then there is a finite sequence $\beta : V' \rightarrow V$ of blowups at closed points outside the successive inverse images of U and a B -morphism $V' \rightarrow R$ extending φ .*

Thus point blowups resolve the indeterminacy while preserving the map to the base; see [Sta26, Tag 0C5H].

Theorem 4.2 (Birational factorization). *A proper birational morphism between integral Noetherian schemes that are regular of dimension two is a finite composite of blowups at closed points.*

In the direction of the morphism, this is a sequence of contractions of exceptional curves of the first kind; see [Sta26, Tag 0C5R].

Theorem 4.3 (Contraction of an exceptional curve). *Let B be Noetherian, let $V \rightarrow B$ be a finite-type morphism, and let $E \subset V$ be an exceptional curve of the first kind contained in a fibre. If $V \rightarrow B$ is projective, there is a contraction $V \rightarrow V_0$ of E such that $V_0 \rightarrow B$ is projective. The same assertion holds with quasi-projective in place of projective. Here an exceptional curve of the first kind is a projective line over its residue field, lying in the regular locus, with normal bundle of degree -1 ; its contraction is the inverse of the blowup of a regular point. Over an algebraically closed field, a smooth rational curve of self-intersection -1 on a smooth surface is such a curve.*

The contraction remains projective over the given base, which is necessary when comparing resolutions over one surface; see [Sta26, Tag 0C2N].

Theorem 4.4 (Descent through a contraction). *Let V be Noetherian, let $E \subset V$ be an exceptional curve of the first kind, and let $b : V \rightarrow V_0$ be a contraction of E . If a morphism $\gamma : V \rightarrow R$ sends E to a point, then there is a unique morphism $\gamma_0 : V_0 \rightarrow R$ with $\gamma = \gamma_0 b$.*

Constancy on the contracted curve is precisely the descent condition in [Sta26, Tag 0C5J].

Theorem 4.5 (Exceptional negativity). *Let $f : V \rightarrow Z$ be a projective birational morphism from a smooth surface to a normal surface over an algebraically closed field. The intersection matrix of the irreducible exceptional curves of f is negative definite.*

This form of exceptional negativity holds in arbitrary characteristic; see [Fuj21, Theorem 1.1].

Theorem 4.6 (Smoothness of a dlt boundary). *Let S be a normal surface over an algebraically closed field, let $p \in S$, and let C be a reduced irreducible curve through p . If S is klt at p , $K_S + C$ is $\mathbb{Q}$ -Cartier near p , and (S, C) is dlt at p , then C is smooth at p .*

This is the local boundary statement in [Lac24, Lemma A.7]. In positive characteristic it is also a consequence of [Tan16, Proposition 5.5 and Corollary 5.6].

Theorem 4.7 (Liu–Xie). *Let $(S \ni p, B)$ be a plt surface germ over an algebraically closed field, with effective boundary B , and let $\pi : V \rightarrow S$ be its minimal resolution. Write B_V for the strict transform of B . Then $B_V \cdot H < 2$ for every exceptional prime H , and at most one exceptional prime has $B_V \cdot H \geq 1$. If such a prime exists, the exceptional divisor is a chain of smooth rational curves and this prime is an end of the chain.*

The statement is [LX23, Lemma 4.1(i)–(iii)], with arbitrary characteristic explicitly allowed by [LX23, Remark 1.7]. The chain may have weights greater than two.

Theorem 4.8 (Connected fibres). *Let $f : V \rightarrow Z$ be a proper birational morphism of integral schemes, with V reduced and Z normal. Then $f_* \mathcal{O}_V = \mathcal{O}_Z$, and the fibres of f are geometrically connected.*

This is the birational case of [Sta26, Tag 0AY8]: the generic fibre is the spectrum of the common function field.

We now use the resolution results to compare a smooth model with the minimal resolution.

Lemma 4.9. *Let S be a normal projective surface over an algebraically closed field, and let $\theta : R \rightarrow S$ be its minimal resolution. If V is a smooth projective surface and $\eta : V \rightarrow S$ is a birational morphism, then there is a unique projective birational morphism $\alpha : V \rightarrow R$ such that $\eta = \theta \alpha$.*

Thus a smooth model mapping to S also maps to its minimal resolution. The assertion concerns morphisms, so it can be used to pull back a fibration without introducing further blowups.

Proof. We resolve the rational map to R and then descend it through each additional blowup.

Step 1. In this step, we resolve the rational map from V to R over S . By Theorem 4.1, there is a sequence of point blowups $\beta : V' \rightarrow V$ and a morphism $\gamma : V' \rightarrow R$ over S with

$$\eta \beta = \theta \gamma.$$

Step 2. In this step, we descend the morphism to R through each additional blowup. Let $b : W' \rightarrow W$ be the last blowup in this sequence, with exceptional curve F , and suppose that $\gamma' : W' \rightarrow R$ has been constructed over S . We show that $\gamma'(F)$ is a point. Otherwise its image is a θ -exceptional curve G . By Theorem 4.2, the morphism γ' is a sequence of contractions of smooth rational (-1) -curves. Since F survives this sequence, the self-intersection of its image increases at each contraction by the square of its intersection with the contracted curve. Consequently

$$G^2 \geq F^2 = -1.$$

Theorem 4.5 gives $G^2 < 0$. Hence $G^2 = -1$, and every increase in the preceding calculation is zero. All the contractions are therefore disjoint from the successive images of F . It follows that G is a smooth rational (-1) -curve. Theorem 4.3 contracts G over S to a smooth point, contradicting the minimality of R .

We have proved that $\gamma'(F)$ is a point. The descent property in Theorem 4.4 now gives a morphism $W \rightarrow R$ through which γ' factors. Repeating this for the remaining blowups gives $\alpha : V \rightarrow R$.

Step 3. In this step, we verify the factorization, uniqueness, and projectivity of α . The identity $\eta = \theta\alpha$ holds on the common birational open set, hence everywhere. The same observation proves uniqueness. The graph of α is closed in $V \times_S R$, and the projection of this product to R is projective. Thus α is projective. $\square$

The comparison also identifies the local and global minimal resolutions. Indeed, restrict a global minimal resolution to an open neighbourhood of one singular point. Its exceptional curves still contain no smooth rational (-1) -curve. The preceding descent argument works over this neighbourhood, since all the blowups and contractions are projective over the base. It gives a morphism from the restricted resolution to the local minimal resolution. If that morphism were not an isomorphism, Theorem 4.2 would supply a smooth rational (-1) -curve in its source contracted over the base. This is impossible. Thus the exceptional fibre and its intersection matrix are the same on the local and global minimal resolutions.

4.2. Adjunction along a dlt boundary.

Proposition 4.10. *Let S be a klt surface over an algebraically closed field, and let C be a reduced irreducible curve with $K_S + C$ $\mathbb{Q}$ -Cartier. Assume (S, C) is dlt at every point of C . Then C is smooth. Over each $p \in C \cap \text{Sing}(S)$ the minimal resolution has a nonempty chain $[b_{p,1}, \dots, b_{p,r_p}]$, with $b_{p,i} \geq 2$, and C_Y meets exactly one end transversely once. Let d_p be the determinant of the negative intersection matrix of this chain. If S is projective, then*

$$(K_S + C) \cdot C = 2p_a(C) - 2 + \sum_{p \in C \cap \text{Sing}(S)} \left(1 - \frac{1}{d_p}\right), \quad d_p \geq 2.$$

Each singular point on the boundary contributes at least $1/2$ to adjunction. The contribution depends only on the determinant of its exceptional chain.

Proof. We determine the local configuration, solve the intersection equations on its chain, and then restrict the global pullback to the boundary.

Step 1. In this step, we identify the exceptional chain and its transverse end intersection with the boundary. First, (S, C) is plt along C . To check the exceptional divisors, take a log resolution witnessing the dlt condition and write its log pullback as

$$K_V + C_V + \sum_j t_j H_j, \quad t_j < 1.$$

Its support has simple normal crossings, and C_V is its only component with coefficient one. Blowing up a crossing of coefficients s, t gives exceptional coefficient $s + t - 1 < 1$; blowing up a point of one component gives coefficient $s - 1 < 1$; blowing up off the support gives coefficient -1 . After any such blowup the strict transform of C_V remains the only component with coefficient one. Every exceptional divisor over V is obtained by a sequence of point blowups, so induction proves the plt assertion. Theorem 4.6 gives that C is smooth.

Fix $p \in C \cap \text{Sing}(S)$, and let $\pi_p : Y_p \rightarrow S$ denote the minimal resolution over a neighbourhood of p . Its strict transform C_{Y_p} meets the exceptional fibre: the map from this strict transform to C is proper with dense image. Theorem 4.7 applies in arbitrary characteristic. It gives

$$C_{Y_p} \cdot H < 2 \quad \text{for every exceptional prime } H, \quad \#\{H : C_{Y_p} \cdot H \geq 1\} \leq 1.$$

These intersections are nonnegative integers. Thus exactly one is one, and all the others are zero. The same theorem says that the exceptional fibre is a chain of smooth rational curves and that the curve with intersection one is an end. Order it as $E_1, \dots, E_m$, beginning at this end. An intersection number one on a smooth surface means one transverse intersection of smooth branches. Since C_{Y_p} meets no other E_i , this point is not a node of the exceptional chain. Minimality gives $E_i^2 = -b_i$ with $b_i \geq 2$. In particular, this minimal resolution of the surface already resolves the pair near p .

Step 2. In this step, we compute the coefficient at the marked end and prove that the chain determinant is at least two. Choose compatible canonical divisors and write, near the fibre over p ,

$$\pi_p^*(K_S + C) = K_{Y_p} + C_{Y_p} + \sum_{i=1}^m c_i E_i.$$

Let $B = -(E_i \cdot E_j)_{i,j}$, let $c = (c_i)_{i=1}^m$, and let ε_i denote the i th standard basis vector in $\mathbb{Q}^m$. Adjunction gives $K_{Y_p} \cdot E_i = b_i - 2$. Intersecting the displayed pullback with each E_i therefore gives

$$Bc = (b_i - 2)_{i=1}^m + \varepsilon_1.$$

For $m = 1$ this says $b_1 c_1 = b_1 - 1$, so $c_1 = 1 - 1/b_1$. For $m \geq 2$, let $\mathbf{1}$ be the vector all of whose entries are one. The first and last entries of $B\mathbf{1}$ are $b_1 - 1$ and $b_m - 1$, while its other entries are $b_i - 2$. Hence

$$B(\mathbf{1} - c) = \varepsilon_m, \quad 1 - c_1 = (B^{-1})_{1m}.$$

The minor obtained by deleting row m and column 1 has determinant $(-1)^{m-1}$. Its cofactor has the additional sign $(-1)^{m+1}$, so

$$(B^{-1})_{1m} = \frac{1}{\det B}.$$

Both cases give $c_1 = 1 - 1/d_p$.

For completeness, let Δ_j be the determinant of the leading $j \times j$ submatrix of B , with $\Delta_0 = 1$. Then

$$\Delta_1 = b_1 \geq 2, \quad \Delta_j = b_j \Delta_{j-1} - \Delta_{j-2} \quad (j \geq 2).$$

If $\Delta_{j-1} > \Delta_{j-2} > 0$, then

$$\Delta_j - \Delta_{j-1} = (b_j - 2)\Delta_{j-1} + \Delta_{j-1} - \Delta_{j-2} > 0.$$

Induction proves $d_p = \Delta_m \geq 2$.

Step 3. In this step, we restrict the global pullback to C_Y and compute the different. Assume now that S is projective, and let $\pi : Y \rightarrow S$ be its global minimal resolution. By the comparison above, its fibre over each p is the chain just used, with the same self-intersections and the same marked end. The strict transform C_Y is smooth: this holds over the smooth locus of S because C is smooth, and over a singular point because its intersection with the exceptional fibre is transverse. The morphism $C_Y \rightarrow C$ is proper and birational between smooth curves, and is therefore an isomorphism.

Write $\pi^*(K_S + C) = K_Y + C_Y + R$, with R exceptional. Fibres over points outside C do not meet C_Y . Over a point $p \in C \cap \text{Sing}(S)$, only the marked end meets C_Y , and its coefficient in R is $1 - 1/d_p$. Restriction and adjunction on Y therefore identify

$$\text{Diff}_C = \sum_{p \in C \cap \text{Sing}(S)} \left(1 - \frac{1}{d_p}\right) p.$$

There is no contribution at a smooth point of S , where ordinary adjunction applies to the smooth Cartier curve C . Taking degrees, or equivalently intersecting the pullback with C_Y , gives

$$\begin{aligned} (K_S + C) \cdot C &= (K_Y + C_Y) \cdot C_Y + R \cdot C_Y \\ &= 2p_a(C) - 2 + \sum_{p \in C \cap \text{Sing}(S)} \left(1 - \frac{1}{d_p}\right). \end{aligned}$$

□

4.3. Singularities on a contracted curve.

Lemma 4.11. *Let Z be a normal projective $\mathbb{Q}$ -Gorenstein surface over an algebraically closed field. Let $\nu : V \rightarrow Z$ be its minimal resolution, with exceptional primes $H_1, \dots, H_n$, and choose compatible canonical divisors such that*

$$K_V = \nu^* K_Z + \sum_{i=1}^n a_i H_i.$$

Then $a_i \leq 0$ for every i . If $n > 0$, the matrix $A = -(H_i \cdot H_j)_{i,j}$ has entrywise nonnegative inverse.

This sign statement holds without a klt hypothesis. It distinguishes a divisor on a minimal resolution from a divisor introduced by further blowups.

Proof. There is nothing to prove when $n = 0$. We prove the sign of A^{-1} and then use the canonical intersection equations to bound the discrepancies.

Step 1. In this step, we prove that A^{-1} is entrywise nonnegative. By Theorem 4.5, A is positive definite, and its off-diagonal entries are nonpositive. If $Ax = u$ with $u \geq 0$ coordinatewise, write $x = x_+ - x_-$ for its positive and negative parts, which have disjoint supports. If $x_- \neq 0$, then

$$0 \leq u^\dagger x_- = x_+^\dagger A x_- - x_-^\dagger A x_- < 0.$$

Here the first term is nonpositive by the disjoint supports and the off-diagonal signs, and the second is strictly negative by positive definiteness. Thus $x \geq 0$. Applying this to the standard basis vectors proves the assertion about A^{-1} .

Step 2. In this step, we prove that every minimal-resolution discrepancy is nonpositive. Put $u_i = K_V \cdot H_i$. Adjunction gives

$$u_i = 2p_a(H_i) - 2 - H_i^2 \geq 0.$$

Indeed, if $p_a(H_i) = 0$, the integral curve H_i is smooth rational, and minimality excludes $H_i^2 = -1$. Thus $H_i^2 \leq -2$ in this case. If $p_a(H_i) \geq 1$, the inequality follows from $H_i^2 < 0$. Intersecting the canonical formula with the H_i gives $A(-a) = u$. Since A^{-1} is entrywise nonnegative, $-a \geq 0$. □

Proposition 4.12. *Let $g : T \rightarrow S$ be a projective birational morphism of normal projective $\mathbb{Q}$ -Gorenstein surfaces over an algebraically closed field. Assume T is $\mathbb{Q}$ -factorial, $\text{Exc}(g) = \Sigma$ is irreducible, and $K_T \cdot \Sigma < 0$. Then*

$$\#(\text{Sing}(T) \cap \Sigma) \leq 2, \quad 0 < -K_T \cdot \Sigma \leq 1.$$

The same intersection calculation bounds both the number of singular points on Σ and its anticanonical degree.

Proof. We first identify the strict transform of Σ as a (-1) -curve. We then assign to each singular point on Σ two nonnegative intersection contributions and bound their sums.

Step 1. In this step, we identify the strict transform of Σ as a smooth rational (-1) -curve. Let $\tau : X \rightarrow T$ and $\sigma : Y \rightarrow S$ be the minimal resolutions, and let C be the strict transform of Σ on X . Lemma 4.9 gives a morphism $h : X \rightarrow Y$ with $\sigma h = g\tau$.

The curve Σ is $\mathbb{Q}$ -Cartier because T is $\mathbb{Q}$ -factorial. Its pullback $\tau^*\Sigma$ is a nonzero divisor supported on the exceptional curves of $g\tau$. Thus Theorem 4.5 gives $\Sigma^2 = (\tau^*\Sigma)^2 < 0$. For compatible canonical divisors,

$$K_T = g^*K_S + d\Sigma, \quad d = \frac{K_T \cdot \Sigma}{\Sigma^2} > 0.$$

Since τ is an isomorphism at the generic point of C , the discrepancy of C over S is d . If h did not contract C , its image on Y would be a σ -exceptional prime with the same discrepancy. This contradicts Lemma 4.11. Therefore C is h -exceptional.

Every other h -exceptional curve is τ -exceptional. Otherwise its image on T would be a curve contracted by g , hence would be Σ , whose unique strict transform is C . By Theorem 4.2, the nontrivial morphism h has a smooth rational (-1) -curve in its exceptional locus. Minimality of τ excludes all the τ -exceptional curves from being such a curve. It follows that

$$C^2 = -1, \quad K_X \cdot C = -1.$$

Step 2. In this step, we prove the inequality $2q_p + r_p \geq 1$ for each singular point on Σ . Put $P = \text{Sing}(T) \cap \Sigma$. For $p \in P$, write the reduced τ -exceptional fibre as $\sum_i H_{p,i}$ and set

$$A_p = -(H_{p,i} \cdot H_{p,j})_{i,j}, \quad v_{p,i} = C \cdot H_{p,i}, \quad u_{p,i} = K_X \cdot H_{p,i}.$$

The vector v_p is a nonzero vector of nonnegative integers. To see nonvanishing, $C \rightarrow \Sigma$ is proper and surjective, so C meets the fibre over p ; the fibre is supported on exceptional curves. As in Lemma 4.11, adjunction and minimality give $u_p \geq 0$, and the inverse of A_p is entrywise nonnegative. Define

$$z_p = A_p^{-1}v_p, \quad q_p = v_p^t z_p, \quad r_p = u_p^t z_p.$$

All these vectors and numbers are nonnegative.

Choose i such that $v_{p,i} \geq 1$, and put $w_i = -H_{p,i}^2$. The i th row of $A_p z_p = v_p$ gives

$$v_{p,i} = w_i z_{p,i} - \sum_{j \neq i} (H_{p,i} \cdot H_{p,j}) z_{p,j} \leq w_i z_{p,i}.$$

Adjunction also gives

$$u_{p,i} + 2v_{p,i} = w_i + 2(p_a(H_{p,i}) + v_{p,i} - 1) \geq w_i.$$

Since all coordinates of $u_p + 2v_p$ and z_p are nonnegative,

$$(4.1) \quad 2q_p + r_p = (2v_p + u_p)^t z_p \geq w_i z_{p,i} \geq 1.$$

Step 3. In this step, we sum the local contributions to bound the singular-point count and anticanonical degree. The pullback of Σ is

$$\tau^*\Sigma = C + \sum_{p \in P} \sum_i z_{p,i} H_{p,i}.$$

Indeed, the equations $A_p z_p = v_p$ express its zero intersections with the exceptional curves. Fibres over points not on Σ have zero coefficients. Different fibres are disjoint, so squaring yields

$$\Sigma^2 = C^2 + \sum_{p \in P} (2v_p^t z_p - z_p^t A_p z_p) = -1 + \sum_{p \in P} q_p.$$

Consequently $\sum_p q_p < 1$.

Write the canonical pullback as

$$\tau^*K_T = K_X + \sum_{x \in \text{Sing}(T)} \sum_i e_{x,i} H_{x,i}.$$

Intersecting with the fibre over $p \in P$ gives $A_p e_p = u_p$. Using symmetry of A_p and $K_X \cdot C = -1$, we obtain

$$K_T \cdot \Sigma = (\tau^*K_T) \cdot C = -1 + \sum_{p \in P} e_p^t v_p = -1 + \sum_{p \in P} r_p.$$

Thus $\sum_p r_p < 1$, and nonnegativity gives $K_T \cdot \Sigma \geq -1$. Finally, summing (4.1) gives

$$\#P \leq 2 \sum_{p \in P} q_p + \sum_{p \in P} r_p < 3.$$

The cardinality is an integer, so $\#P \leq 2$. The canonical intersection formula gives the asserted anticanonical degree bound. $\square$

4.4. Counting points and following exceptional curves.

Lemma 4.13. *Let S, T, S' be normal projective surfaces with finite singular loci, and let $S \xleftarrow{f} T \xrightarrow{g} S'$ be birational morphisms. Assume f contracts exactly one curve E to a singular point x , $f^{-1}(x) = E$ as sets, and f is an isomorphism off x . Assume g contracts exactly one curve Σ to q and is an isomorphism off q . Put $r = \#(\text{Sing}(T) \cap E)$, $\kappa = \#(\text{Sing}(T) \cap \Sigma)$, and $\delta = 1$ if $q \in \text{Sing}(S')$, with $\delta = 0$ otherwise. Then*

$$\# \text{Sing}(S) = \# \text{Sing}(S') + 1 - r + \kappa - \delta.$$

If $E \neq \Sigma$, $C = g(E)$, $m = \#(C \cap \text{Sing}(S'))$, and $\kappa \leq 2$, then $\# \text{Sing}(S) \leq \# \text{Sing}(S') + 3 - m$.

The identity counts points on the open sets where the two maps are isomorphisms. In the inequality, singular points on the marked image curve reduce the possible increase in the total count.

Proof. We count singular points on the two isomorphism loci and then compare this count with the singular points on C .

Step 1. In this step, we prove the singular-point identity for the link. The fibre $g^{-1}(q)$ is Σ as a set. Indeed, it is connected by Theorem 4.8, since S' is normal. Its only possible one-dimensional component is Σ . Any point outside that component would be an isolated component of the fibre, contrary to connectedness.

The isomorphism loci give bijections of finite sets

$$\text{Sing}(T) \setminus E \longleftrightarrow \text{Sing}(S) \setminus \{x\}, \quad \text{Sing}(T) \setminus \Sigma \longleftrightarrow \text{Sing}(S') \setminus \{q\}.$$

Since x is singular, these bijections yield

$$\# \text{Sing}(T) = \# \text{Sing}(S) - 1 + r, \quad \# \text{Sing}(S') = \# \text{Sing}(T) - \kappa + \delta.$$

Eliminating $\# \text{Sing}(T)$ proves the identity. No disjointness between $E \cap \text{Sing}(T)$ and $\Sigma \cap \text{Sing}(T)$ is needed.

Step 2. In this step, we use the singular points on C to bound the increase in the total count. Now assume $E \neq \Sigma$ and put $C = g(E)$. Every point of $C \cap \text{Sing}(S')$ other than q has a unique singular preimage in $E \setminus \Sigma$. This gives an injection

$$(C \cap \text{Sing}(S')) \setminus \{q\} \hookrightarrow E \cap \text{Sing}(T).$$

The point q contributes at most δ to m , whether or not it lies on C . Hence $m \leq r + \delta$. Combining this with $\kappa \leq 2$ and the identity gives

$$\# \text{Sing}(S) \leq \# \text{Sing}(S') + 3 - r - \delta \leq \# \text{Sing}(S') + 3 - m.$$

$\square$

Remark 4.14. We describe the morphisms needed for the next comparison. Suppose that $p : Y \rightarrow S_0$ is a minimal resolution and that a partial resolution $f_1 : T_1 \rightarrow S_0$ is constructed by retaining one exceptional prime E_Y on Y and contracting the other exceptional primes over S_0 . This construction supplies a morphism $\tau : Y \rightarrow T_1$ with $p = f_1 \tau$. It maps E_Y birationally onto its image and contracts every other component of $\text{Exc}(p)$.

Let $g_1 : T_1 \rightarrow S_1$ be birational, and let $\theta : R \rightarrow S_1$ be the minimal resolution. Since $g_1 \tau : Y \rightarrow S_1$ is a morphism from a smooth projective surface, Lemma 4.9 gives $\alpha : Y \rightarrow R$ with

$\theta\alpha = g_1\tau$. If a second partial resolution $f_2 : T_2 \rightarrow S_1$ is constructed from R by retaining one exceptional prime, write $c : R \rightarrow T_2$ for that contraction, so $\theta = f_2c$. Then

$$\mu = c\alpha : Y \longrightarrow T_2, \quad f_2\mu = g_1\tau.$$

Thus these partial-resolution constructions provide the required morphisms. The existence of c must be checked from the construction of the second extraction; it is not an assertion about an arbitrary birational model of S_1 .

Proposition 4.15. *Let $S_0 \xleftarrow{f_1} T_1 \xrightarrow{g_1} S_1 \xleftarrow{f_2} T_2$ be birational morphisms of normal projective surfaces. Let $p : Y \rightarrow S_0$ be a minimal resolution with reduced exceptional divisor D , and suppose that morphisms $\tau : Y \rightarrow T_1$ and $\mu : Y \rightarrow T_2$ satisfy $p = f_1\tau$ and $f_2\mu = g_1\tau$. Assume that exactly one component E_Y of D is not contracted by τ , that $E_1 = \tau(E_Y)$ maps birationally under g_1 to a curve A_1 , and that f_2 extracts exactly one curve E_2 over a point and is an isomorphism at the generic point of A_1 . Write A_T for the strict transform of A_1 on T_2 . Then every component of D maps under μ to a point, A_T , or E_2 . For any fibration on T_2 , the horizontal components of D are among the strict transforms of its horizontal curves A_T, E_2 .*

Once the comparison morphisms exist, only two curves on T_2 can carry a horizontal component of the original exceptional divisor.

Proof. The equalities $p = f_1\tau$ and $f_2\mu = g_1\tau$ show on function fields that τ and μ are birational. Let G be an irreducible component of D . If $G = E_Y$, then

$$f_2(\mu(G)) = g_1(\tau(G)) = A_1.$$

Its image under μ is therefore a curve. Because f_2 is an isomorphism at the generic point of A_1 , the unique curve mapping onto A_1 is its strict transform A_T . Hence $\mu(G) = A_T$.

If $G \neq E_Y$, then $\tau(G)$ is a point. Thus $f_2(\mu(G)) = g_1(\tau(G))$ is a point. Either $\mu(G)$ is already a point, or it is a curve contracted by f_2 . In the latter case it is E_2 , the unique exceptional curve of f_2 .

Finally let $h : T_2 \rightarrow B$ be a fibration onto a curve. If G is horizontal for $h\mu$, then $\mu(G)$ is a curve and is horizontal for h . The preceding alternatives leave only A_T and E_2 . A birational morphism between normal surfaces has a unique strict transform of each curve on its target, so G is the strict transform of one of these two curves. Every μ -exceptional component maps to a point under $h\mu$ and is vertical. $\square$

4.5. Retaining an end of an exceptional chain. We now construct the partial resolution used when a curve meets just one end of an exceptional chain. We recall the two surface theorems needed for the construction.

Theorem 4.16 (Tanaka). *Let S be a normal surface over k , and let Δ be an effective $\mathbb{Q}$ -divisor such that (S, Δ) is klt. Then S has rational singularities and is $\mathbb{Q}$ -factorial.*

The two conclusions are the successive assertions of [Tan15, Fact 3.4].

Theorem 4.17 (Surface minimal model program). *Let $\pi : V \rightarrow B$ be a projective morphism, where V is a normal $\mathbb{Q}$ -factorial surface over k and B is a quasi-projective k -scheme. Let Δ be an effective $\mathbb{Q}$ -divisor with coefficients in $[0, 1]$. Every $(K_V + \Delta)$ -negative extremal ray over B has a projective contraction. A birational step contracts one irreducible curve and preserves normality and $\mathbb{Q}$ -factoriality. Successive negative birational contractions terminate with a model (V_r, Δ_r) for which either $K_{V_r} + \Delta_r$ is nef over B , or there is a projective contraction $V_r \rightarrow Z$ over B with connected fibres, $\dim Z < 2$, relative Picard number one and $-(K_{V_r} + \Delta_r)$ relatively ample. Here Δ_r is the successive pushforward of Δ .*

This is the $\mathbb{Q}$ -factorial case of [Tan18, Theorems 1.1, 4.4 and 4.5]. The excellent regular ground scheme in that result is $\text{Spec } k$; the relative base B need not be regular. In particular, the theorem applies in characteristic three.

Proposition 4.18. *Let S be a normal projective $\mathbb{Q}$ -factorial klt surface over k , let $p : Y \rightarrow S$ be its minimal resolution, and let $y \in \text{Sing}(S)$. Suppose the reduced exceptional fibre over y is a nonempty simple normal crossings chain $H_1, \dots, H_l$ of smooth rational curves, with $H_i^2 \leq -2$. Let B_Y be a reduced irreducible curve not exceptional for p . Assume that B_Y meets H_1 transversely once and is disjoint from every other p -exceptional prime. Then there exist a normal projective $\mathbb{Q}$ -factorial klt surface Z and projective birational morphisms*

$$Y \xrightarrow{\tau} Z \xrightarrow{h} S, \quad p = h\tau,$$

such that τ contracts exactly the p -exceptional primes other than H_1 . The exceptional locus of h is $G = \tau(H_1) \simeq \mathbb{P}^1$, with $h(G) = y$, and h is an isomorphism over $S \setminus \{y\}$. The morphism τ is an isomorphism near B_Y ; its image B_Z lies in Z^{reg} and satisfies $B_Z \cdot G = 1$.

Proof. We construct the morphism over S by an effective perturbation, show that it contracts precisely the unwanted exceptional curves, and then prove that the retained curve is smooth.

Let C be the reduced sum of the p -exceptional primes other than H_1 . If $C = 0$, take $Z = Y$, $\tau = \text{id}_Y$ and $h = p$. All the assertions then follow from the hypotheses and the smoothness of Y . Assume henceforth that $C \neq 0$.

Step 1. In this step, we construct an effective perturbation supported on the exceptional curves other than H_1 . Choose compatible canonical divisors and write

$$K_Y + \Theta = p^*K_S.$$

Lemma 4.11 gives $\Theta \geq 0$. Since S is klt, every coefficient of Θ is less than one, and the pullback equality shows that (Y, Θ) is klt.

Write $C = C_1 + \dots + C_s$ and put $M = (C_i \cdot C_j)_{i,j}$. Theorem 4.5 makes $-M$ positive definite, with nonpositive off-diagonal entries. The inverse-matrix argument in Lemma 4.11 applies to this matrix as well. Thus the rational vector

$$w = (-M)^{-1}(1, \dots, 1)^t$$

is nonnegative. Its i th equation gives $1 \leq (-C_i^2)w_i$, so every w_i is positive. Consequently

$$F = \sum_{i=1}^s w_i C_i \quad \text{satisfies} \quad \text{Supp } F = C, \quad F \cdot C_i = -1.$$

Choose a rational $\varepsilon > 0$ sufficiently small that $\Delta = \Theta + \varepsilon F$ makes (Y, Δ) klt. To do so, take a log resolution of $\text{Supp}(\Theta + F)$: its finitely many boundary coefficients are strictly below one at $\varepsilon = 0$ and vary linearly with ε . Thus Δ is effective, has coefficients less than one, and

$$L := K_Y + \Delta = p^*K_S + \varepsilon F.$$

Step 2. In this step, we run the surface minimal model program over S while retaining H_1 and klt singularities. Apply Theorem 4.17 to (Y, Δ) over S . Write $h_i : Y_i \rightarrow S$ for the successive models, with $Y_0 = Y$ and $h_0 = p$, and let Δ_i, F_i be the pushforwards of Δ, F . Compatible canonical divisors give

$$L_i := K_{Y_i} + \Delta_i = h_i^*K_S + \varepsilon F_i$$

at every stage. The surfaces Y_i are normal and $\mathbb{Q}$ -factorial, and F_i is effective and $\mathbb{Q}$ -Cartier. If an irreducible curve D is not in $\text{Supp } F_i$, then $F_i \cdot D \geq 0$: a positive Cartier multiple of F_i restricts to an effective divisor on the normalization of D . For a curve contracted over S this gives $L_i \cdot D \geq 0$. Hence each negative contraction removes a component of $\text{Supp } F_i$. In particular, the strict transform of H_1 is never contracted.

The pairs remain klt. Indeed, for a birational step $\varphi_i : Y_i \rightarrow Y_{i+1}$ with exceptional curve E_i , write

$$L_i = \varphi_i^*L_{i+1} + a_i E_i.$$

The self-intersection E_i^2 is negative: pull E_i back to a smooth resolution of Y_i and apply Theorem 4.5 to its nonzero exceptional pullback over Y_{i+1} . Therefore $a_i = (L_i \cdot E_i)/E_i^2 > 0$. On any

higher smooth model the log pullback boundary from (Y_{i+1}, Δ_{i+1}) is the one from (Y_i, Δ_i) minus the effective pullback of $a_i E_i$. Its coefficients remain less than one. This proves the assertion by induction. Since $\Delta_i \geq 0$, each Y_i is itself klt; Theorem 4.16 also gives rational singularities.

Every step is over S . A Mori fibre contraction at any stage would factor the surjective birational map $h_i : Y_i \rightarrow S$ through a scheme of dimension less than two. Such a scheme cannot dominate the surface S . Thus termination gives a projective birational factorization

$$Y \xrightarrow{\tau} Z \xrightarrow{h} S$$

with $K_Z + \tau_* \Delta$ nef over S . In particular, Z is projective over k , normal, $\mathbb{Q}$ -factorial and klt.

Step 3. In this step, we identify the exact contraction locus and prove that τ is an isomorphism near B_Y . Set $F_Z = \tau_* F$. It is effective, h -exceptional and nef over S , since $K_Z + \tau_* \Delta = h^* K_S + \varepsilon F_Z$. If $F_Z \neq 0$, the pullback $P = \tau^* F_Z$ is a nonzero effective p -exceptional divisor on the smooth surface Y . For every p -exceptional prime J , the projection formula gives

$$P \cdot J = F_Z \cdot \tau_* J \geq 0.$$

The right side is zero if τ contracts J , and is nonnegative otherwise because $\tau(J)$ is contracted by h . Writing P as an effective sum of these primes gives $P^2 \geq 0$, contrary to Theorem 4.5. Hence $F_Z = 0$. Every component of C had a positive coefficient in F , so all of them are contracted. The support argument above excludes every other curve.

It follows that h has precisely the exceptional curve $G = \tau(H_1)$, and $h(G) = y$. Connectedness of fibres (Theorem 4.8) excludes isolated points in $h^{-1}(y)$ outside G . Over $S \setminus \{y\}$ its fibres contain no curve, so it is proper and quasi-finite, hence finite. A finite birational morphism to a normal surface is an isomorphism.

We also have $\tau^{-1}(\tau(C)) = C$ set-theoretically. Otherwise a fibre containing a contracted component would contain an isolated point outside C , contradicting Theorem 4.8. Outside $\tau(C)$, the same proper quasi-finite argument shows that τ is an isomorphism. Since B_Y is disjoint from C , this is an isomorphism on a neighbourhood of B_Y . The image B_Z is therefore contained in Z^{reg} , and its sole intersection with G is the transverse intersection inherited from $B_Y \cap H_1$. Thus $B_Z \cdot G = 1$.

Step 4. In this step, we prove that the retained curve G is smooth and isomorphic to $\mathbb{P}^1$. If $l = 1$, the exceptional locus of τ is disjoint from H_1 , so τ is an isomorphism near H_1 , and $G \simeq H_1 \simeq \mathbb{P}^1$. It remains to treat $l \geq 2$. Put $m = l - 1$ and write

$$T_j = H_{j+1}, \quad T_j^2 = -b_j, \quad b_j \geq 2 \quad (1 \leq j \leq m).$$

Near the fibre over y , the morphism τ contracts exactly these tail components. Since $K_Z + G$ is $\mathbb{Q}$ -Cartier, write there

$$\tau^*(K_Z + G) = K_Y + H_1 + \sum_{j=1}^m d_j T_j, \quad u_j = 1 - d_j.$$

Intersection with each T_j is zero. Using $K_Y \cdot T_j = b_j - 2$ and the chain intersections, for $m = 1$ we obtain $b_1 u_1 = 1$. For $m > 1$ the equations are

$$\begin{aligned} b_1 u_1 - u_2 &= 0, \\ b_j u_j - u_{j-1} - u_{j+1} &= 0 \quad (1 < j < m), \\ b_m u_m - u_{m-1} &= 1. \end{aligned}$$

Define $s_1 = b_1$ and $s_j = b_j - 1/s_{j-1}$ for $2 \leq j \leq m$. Since $b_j \geq 2$, induction gives $s_j > 1$. The equations then imply $u_{j+1} = s_j u_j$ for $1 \leq j < m$ and $s_m u_m = 1$. Hence every $u_j > 0$, also when $m = 1$, and every $d_j < 1$.

On Y the chain has simple normal crossings, with coefficient one only on H_1 in this log pullback. Blowing up a point introduces a boundary coefficient equal to the sum of the coefficients of the components through the point minus one. At most two components pass through any

such centre, and at most one has coefficient one; every new exceptional coefficient is therefore less than one. Induction through point blowups, using Theorem 4.2, proves that (Z, G) is plt at the image of the tail. Elsewhere along G , the morphism is an isomorphism from the smooth pair (Y, H_1) . Thus (Z, G) is dlt along G . Since Z is klt, Theorem 4.6 shows that G is smooth. Finally, $H_1 \rightarrow G$ is a proper birational morphism of integral curves and hence finite. The normality of G makes it an isomorphism, so $G \simeq \mathbb{P}^1$. $\square$

5. OPEN PAIRS AND BOUNDARY CONTAINMENT

We retain the notation of Section 2. We compare the exceptional divisor of a minimal resolution with the boundary of an open pair. Once the comparison morphism and boundary containment are established, a ruling of the resolved pair gives a ruling of the original resolution with no larger exceptional degree. In the other case, the common boundary produces an elliptic-boundary descendant. For this second argument, we must show that some exceptional curve survives as a curve. We also record rationality and singularity counts for the canonical surfaces that occur with cuspidal anticanonical boundaries.

For a singular boundary curve, adjunction must also account for its conductor. We use adjunction on the normalization in the form of [Kee99, Adjunction Formula 5.3].

Theorem 5.1 (Keel). *Let S be a normal projective surface over an algebraically closed field, and let $B \subset S$ be a reduced irreducible curve with $K_S + B$ $\mathbb{Q}$ -Cartier. Let $\nu : B^\nu \rightarrow B$ be the normalization, and let J be the divisor on B^ν defined by the conductor ideal. Then there is an effective $\mathbb{Q}$ -divisor Θ on B^ν , supported over $\text{Sing}(S)$, such that*

$$\nu^*((K_S + B)|_B) \sim_{\mathbb{Q}} K_{B^\nu} + J + \Theta.$$

When $B \subset S^{\text{reg}}$, the divisor Θ vanishes. Singularities of the boundary itself still contribute through the conductor divisor J .

We use the following form of [PP24b, Theorem 1.2(a)] in characteristic three.

Theorem 5.2 (Palka–Pelka). *Let $S \not\cong \mathbb{P}^2$ be a normal projective surface over k with $\rho(S) = 1$, and let B be a nonzero reduced divisor such that $-(K_S + B)$ is ample. Let $\pi : V \rightarrow S$ be its minimal log resolution and let $D_V = B_V + \text{Exc}(\pi)_{\text{red}}$. Then either V admits a $\mathbb{P}^1$ -fibration whose general fibre has intersection at most two with D_V , or the same log smooth pair (V, D_V) is the minimal log resolution of (W, C) , where W is a canonical rank-one surface and $C \subset W^{\text{reg}}$ is cuspidal with $p_a(C) = 1$.*

The words “the same log smooth pair” specify both the surface and its reduced boundary. If we denote the cuspidal curve in the second alternative by T , there is a birational morphism $\tau : V \rightarrow W$ with

$$(5.1) \quad B_V + \text{Exc}(\pi)_{\text{red}} = T_V + \text{Exc}(\tau)_{\text{red}}.$$

The equality is an equality of reduced divisors on V .

When the ruling alternative fails, the classification gives more precise data. We record [PP24b, Theorem 1.2(a), Corollary 3.6 and Proposition 9.1], together with the characteristic-three column of [PP24b, Proposition 1.5(a),(d) and Table 1]. By [PP24b, Theorem 1.2(a) and Proposition 9.1], the canonical surface in the second alternative is not of type E_6 , E_7 or E_8 , its minimal resolution has at least six exceptional curves, and the height of the pair exceeds the height of that surface by two. In characteristic three, the rows of [PP24b, Table 1] satisfying these conditions and carrying a cuspidal curve are the height-two row $3A_2$, $A_2 + A_5$, A_8 , $A_2 + E_6$ and the height-four row $4A_2$. All five types are included below; [PP24b, Proposition 3.2(6.2)] names the types $3A_2$ and $4A_2$.

Theorem 5.3 (Palka–Pelka). *Let $S \not\cong \mathbb{P}^2$ be a rank-one $\mathbb{Q}$ -factorial klt surface over k , and let C be a nonzero reduced irreducible curve such that $-(K_S + C)$ is ample. Let $\pi : V \rightarrow S$ be the minimal log resolution, and put $Z = C_V + \text{Exc}(\pi)_{\text{red}}$. Assume $\text{ht}(S, C) > 2$. Then there*

exist a canonical rank-one del Pezzo surface W , an irreducible cuspidal curve $T \subset W^{\text{reg}}$ with $p_a(T) = 1$ and $T \sim -K_W$, and a birational morphism $\tau : V \rightarrow W$ such that the following hold. Let $\mu : U \rightarrow W$ be the minimal surface resolution, and let r be the number of its exceptional curves.

(i) The morphism τ is the minimal log resolution of (W, T) , and $Z = T_V + \text{Exc}(\tau)_{\text{red}}$ as reduced divisors.

(ii) The singularity type of W is one of

$$3A_2, \quad A_2 + A_5, \quad A_8, \quad A_2 + E_6, \quad 4A_2,$$

so that $r = 6, 7, 8, 8, 8$, respectively. Moreover $\text{ht}(S, C) = 4$ for the first four types and $\text{ht}(S, C) = 6$ for the type $4A_2$.

(iii) The strict transform T_U satisfies $T_U^2 = 9 - r$.

The numerical value in (iii) will determine the marked tip of the cusp resolution. We first establish the transfer argument, which only uses the common boundary and the genus of its image.

Lemma 5.4. *Let S be a normal projective surface, let $C \subset S$ be an irreducible reduced curve, and let $\pi : V \rightarrow S$ and $\rho : Y \rightarrow V$ be projective birational morphisms with V and Y smooth. Let D be a reduced divisor on Y , and put $Z = C_V + \text{Exc}(\pi)_{\text{red}}$. Suppose that every component of D maps under $\pi\rho$ to a point or to C . Then the following hold.*

(i) The divisorial pushforward satisfies $0 \leq \rho_* D \leq Z$.

(ii) If $f : V \rightarrow \Lambda$ is a $\mathbb{P}^1$ -fibration over a smooth projective curve, with general fibre F , then $f\rho : Y \rightarrow \Lambda$ is a $\mathbb{P}^1$ -fibration whose general fibre G satisfies $G \cdot D \leq F \cdot Z$.

Thus containment need only be checked on the endpoint S . Curves contracted on the way to V do not contribute to the general fibre.

Proof. Let D_j be a component of D whose image under ρ is a curve L . If $\pi(L)$ is a point, then L is a component of $\text{Exc}(\pi)_{\text{red}}$. Otherwise $\pi(L) = C$. A birational morphism to a normal surface is an isomorphism at the generic point of every curve, so $L = C_V$. Moreover, distinct noncontracted components of D have distinct images on V , and each maps birationally onto its image. This proves (i), including the assertion that $\rho_* D$ is reduced.

The exceptional curves of ρ have finitely many images on V . A general fibre F of f avoids those points, so ρ is an isomorphism near F and its inverse image G is a smooth projective line. The composite has connected fibres: indeed, $\rho_* \mathcal{O}_Y = \mathcal{O}_V$ and $f_* \mathcal{O}_V = \mathcal{O}_\Lambda$, and its Stein factorization therefore has trivial finite part. It is a $\mathbb{P}^1$ -fibration. For a general fibre $G = \rho^* F$, the projection formula gives

$$G \cdot D = F \cdot \rho_* D \leq F \cdot Z.$$

The inequality follows from (i), since a fibre is nef. $\square$

Lemma 5.5. *Let X and W be normal projective surfaces of Picard number one. Let $p : Y \rightarrow X$ be the minimal resolution, and put $D = \text{Exc}(p)_{\text{red}}$. Let $T \subset W^{\text{reg}}$ be an irreducible reduced singular curve, and let $\tau : V \rightarrow W$ be a log resolution of (W, T) . Suppose that there is a projective birational morphism $\rho : Y \rightarrow V$. Put $f = \tau\rho$, and suppose that every component of D maps under f to a point or to T . Then the reduced divisorial image satisfies $f_* D = T$.*

The minimality of $Y \rightarrow X$ rules out an empty image. It is essential here that $V \rightarrow W$ resolves a singular curve at a smooth point of W .

Proof. We first show that an empty divisorial image would identify X with W . We then use the resolution over a singular point of T to contradict minimality.

Step 1. In this step, we show that an empty divisorial image would induce an isomorphism $X \simeq W$. Suppose $f_* D = 0$. Every curve in a fibre of p is then contracted by f . The fibres of p

are connected, since X is normal, so f is constant on each fibre. To see the resulting factorization directly, let $\Gamma \subset X \times W$ be the reduced image of (p, f) . The projection $\Gamma \rightarrow X$ is proper and birational, and each of its fibres has a single point. It is therefore finite. Since X is normal, it is an isomorphism. The other projection gives a birational morphism $\eta : X \rightarrow W$ with $f = \eta p$.

Choose an ample Cartier divisor H on W . The numerical class of $\eta^* H$ is nonzero: it has positive intersection with every curve on X not contracted by η . Since $\rho(X) = 1$, this class is a positive multiple of an ample class on X . Thus it has positive intersection with every curve on X , whereas any η -exceptional curve would have intersection zero. Hence η contracts no curve. It is finite and birational onto the normal surface W , and therefore is an isomorphism.

Step 2. In this step, we use an exceptional curve over a singular point of T to contradict minimality. It follows that $f = \eta p$ is the minimal resolution of W . In particular, f is an isomorphism over W^{reg} . Let q be a singular point of T . The morphism τ cannot be an isomorphism near q : otherwise the strict transform of T would still be singular there, contrary to the log resolution condition. Thus τ has an exceptional curve over q . Its strict transform on Y is contracted by $f = \tau \rho$ to $q \in W^{\text{reg}}$, a contradiction.

Consequently, some component of D maps onto T . Every other component maps to a point or to the same curve, so the reduced divisorial image is exactly T . $\square$

Proposition 5.6. *Let X be a rank-one klt del Pezzo surface over k , let $p : Y \rightarrow X$ be its minimal resolution, and put $D = \text{Exc}(p)_{\text{red}}$. Let $S \not\cong \mathbb{P}^2$ be a rank-one klt surface with a nonzero reduced irreducible curve C such that $-(K_S + C)$ is ample. Let $\pi : V \rightarrow S$ be the minimal log resolution of (S, C) . Suppose that there is a projective birational morphism $\rho : Y \rightarrow V$ such that every component of D maps under $\pi \rho$ to a point or to C . Then $\# \text{Sing}(X) \leq 7$.*

The hypothesis controls images of curves under a morphism from the original minimal resolution. No equality between D and the total boundary of the endpoint is assumed.

Proof. Put $Z = C_V + \text{Exc}(\pi)_{\text{red}}$. By Lemma 5.4, we have $\rho_* D \leq Z$. Apply Theorem 5.2 to (S, C) .

Step 1. In this step, we prove the bound when the resolved pair admits a ruling of boundary degree at most two. If V admits a $\mathbb{P}^1$ -fibration with general fibre F satisfying $F \cdot Z \leq 2$, Lemma 5.4 gives a $\mathbb{P}^1$ -fibration on Y whose general fibre G satisfies $G \cdot D \leq 2$. Corollary 3.17 then gives $\# \text{Sing}(X) \leq 7$.

Step 2. In this step, we construct an elliptic-boundary descendant in the other open-pair alternative. In the other case, write $\tau : V \rightarrow W$ for the second minimal log resolution, and $T \subset W^{\text{reg}}$ for its cuspidal curve. The common boundary equality (5.1) reads

$$Z = C_V + \text{Exc}(\pi)_{\text{red}} = T_V + \text{Exc}(\tau)_{\text{red}}.$$

Hence every component of D maps under $f = \tau \rho$ to a point or to T . The surface W has Picard number one, and T is singular. Lemma 5.5 therefore gives $f_* D = T$, including nonemptiness. Since $p_a(T) = 1$ and $T \subset W^{\text{reg}}$, the morphism f makes W an elliptic-boundary descendant of X in Definition 3.1. Theorem 3.5(i) also gives $T \sim -K_W$. Theorem 3.6 gives $\# \text{Sing}(X) \leq 7$.

The ruling case is controlled by the exceptional degree, and the common-resolution case by the elliptic-boundary descendant. These are the alternatives of Theorem 5.2. $\square$

Remark 5.7. Here is a sufficient condition for the morphism ρ in Proposition 5.6. Suppose a projective birational morphism $\psi : Y \rightarrow S$ is already given. Let $q : U \rightarrow S$ be the minimal surface resolution. If $C_U + \text{Exc}(q)_{\text{red}}$ is simple normal crossing, then q is the minimal log resolution of (S, C) . Indeed, it is a log resolution, and Lemma 4.9 shows that every resolution of S factors through q . Thus $U = V$ over S , and the same lemma applied to ψ supplies the required projective birational morphism $\rho : Y \rightarrow V$.

In particular, this condition holds when S is a rank-one klt del Pezzo surface and (S, C) is dlt along C . Theorem 3.7 gives simple normal crossings for $\text{Exc}(q)_{\text{red}}$, including over $\text{Sing}(S) \setminus C$.

Proposition 4.10 shows that C is smooth and that C_U meets an end of each exceptional chain over C transversely once. These checks prove that $C_U + \text{Exc}(q)_{\text{red}}$ is simple normal crossing everywhere.

More generally, if ψ itself is a log resolution of (S, C) , it factors through the minimal log resolution directly. A birational morphism to S alone does not give that conclusion. For example, a cusp of C in S^{reg} requires blowups that the minimal surface resolution of S does not perform.

We next give the local calculation that marks the cuspidal component in the exceptional graphs. The marking is needed when two tips have the same self-intersection.

Lemma 5.8. *Let U be a smooth projective surface, and let $T \subset U$ be an integral curve with $p_a(T) = 1$, smooth except for one ordinary cusp. Put $d = T^2$, and let $\beta : V \rightarrow U$ be the minimal log resolution of (U, T) . Then $T_V + \text{Exc}(\beta)_{\text{red}}$ is a four-component fork. Its central component has square -1 , the two exceptional tips have squares -2 and -3 , and the remaining tip is T_V , with square $d - 6$.*

This identifies the strict transform by its origin, even if its self-intersection agrees with that of an exceptional tip.

Proof. At the cusp we use completed local coordinates with equation $v^2 = u^3$. After the first blowup, the chart $v = uv_1$ gives the strict transform $v_1^2 = u$. It is smooth and tangent to the first exceptional curve $u = 0$.

Blow up their point of tangency. In the chart $u = v_1u_2$, the new strict transform has equation $v_1 = u_2$. The new exceptional curve is $v_1 = 0$, and the strict transform of the first exceptional curve is $u_2 = 0$. These three smooth curves have distinct tangent directions at their common point. A third blowup separates them. Its exceptional curve R meets each of the three strict transforms transversely, at three distinct points. The total reduced divisor is now simple normal crossing. Neither of the first two stages has that property, so these are precisely the blowups of the minimal log resolution. The only exceptional (-1) -curve is R , and contracting it restores the triple point.

The first exceptional curve is blown up twice more, and the second once more. Their final squares are therefore -3 and -2 , respectively, while $R^2 = -1$. The successive multiplicities of the strict transform of T at the three centres are $2, 1, 1$. Consequently,

$$T_V^2 = T^2 - 2^2 - 1^2 - 1^2 = d - 6.$$

The normalization of T is rational: its cusp has delta invariant one and $p_a(T) = 1$. Thus all four components are smooth rational curves, with the stated incidences and marking. $\square$

Proposition 5.9. *Let $S \not\cong \mathbb{P}^2$ be a rank-one $\mathbb{Q}$ -factorial klt surface over k , and let C be a nonzero reduced irreducible curve such that $-(K_S + C)$ is ample. Let $\pi : V \rightarrow S$ be the minimal log resolution, put $Z = C_V + \text{Exc}(\pi)_{\text{red}}$, and assume $\text{ht}(S, C) > 2$. Let W, T, τ, r be as in Theorem 5.3, and put $\ell = r - 3$, so $\ell \in \{3, 4, 5\}$. Then Z has exactly two kinds of connected components. The first kind consists of the strict transforms of the r exceptional curves of the minimal surface resolution of W ; they have square -2 and the same weighted graph as on that resolution. The remaining connected component is a four-component fork. The fork has central curve of square -1 and three tips of squares $-2, -3, -\ell$. The strict transform T_V is the marked tip of square $-\ell$. The height of the resolved pair is 6 when W has type $4A_2$, and 4 otherwise.*

The (-2) -curves resolve the surface singularities. The remaining connected component resolves the cuspidal boundary at a smooth point.

Proof. We compare τ with the minimal surface resolution of W and then compute the marked boundary graph.

Step 1. In this step, we identify the blowups relating the minimal surface resolution to τ . Let $\mu : U \rightarrow W$ be the minimal surface resolution, and put $E_U = \text{Exc}(\mu)_{\text{red}}$. The surface W

is canonical, so E_U consists of r curves of square -2 . Since $T \subset W^{\text{reg}}$, the morphism μ is an isomorphism near T , and T_U is disjoint from E_U . In particular, T_U is cuspidal and has arithmetic genus one. The genus formula for normalization shows that its cusp has delta invariant one and is its only singularity. A unibranch plane singularity of delta invariant one is an ordinary cusp: its completed local ring has colength one in its normalization $k[[t]]$ and contains no element of order one, so it is $k + t^2 k[[t]] = k[[t^2, t^3]]$.

By Lemma 4.9, the morphism $\tau : V \rightarrow W$ factors as $\tau = \mu\beta$ with $\beta : V \rightarrow U$ projective and birational. The divisor E_U is already simple normal crossing. Consequently, the three blowups resolving the cusp of T_U give a log resolution of (W, T) . They are necessary at the cusp, and the resolution over each singular point of W is already minimal. Thus, by the minimality of τ , these are exactly the blowups in β .

Step 2. In this step, we compute the fork, identify its marked tip, and record the height. Apply Lemma 5.8 to $T_U \subset U$. By Theorem 5.3(iii),

$$T_U^2 = 9 - r, \quad T_V^2 = (9 - r) - 6 = -(r - 3) = -\ell.$$

The reduced divisor $T_V + \text{Exc}(\beta)_{\text{red}}$ is therefore the four-component fork in the statement, with T_V as its marked tip. The blowups take place away from E_U , so its strict transform E_V retains the weighted graph of E_U and is disjoint from that fork. The common-boundary equality gives

$$Z = T_V + \text{Exc}(\tau)_{\text{red}} = E_V + T_V + \text{Exc}(\beta)_{\text{red}},$$

which identifies the entire reduced boundary.

For $\ell = 3$, one (-3) -tip comes from the first blowup of the cusp; the other is T_V and is the marked tip. For $\ell = 4$ and $\ell = 5$, the marked tip is the unique tip of square $-\ell$. Finally, Theorem 5.3(ii) gives the height $\text{ht}(S, C)$, which is the height of its resolved pair (V, Z) . $\square$

We close with rationality and numerical consequences of the classification. The rationality criterion is [PP24b, Lemma 2.7].

Theorem 5.10 (Palka–Pelka). *A normal projective del Pezzo surface of Picard number one over an algebraically closed field is rational if and only if all its singularities are rational.*

For klt del Pezzo surfaces, the condition on the singularities follows from the earlier local result.

Theorem 5.11. *Let S be a rank-one klt del Pezzo surface over k . Every smooth projective resolution of S is a rational surface.*

Thus rationality is available on every smooth model used to resolve the surface and its boundary.

Proof. Apply Theorem 4.16 to $(S, 0)$. It shows that S has rational singularities. Since S is del Pezzo of Picard number one, Theorem 5.10 implies that S is rational. Every resolution of S is birational to S , and hence is rational. $\square$

For canonical surfaces carrying a cuspidal anticanonical curve, the characteristic-three column of [PP24b, Proposition 1.5(a),(d) and Table 1] gives the following list. This list concerns canonical surfaces with the specified cuspidal curve; it is distinct from the height classification used earlier. The plane and the quadric cone are recorded separately.

Theorem 5.12 (Palka–Pelka). *Let W be a canonical rank-one del Pezzo surface over k with an integral rational anticanonical curve contained in W^{reg} whose only singularity is an ordinary cusp. Then either W is the projective plane or the quadric cone, or its singularity type belongs to the following list:*

$$A_1 + A_2, A_4, D_5, E_6, E_7, E_8, 3A_2, A_2 + A_5, A_8, A_2 + E_6, 4A_2.$$

If r is the number of exceptional curves of its minimal resolution, then $K_W^2 = 9 - r$.

The connected components of a listed graph count the singular points; its vertices count the exceptional curves entering the degree formula.

Theorem 5.13. *Let W be a canonical rank-one del Pezzo surface over k with an integral rational anticanonical curve in W^{reg} whose only singularity is an ordinary cusp. Then $\#\text{Sing}(W) \leq 4$. If $\#\text{Sing}(W) = 3$, the singularity type is $3A_2$ and $K_W^2 = 3$; if $\#\text{Sing}(W) = 4$, the type is $4A_2$ and $K_W^2 = 1$. If $K_W^2 = 4$, the singularity type is D_5 .*

The cases with three or four singular points are therefore determined by their counts, while degree four determines a different graph.

Proof. The projective plane is smooth and has $K_W^2 = 9$. The quadric cone has one A_1 singularity, with one exceptional curve in its minimal resolution. Its degree is $9 - 1 = 8$ by Theorem 5.12. Thus both satisfy the bound and neither enters any of the three further cases.

For every other W , use the list in Theorem 5.12. It has at most four connected components. The only listed type with three components is $3A_2$, and the only type with four is $4A_2$. Each A_2 graph has two vertices, so their minimal resolutions have respectively $r = 6$ and $r = 8$ exceptional curves. The same theorem gives

$$K_W^2 = 9 - 6 = 3 \quad \text{for type } 3A_2, \quad K_W^2 = 9 - 8 = 1 \quad \text{for type } 4A_2.$$

Finally, if $K_W^2 = 4$, then $r = 5$. Among the listed types, only D_5 has five vertices. This proves all the assertions. $\square$

6. HUNT CONSTRUCTIONS

We retain the coefficient convention of Section 2. A hunt replaces a rank-one log del Pezzo surface by extracting one exceptional curve and contracting the other extremal ray. The extracted curve becomes a boundary curve on the next surface, or a horizontal curve in a fibration. We describe this construction and the geometric alternatives furnished by Nagaoka's results. These alternatives retain the boundary coefficients and marked points needed to compare singularities across the contractions.

6.1. Extraction and boundary adjustment.

Definition 6.1. Let (S, Δ) be a log pair and let $Y \rightarrow S$ be a birational model. A *tiger* on Y is a prime divisor E on Y for which there is an effective $\mathbb{Q}$ -Cartier divisor α on S such that $K_S + \Delta + \alpha \equiv 0$ and $e(E; S, \Delta + \alpha) \geq 1$. It is exceptional if it is exceptional over S . With $\Delta = \sum_i a_i D_i$ a boundary, the pair is *flush*, respectively *level*, if every exceptional divisor F over S satisfies $e(F; S, \Delta) < \min(1, a_i)$, respectively $e(F; S, \Delta) \leq \min(1, a_i)$. For the empty boundary the minimum is understood to be 1.

Here a boundary has coefficients in $[0, 1]$, and its displayed decomposition consists of its nonzero prime components. The minimum in the definition is taken over 1 and all its component coefficients. Thus flushness compares every exceptional coefficient with every boundary coefficient. A nonexceptional tiger is the strict transform on Y of a curve on S . The model Y is part of the no-tiger hypothesis.

Definition 6.2. Let (S, Δ) have S rank-one log del Pezzo and $-(K_S + \Delta)$ ample. A *hunt step* starts with a relative Picard-number-one extraction $f : T \rightarrow S$ of a divisor E from the minimal resolution, chosen to have maximal coefficient among its exceptional divisors. When $\Delta = 0$, at a chain singularity choose E not to be a (-2) -curve, and at a nonchain singularity choose the central component. Write $K_T + \Gamma = f^*(K_S + \Delta)$, take $0 < \varepsilon \ll 1$, and set $\Gamma' = \lambda(\Gamma + \varepsilon E)$, where λ makes $K_T + \Gamma'$ trivial on the other edge of $\text{NE}(T)$. Let g contract that edge. If g is a fibration we call the output a *net*; otherwise put $S' = g(T)$ and $\Delta' = g_*\Gamma'$. Starting with $(S_0, 0)$ we use

$$S_{i-1} \xleftarrow{f_i} T_i \xrightarrow{g_i} S_i, \quad E_i = \text{Exc}(f_i), \quad \Sigma_i = \text{Exc}(g_i), \quad x_i = f_i(E_i), \quad q_i = g_i(\Sigma_i)$$

when the output is birational. Put $A_1 = g_1(E_1)$, $\Delta_1 = a_1 A_1$, $B_2 = g_2(E_2)$, and, if the old curve survives, $A_2 = g_2((A_1)_{T_2})$ and $\Delta_2 = a_2 A_2 + b_2 B_2$. The coefficient of E_2 in the unscaled second pullback is denoted e_1 .

In the no-tiger hunt, whenever several singular points attain the maximal coefficient, we choose a chain singularity if one is available; among the tied chain choices, we extract a component of least self-intersection on the minimal resolution.

In the second step, $(A_1)_{T_2}$ is the old boundary and E_2 is the newly extracted curve. When both survive, their images are A_2 and B_2 , respectively. The coefficient e_1 belongs to the unscaled pullback, whereas a_2, b_2 belong to the output boundary.

The maximal coefficient is measured with respect to the input pair (S, Δ) . The chain and central-component conditions concern the exceptional graph of its minimal resolution. Since f extracts a component of that resolution, the resolution factors through T , and f is an isomorphism away from $x = f(E)$. The intermediate surface has Picard number two.

The rule for equal maximal coefficients is the convention in [Lac24, Definition 3.13]. All no-tiger hunts in this paper use this choice. It fixes the extraction choices used in the boundary coefficient estimates.

We choose the perturbation rational and sufficiently small for the given hunt step. In a sequence we denote the perturbation and scaling at the i th step by ε_i and λ_i . The convention is to perturb the exceptional coefficient first and then multiply the whole boundary by λ_i . In particular,

$$\begin{aligned} K_{T_i} + \Gamma_i &= f_i^*(K_{S_{i-1}} + \Delta_{i-1}), \\ \Gamma'_i &= \lambda_i(\Gamma_i + \varepsilon_i E_i). \end{aligned}$$

The choice $0 < \varepsilon_i \ll 1$ remains part of the construction whenever we use its geometric properties.

We use the following parts of [Nag25a, Lemma 2.7(1), (4), (6)–(8)].

Theorem 6.3 (Nagaoka). *In a hunt step over an algebraically closed field, the opposite edge is K_T -negative and contractible, $\lambda > 1$, and its contraction is either a $\mathbb{P}^1$ -fibration or birational. In the birational case S' is a rank-one log del Pezzo surface and $-(K_{S'} + \Delta')$ is ample. If (S, Δ) has no tiger on a model Y dominating T , then the induced pairs on T and S' have no tiger on that model.*

In the last assertion the induced pair on T is (T, Γ') ; in the birational case the endpoint pair is (S', Δ') . The morphism $Y \rightarrow S'$ is the composition $Y \rightarrow T \rightarrow S'$. Thus each application of no-tiger persistence uses a specified model dominating the extraction. The lemma requires a log pair, ampleness of $-(K_S + \Delta)$, an extraction from the minimal resolution of relative Picard number one, and the small perturbation above. Flushness, when needed, will be supplied by the no-tiger theorem below.

The birational step also identifies the adjusted adjoint divisors:

$$K_T + \Gamma' = g^*(K_{S'} + \Delta').$$

Indeed, their difference is g -exceptional, so it is a rational multiple of $\Sigma = \text{Exc}(g)$. Its intersection with Σ is zero by the choice of λ , whereas $\Sigma^2 < 0$. The multiple is therefore zero. At the first birational step this gives

$$\Gamma'_1 = a_1 E_1, \quad a_1 = \lambda_1(e(E_1; S_0, 0) + \varepsilon_1), \quad K_{T_1} + a_1 E_1 = g_1^*(K_{S_1} + a_1 A_1).$$

The boundary A_1 consequently records the curve extracted from S_0 on the next rank-one surface.

6.2. Geometric alternatives. The second birational step may retain both this old boundary and the newly extracted curve. We use the following terminology for their intersections.

Definition 6.4. Let A, B be distinct rational curves on a klt surface S , and assume $(S, A + B)$ is dlt at the singular points of S . The pair is a *fence* if A and B meet at exactly one point and transversely; it is a *banana* if they meet at exactly two points and transversely at each; it is a *tacnode configuration* if they meet in at most two points, with smooth branches tangent to

order at least two at one point and, if there is another intersection, transverse branches there. The indicated intersection points are smooth points of S .

These are the configurations of [Nag25a, Definition 2.5]. At a tacnode the order is the local intersection multiplicity of the two smooth branches; it may exceed two. The dlt condition at the surface singularities and the conditions at the intersection points are both part of the definition.

When a tiger exists on the minimal resolution, a single extraction suffices to reach a fibration or a surface with a distinguished reduced curve. The choice of exceptional tiger, including its possible replacement, is supplied by [Nag25b, Theorem 5.3], restated in [Nag25a, Theorem 1.1].

Theorem 6.5 (Nagaoka). *Let X be a rank-one log del Pezzo surface over k with a tiger on its minimal resolution. If an exceptional tiger exists, it can be chosen so that its extraction $f : T \rightarrow X$ and the opposite contraction g have one of the following outputs:*

- (i) $g : T \rightarrow B$ is a net and the extracted curve E is a section or bisection, with (T, E) plt;
- (ii) $g : T \rightarrow S$ is birational, S is a rank-one Du Val del Pezzo surface, and $C = g(E)$ is a reduced irreducible singular anticanonical curve in S^{reg} , with $g(\text{Exc}(g))$ its singular point;
- (iii) $g : T \rightarrow S$ is birational, S is rank-one log del Pezzo, (S, C) is plt for $C = g(E)_{\text{red}}$, and $K_S + C$ is anti-nef, with $g(\text{Exc}(g)) \in C$.

If there is no exceptional tiger, then X itself contains a reduced irreducible curve C with (X, C) plt and $K_X + C$ anti-nef.

The fibration alternative controls the degree of the extracted curve on a general fibre. The anticanonical alternative supplies an entire singular curve in the smooth locus of the endpoint. In the plt alternative, Picard number one leaves two possibilities: $K_S + C$ is anti-ample or $K_S + C \equiv 0$. To see this, write its numerical class as tH for an ample class H ; anti-nefness gives $t \leq 0$. The same distinction applies to the curve on X in the final assertion, whose hypothesis includes the absence of exceptional tigers on the minimal resolution.

For the no-tiger case we use [Nag25a, Proposition 2.10]. A second step is considered after a birational first output. Our indices place the extraction centre x_i on S_{i-1} and the contraction point q_i on S_i . In particular,

$$x_2 = f_2(E_2) \in S_1, \quad q_2 = g_2(\Sigma_2) \in S_2.$$

The notation of [Nag25a, Notation 2.9] indexes the extraction by its source surface. Its f_{i-1} and x_{i-1} correspond to our f_i and x_i , and its π_i is our g_i . Thus the centre denoted by x_1 in the incidence assertions of that proposition is our x_2 .

We also specify how an intersection associated with this centre is followed to S_2 . When both curves survive, put $A_T = (A_1)_{T_2}$. Away from q_2 , the contraction is an isomorphism, so, as sets of closed points,

$$(6.1) \quad (A_2 \cap B_2) \setminus \{q_2\} = g_2((A_T \cap E_2) \setminus \Sigma_2).$$

Every point of $(A_T \cap E_2) \setminus \Sigma_2$ maps to x_2 under f_2 . This is the meaning of the image of the extraction centre in the incidence statement: it is followed through the intersection of the two curves on T_2 .

Theorem 6.6 (Nagaoka). *Let S_0 be a rank-one log del Pezzo surface over k without a tiger on its minimal resolution. Run the first two hunt steps with initial boundary zero. Then (T_1, E_1) is dlt and (T_1, Γ'_1) is flush. Either the first output is a net, or $(S_1, a_1 A_1)$ is flush and precisely one of the following types occurs: $p_a(A_1) > 1$; arithmetic genus one with an ordinary node at q_1 ; arithmetic genus one with an ordinary cusp at q_1 ; or a rational curve with (S_1, A_1) dlt. For the second step there are three alternatives:*

- (i) *the output is a net;*
- (ii) *g_2 contracts $(A_1)_{T_2}$, (T_2, Γ'_2) is flush, (S_1, A_1) is dlt, $q_2 \in S_2^{\text{reg}}$, and B_2 is unibranch and singular at q_2 ; the endpoint pair is flush away from q_2 , is not level at q_2 , and Σ_2 is its only exceptional divisor at which flushness fails;*

(iii) both A_2 and B_2 survive.

In (iii), if $a_2 + b_2 \geq 1$, the opposite curve is disjoint from the singular locus of the old boundary, the intermediate and endpoint pairs are flush away from that old singularity and its image, and at least one of $-(K_{S_2} + A_2)$ and $-(K_{S_2} + B_2)$ is ample. The pair $(S_2, A_2 + B_2)$ is a fence, banana, or tacnode configuration. In the banana and tacnode cases (S_2, B_2) is plt. In the banana case the second extraction centre lies on A_1 . In the tacnode case q_2 is the tangency point; the only possible other intersection is the image of the second extraction centre, and it occurs exactly when that centre lies on A_1 .

All the conclusions concerning two surviving curves, including the configuration and incidence assertions, have the hypothesis $a_2 + b_2 \geq 1$. In the tacnode case with $x_2 \in A_1$, let z_2 be the unique point of the set in (6.1). The incidence assertions read explicitly

$$A_2 \cap B_2 = \begin{cases} \{q_2, z_2\}, & x_2 \in A_1, \\ \{q_2\}, & x_2 \notin A_1, \end{cases}$$

In the first case $z_2 \neq q_2$. In the banana case the centre condition is $x_2 \in A_1$. The singular locus of the old boundary in the theorem means its curve singularities, viewed on T_2 and then on S_2 ; the flushness assertions retain these specified exceptional locations.

The coefficient condition can be checked directly on the pullback before the second contraction. The following elementary consequence of the scaling convention will be useful.

Lemma 6.7. *Let S_0 be a rank-one log del Pezzo surface over k without a tiger on its minimal resolution. Suppose that its first two hunt steps are birational and that both $(A_1)_{T_2}$ and E_2 survive the second contraction. Let ε_2 and λ_2 be the perturbation and scaling in that step, and assume $a_1 \geq 2/3$ and $e_1 \geq 1/3$. Then*

$$a_2 = \lambda_2 a_1 > \frac{2}{3}, \quad b_2 = \lambda_2(e_1 + \varepsilon_2) > \frac{1}{3},$$

and $a_2 + b_2 > 1$.

Proof. Put $A_T = (A_1)_{T_2}$. The extraction has a single exceptional component, and the coefficient on a strict transform is unchanged by log pullback. Hence

$$\Gamma_2 = a_1 A_T + e_1 E_2, \quad \Gamma'_2 = \lambda_2 a_1 A_T + \lambda_2(e_1 + \varepsilon_2) E_2.$$

A birational pushforward preserves the coefficient of every noncontracted prime divisor. Applying g_{2*} gives the two claimed equalities. By Theorem 6.3, $\lambda_2 > 1$, and the perturbation satisfies $\varepsilon_2 > 0$. Under the two lower bounds we obtain

$$a_2 \geq \lambda_2 \frac{2}{3} > \frac{2}{3}, \quad b_2 \geq \lambda_2 \left(\frac{1}{3} + \varepsilon_2 \right) > \frac{1}{3}.$$

Adding proves the last inequality. □

Thus suitable lower bounds on the first boundary coefficient and the unscaled second exceptional coefficient make the two-curve conclusions of Theorem 6.6 applicable. The required coefficient bounds will be obtained from the geometry of the first boundary.

Remark 6.8. Theorem 6.3 and Theorem 6.6 are the preliminary hunt results of [Nag25a], whose standing field convention in Section 1.1 allows arbitrary characteristic. Theorem 5.3 of [Nag25b] is stated under that paper's positive-characteristic convention and gives Theorem 6.5. These source statements therefore apply over k . Their use here is confined to the extraction construction and the geometric alternatives just stated. Each subsequent argument works with those alternatives and their stated hypotheses.

7. TIGER REDUCTIONS WITH NEGATIVE ADJOINT BOUNDARY

We retain the notation of Section 6. We treat the net and anticanonical outputs of the tiger reduction, and then its plt outputs with negative adjoint class. The comparison with the original surface takes place on its minimal resolution: only the extracted exceptional component can have a divisorial image on the endpoint.

Lemma 7.1. *Let X be a normal projective surface over k , let $p : Y \rightarrow X$ be its minimal resolution, and put $D = \text{Exc}(p)_{\text{red}}$. Let $f : T \rightarrow X$ be the extraction of one component E_Y of D from this resolution, and denote the extracted curve on T by E . Then there is a projective birational morphism $\tau : Y \rightarrow T$ such that $p = f\tau$, $\tau(E_Y) = E$ birationally, and τ contracts every component of $D - E_Y$. If $g : T \rightarrow S$ is a birational morphism of normal projective surfaces that does not contract E , and $C = g(E)_{\text{red}}$, then the reduced divisorial image of D under $\eta = g\tau$ is exactly C .*

Thus the exceptional divisor on the original resolution contributes only one curve to any birational endpoint of the extraction.

Proof. The extraction of E_Y from the minimal resolution retains E_Y and contracts the other p -exceptional curves over X . This gives τ and the equality $p = f\tau$. A birational morphism to a normal surface is an isomorphism at the generic point of each curve on the target. Consequently E_Y maps birationally onto E .

For the last assertion, $\eta(E_Y) = C$, whereas every component of $D - E_Y$ already has point image under τ . Its image under η is therefore also a point. The only one-dimensional image is C , as claimed. $\square$

Proposition 7.2. *Let X be a rank-one klt del Pezzo surface over k with an exceptional tiger. If its chosen reduction in Theorem 6.5 has output (i) or (ii), then $\#\text{Sing}(X) \leq 7$.*

The net gives a ruling of small exceptional degree; the anticanonical output gives an elliptic-boundary descendant.

Proof. We pull the net back to the minimal resolution in the first case and identify the divisorial image of its exceptional divisor in the second. Let $p : Y \rightarrow X$ be the minimal resolution and put $D = \text{Exc}(p)_{\text{red}}$. Write $f : T \rightarrow X$ for the tiger extraction and E for its exceptional curve. Lemma 7.1 gives a morphism $\tau : Y \rightarrow T$ with $p = f\tau$ and a unique component E_Y of D that survives on T .

Step 1. In this step, we pull the net back to a ruling of Y with exceptional degree at most two. In output (i), let $g : T \rightarrow B$ be the net. The composite $h = g\tau : Y \rightarrow B$ is a morphism. By Theorem 4.8 ([Sta26, Tag 0AY8]), $\tau_*\mathcal{O}_Y = \mathcal{O}_T$. Since $g_*\mathcal{O}_T = \mathcal{O}_B$, the composite has connected fibres. A general fibre F of g avoids the finitely many images of curves contracted by τ . Its inverse image G is consequently isomorphic to F , so h is a $\mathbb{P}^1$ -fibration. Every component of $D - E_Y$ is vertical for h , and the projection formula gives

$$G \cdot D = G \cdot E_Y = F \cdot E \leq 2,$$

because E is a section or a bisection. Corollary 3.17 now gives $\#\text{Sing}(X) \leq 7$.

Step 2. In this step, we identify the anticanonical output as an elliptic-boundary descendant of X . In output (ii), write $g : T \rightarrow S$ for the birational contraction and $C = g(E)$. By Theorem 6.5, the surface S is Du Val and C is a reduced irreducible anticanonical curve contained in S^{reg} . In particular, C is Cartier, and adjunction gives

$$2p_a(C) - 2 = (K_S + C) \cdot C = 0.$$

The morphism $\eta = g\tau : Y \rightarrow S$ is birational, and Lemma 7.1 gives $\eta_*D = C$ as a reduced divisorial image. Thus S is an elliptic-boundary descendant in the sense of Definition 3.1. Theorem 3.6 gives $\#\text{Sing}(X) \leq 7$. $\square$

For the plt outputs, we use Lemma 4.9 to factor a resolution through the minimal surface resolution. The plane case also requires domination of the blowup at a specified point.

Lemma 7.3. *Let Y and S be smooth projective surfaces over an algebraically closed field, let $\eta : Y \rightarrow S$ be a projective birational morphism, and let $q \in S$ be a closed point. Let $\sigma : V = \text{Bl}_q S \rightarrow S$ be the blowup at q . If η contracts an irreducible curve to q , then there is a projective birational morphism $\rho : Y \rightarrow V$ such that $\eta = \sigma\rho$.*

This identifies the blowup on which a pencil centred at q can be resolved while retaining a morphism from Y .

Proof. By Theorem 4.2 ([Sta26, Tag 0C5R]), η is a composite of point blowups. If none of their centres lay over q , this composite would be an isomorphism near q , contrary to the contracted curve in its fibre. The first centre over q is q itself, since all preceding blowups are isomorphisms near that point. All preceding centres lie over other points of S , so those blowups commute with the blowup of q . We may therefore place σ first in the factorization. The remaining point blowups give the projective birational morphism ρ and the equality $\eta = \sigma\rho$. $\square$

Lemma 7.4. *Let S be a rank-one klt del Pezzo surface over k and let C be a nonzero reduced irreducible curve such that (S, C) is plt. Then the minimal surface resolution $\pi : V \rightarrow S$ is the minimal log resolution of (S, C) .*

In this situation resolving the boundary requires no additional point blowups.

Proof. By Theorem 3.7 ([PP24a, Lemma 2.7(c)]), the reduced exceptional divisor $\text{Exc}(\pi)_{\text{red}}$ has simple normal crossing support. Proposition 4.10 shows that C is smooth. Over a singular point on C , its strict transform C_V meets just one end of the exceptional chain, transversely once. Away from those fibres, C_V is disjoint from the exceptional divisor, and over the smooth locus π is an isomorphism. Thus $C_V + \text{Exc}(\pi)_{\text{red}}$ has simple normal crossing support on V .

It follows that π is a log resolution of (S, C) . Every log resolution is a smooth resolution of S , so it factors through π by Lemma 4.9. No proper contraction of V over S can remain a smooth resolution, by minimality. Hence π is the minimal log resolution of the pair. $\square$

Proposition 7.5. *Let X be a rank-one klt del Pezzo surface over k , and let C be a nonzero reduced irreducible curve such that (X, C) is plt and $-(K_X + C)$ is ample. Then $\#\text{Sing}(X) \leq 7$.*

Here the original surface already carries the boundary needed for the open-pair comparison.

Proof. If $X \simeq \mathbb{P}^2$, then X is smooth. Otherwise let $p : Y \rightarrow X$ be its minimal resolution and put $D = \text{Exc}(p)_{\text{red}}$. Lemma 7.4 identifies p with the minimal log resolution of (X, C) . We apply Proposition 5.6 with $S = X$, $V = Y$, $\pi = p$, and ρ the identity. Every component of D maps to a point under p , so its containment hypothesis holds. This gives $\#\text{Sing}(X) \leq 7$. $\square$

The plane endpoint requires a pencil with a specified centre. The following lemma keeps track of its resolution and of the whole original exceptional divisor.

Lemma 7.6. *Let X be a rank-one klt del Pezzo surface over k , let $p : Y \rightarrow X$ be its minimal resolution, and put $D = \text{Exc}(p)_{\text{red}}$. Let $\eta : Y \rightarrow \mathbb{P}^2$ be a projective birational morphism. Let $C \subset \mathbb{P}^2$ be a reduced irreducible curve of degree at most two, and let $q \in C$. Suppose that η contracts a curve to q and sends every component of D to a point or to C . Then Y admits a $\mathbb{P}^1$ -fibration with general fibre G satisfying $G \cdot D \leq 2$. In particular, $\#\text{Sing}(X) \leq 7$.*

The curve contracted over q ensures that the line pencil is resolved on Y itself.

Proof. We factor η through the blowup of q , then bound the intersection of its line pencil with the image of D .

Step 1. In this step, we factor η through the blowup of q and locate the images of the components of D . Put $V = \text{Bl}_q \mathbb{P}^2$, let $\sigma : V \rightarrow \mathbb{P}^2$ be the blowup, and denote its exceptional curve by L . By Lemma 7.3, there is a projective birational morphism $\rho : Y \rightarrow V$ such that $\eta = \sigma\rho$.

Let D_j be a component of D with curve image $R = \rho(D_j)$. If $\sigma(R)$ is a point, then $R = L$. Otherwise the containment hypothesis gives $\sigma(R) = C$, and R is the strict transform C_V . Thus every component of D maps under ρ to a point or to a component of $C_V + L$.

Step 2. In this step, we pull the line pencil back to Y and bound its exceptional degree. The pencil of lines through q defines a morphism $h : V \rightarrow \mathbb{P}^1$. Let H denote the pullback of the class of a line, put $d = \deg C$, and put $m = \text{mult}_q C \geq 1$. For a general fibre F of h , we have

$$F \equiv H - L, \quad C_V \equiv dH - mL, \quad F \cdot (C_V + L) = d - m + 1 \leq 2.$$

Theorem 4.8 gives $\rho_* \mathcal{O}_Y = \mathcal{O}_V$. Since $h_* \mathcal{O}_V = \mathcal{O}_{\mathbb{P}^1}$, the composite $h\rho : Y \rightarrow \mathbb{P}^1$ has connected fibres. Choose F away from the finitely many images of ρ -exceptional curves. Its inverse image G is isomorphic to F , so the composite is a $\mathbb{P}^1$ -fibration. Components of D contracted by ρ contribute zero to $G \cdot D$. Each remaining component is the unique strict transform of C_V or L . The projection formula therefore gives

$$G \cdot D \leq F \cdot (C_V + L) \leq 2.$$

Corollary 3.17 proves the last assertion. $\square$

Proposition 7.7. *Let X be a rank-one klt del Pezzo surface over k . Suppose its exceptional-tiger reduction is output (iii) of Theorem 6.5, with endpoint (S, C) . If $-(K_S + C)$ is ample, then $\#\text{Sing}(X) \leq 7$.*

The endpoint controls X through a morphism from the original minimal resolution, even when the two surfaces have different singularities.

Proof. We first identify the image of the original exceptional divisor. We then use the minimal log resolution when $S \not\simeq \mathbb{P}^2$, and the line pencil when $S \simeq \mathbb{P}^2$.

Step 1. In this step, we identify C as the reduced divisorial image of the original exceptional divisor. Let $p : Y \rightarrow X$ be the minimal resolution, put $D = \text{Exc}(p)_{\text{red}}$, and write the reduction as

$$X \xleftarrow{f} T \xrightarrow{g} S.$$

Let $E = \text{Exc}(f)$ and $\Sigma = \text{Exc}(g)$ be the two contracted curves, and put $q = g(\Sigma)$. Theorem 6.5 gives $C = g(E)_{\text{red}}$ and $q \in C$. Lemma 7.1 gives the morphism $\tau : Y \rightarrow T$ with $p = f\tau$; let E_Y be the component of D that maps onto E . For $\eta = g\tau : Y \rightarrow S$, every component of D maps to a point or to C , and E_Y maps onto C .

Step 2. In this step, we prove the bound for a nonplane endpoint by the open-pair comparison. Suppose $S \not\simeq \mathbb{P}^2$, and let $\pi : V \rightarrow S$ be its minimal surface resolution. Lemma 4.9 gives a projective birational morphism $\rho : Y \rightarrow V$ such that $\eta = \pi\rho$. The pair (S, C) is plt by Theorem 6.5, so Lemma 7.4 identifies π with its minimal log resolution. All hypotheses of Proposition 5.6 now hold: S has rank one, $-(K_S + C)$ is ample, ρ is a morphism, and the preceding paragraph proves the required image containment. Hence $\#\text{Sing}(X) \leq 7$.

Step 3. In this step, we prove the bound for a plane endpoint using the pencil through q . Suppose $S \simeq \mathbb{P}^2$. If $d = \deg C$, then the ampleness of $-(K_S + C)$ says $3 - d > 0$. Since C is nonzero and reduced, $1 \leq d \leq 2$. The strict transform Σ_Y of Σ on Y is a curve, and $\eta(\Sigma_Y) = q \in C$. Thus η contracts a curve over q . Together with the image containment already proved, these are the hypotheses of Lemma 7.6. It gives $\#\text{Sing}(X) \leq 7$ also in this case. $\square$

8. NUMERICALLY TRIVIAL BOUNDARIES

We retain the notation for discrepancies and one-curve links from Sections 4 and 6. For a numerically trivial plt boundary, adjunction controls the singular points on the boundary. A further hunt controls the singular points away from it. We keep track of both sets, since an exceptional-tiger reduction requires transferring the estimate through an earlier link.

Definition 8.1. Let (S, C) be plt with S rank-one log del Pezzo, C reduced and irreducible, and $K_S + C$ anti-nef. Assume C contains at least three singular points of S . A *pair hunt near coefficient one* is Nagaoka's Case 1 construction: choose a rational $a < 1$ sufficiently close to 1, extract E for (S, aC) as in [Nag25a, Remark 4.2], and contract the opposite extremal ray. The scaled boundary is $aC_T + bE$, where b is chosen to make $K_T + aC_T + bE$ trivial on that ray. The coefficient of C_T remains a in this construction.

Keeping the old coefficient fixed will be essential when the old curve is contracted. The unscaled coefficient of E will be denoted by e ; it is different from the coefficient b in the definition.

Theorem 8.2 (Nagaoka). *Let (S, C) satisfy the hypotheses of Definition 8.1. The output is a net, a birational contraction of C_T , or a birational output with surviving curves C_U and $A = g(E)$ for which $(U, C_U + A)$ is dlt at $\text{Sing}(U)$. If the extraction centre is off C , the surviving-boundary output is a fence. If the centre is on C , it is a fence or banana. In a birational output the choice of extracted curve gives $b < a$.*

This is [Nag25a, Definition and Notation 4.1, Remark 4.2, and Lemma 4.3]. The three surviving-boundary possibilities, together with the net and the contraction of C_T , give five cases.

8.1. The boundary and the extraction.

Lemma 8.3. *Let S be a rank-one klt del Pezzo surface over k , and let C be a reduced irreducible curve such that (S, C) is plt and $K_S + C \equiv 0$. Suppose one of the following holds:*

- (i) C is rational;
- (ii) S is singular and has no exceptional tiger on its minimal resolution $\mu : Y \rightarrow S$.

Then C is smooth and rational. The determinants of the exceptional chains over $C \cap \text{Sing}(S)$, with repetitions, are one of

$$(2, 3, 6), \quad (2, 4, 4), \quad (3, 3, 3), \quad (2, 2, 2, 2).$$

In particular, C contains at least three singular points of S .

The second alternative is the place where the absence of an exceptional tiger matters. In the birational realization the image of the extracted rational curve gives the first alternative instead.

Proof. We determine the genus of C and then solve the adjunction equation for the boundary determinants.

Step 1. In this step, we prove that C is smooth and rational under either hypothesis. Proposition 4.10 makes C smooth and gives

$$(8.1) \quad 0 = 2p_a(C) - 2 + \sum_{p \in C \cap \text{Sing}(S)} \left(1 - \frac{1}{d_p}\right), \quad d_p \geq 2.$$

Thus $p_a(C) \leq 1$. Under (i), smoothness and rationality give $p_a(C) = 0$.

Under (ii), suppose that $p_a(C) = 1$. Equation (8.1) implies $C \cap \text{Sing}(S) = \emptyset$, so C is Cartier. Since $C \equiv -K_S$, its self-intersection is a positive integer. Theorem 5.11 makes Y rational, so $H^1(Y, \mathcal{O}_Y) = 0$. The rational-singularity assertion in Theorem 4.16 gives $\mu_* \mathcal{O}_Y = \mathcal{O}_S$ and $R^1 \mu_* \mathcal{O}_Y = 0$. Leray therefore gives $H^1(S, \mathcal{O}_S) = 0$. The sequence

$$0 \longrightarrow \mathcal{O}_S \longrightarrow \mathcal{O}_S(C) \longrightarrow \mathcal{O}_C(C) \longrightarrow 0$$

and this vanishing give

$$h^0(S, \mathcal{O}_S(C)) = 1 + h^0(C, \mathcal{O}_C(C)) = 1 + C^2 \geq 2.$$

Here Riemann–Roch on the smooth genus-one curve gives the middle equality. At any $p \in \text{Sing}(S)$, the section defining C does not vanish. Evaluation at p is therefore onto a one-dimensional space, and its kernel contains a nonzero section. Its zero divisor H is effective

Cartier, satisfies $H \sim C$, and contains p . By Theorem 3.7, $\mu^*K_S - K_Y$ is effective. For every exceptional prime F above p , the Cartier pullback of H has a positive integral coefficient along F . Adding the nonnegative coefficient of F in $\mu^*K_S - K_Y$ gives $e(F; S, H) \geq 1$. Since $K_S + H \equiv 0$, this is an exceptional tiger, a contradiction. Hence $p_a(C) = 0$ also under (ii).

Step 2. In this step, we determine all possible multisets of boundary determinants. The sum in (8.1) is now 2. Every term lies in $[1/2, 1)$, so its number of terms is three or four. Four terms must all equal $1/2$. For three terms, order the determinants as $d_1 \leq d_2 \leq d_3$. Their reciprocals sum to 1, whence $d_1 \leq 3$. If $d_1 = 3$, all three determinants equal 3. If $d_1 = 2$, then $1/d_2 + 1/d_3 = 1/2$, so $d_2 = 3$ or 4 , giving $(2, 3, 6)$ or $(2, 4, 4)$. $\square$

Set-up 8.4. Let S be a singular rank-one $\mathbb{Q}$ -factorial klt del Pezzo surface over k , and let C be a smooth rational curve such that (S, C) is plt and $K_S + C \equiv 0$. Run a pair hunt near coefficient one. Write $f : T \rightarrow S$ for its extraction, $E = \text{Exc}(f)$, $x = f(E)$, and

$$(8.2) \quad f^*(K_S + C) = K_T + C_T + eE \equiv 0.$$

In a birational output write $g : T \rightarrow U$, $\Sigma = \text{Exc}(g)$, and $q = g(\Sigma)$. Both maps are isomorphisms away from their exceptional curves and their image points. The surviving images are denoted by $C_U = g(C_T)$ and $A = g(E)$.

The point x is singular, since the extraction retains a component of the minimal resolution. The intermediate surface has Picard number two. For a sufficiently close to one, the extracted coefficient for (S, aC) is positive by the centre choice in Definition 8.1. Its crepant pullback to T is therefore a klt pair with effective boundary. Theorem 4.16 makes T $\mathbb{Q}$ -factorial; it gives the same conclusion for the klt birational endpoint U . In a birational output $\Sigma^2 < 0$, the surface U is rank-one klt del Pezzo, and $K_T \cdot \Sigma < 0$. These are the one-curve geometry and the opposite-ray property of Theorem 6.3. In particular, Proposition 4.12 gives

$$(8.3) \quad 0 < -K_T \cdot \Sigma \leq 1, \quad \#(\text{Sing}(T) \cap \Sigma) \leq 2.$$

Lemma 8.5. *In Set-up 8.4, one has $1/2 \leq e < 1$. If $x \in C$, its chain determinant d_x is maximal among the boundary determinants, f retains the end adjacent to C , and*

$$e = 1 - \frac{1}{d_x}.$$

The curves E and C_T then meet transversely once in T^{reg} , and C_T has no surface singularity over x . The remaining boundary determinants are, respectively,

$$(2, 3), \quad (2, 4), \quad (3, 3), \quad (2, 2, 2).$$

If both curves survive g , then

$$(8.4) \quad K_U + C_U + eA \equiv 0.$$

The coefficient bound makes the net case short. The information about the retained end also identifies exactly which old singular point is removed from the boundary.

Proof. The choice incorporated in Definition 8.1 uses a point of maximal boundary determinant if the centre is on C . If the centre is off C , it is chosen only when its coefficient is at least $1 - 1/d$, where d is the largest boundary determinant. This is the choice in [Nag25a, Remark 4.2], referring to the proof of [KM97, Proposition 23.5].

Step 1. We check which component is retained at a boundary centre. Label its chain $F_1, \dots, F_r$ from the end met by C_Y , and let B be the negative intersection matrix. Write $c_i(a)$ for its pullback coefficients at boundary coefficient a . At $a = 1$ the intersection equations give

$$B(\mathbf{1} - c(1)) = (0, \dots, 0, 1)^t.$$

If Δ_j denotes the determinant of the first j vertices and $\Delta_0 = 1$, the inverse matrix gives

$$1 - c_i(1) = \frac{\Delta_{i-1}}{\Delta_r}.$$

The recurrence $\Delta_j = b_j \Delta_{j-1} - \Delta_{j-2}$, with $b_j \geq 2$, makes these leading determinants strictly increasing. Thus $c_1(1) > c_i(1)$ for $i > 1$. Each $c_i(a)$ is affine in a , so the strict inequalities persist near 1. The maximal-coefficient extraction retains F_1 . Its unscaled coefficient is $1 - 1/d_x$ by the determinant calculation in Proposition 4.10. Contracting the other vertices does not affect the transverse point $C_Y \cap F_1$.

Step 2. In this step, we prove the coefficient bounds and determine the residual boundary determinants. This proves $e \geq 1/2$ at a boundary centre. At an off-boundary centre the coefficients for $K_S + aC$ and $K_S + C$ agree, and the centre choice gives the same lower bound. The inequality $e < 1$ follows from plt. Deleting the maximal entry from the four signatures in Lemma 8.3 gives the displayed residual lists.

Step 3. In this step, we push the numerically trivial adjoint relation to U when both curves survive. Finally set $L = K_U + C_U + eA$ when both curves survive. The divisor $K_T + C_T + eE - g^*L$ is a rational multiple of Σ . Both its summands have zero intersection with Σ , so $\Sigma^2 < 0$ makes that multiple zero. Equation (8.2) then implies $L \equiv 0$. $\square$

8.2. Nets and transverse surviving boundaries.

Lemma 8.6. *In Set-up 8.4, if the output is a net, then $\#\text{Sing}(S) \leq 7$.*

Only the extracted divisor can contribute a horizontal exceptional component on the minimal resolution of S .

Proof. Let $h : T \rightarrow B$ be the net, and choose a general fibre F disjoint from $\text{Sing}(T)$. Adjunction gives $K_T \cdot F = -2$, so (8.2) gives

$$C_T \cdot F + e(E \cdot F) = 2.$$

Both intersections are nonnegative. Since $e \geq 1/2$, we get $E \cdot F \leq 4$. The minimal resolution $Y \rightarrow S$ factors through T , and every exceptional component except the one retained as E maps to a point on T . Those components are vertical for $Y \rightarrow B$. Thus the reduced exceptional divisor of $Y \rightarrow S$ has degree at most four on a general fibre. Corollary 3.17 gives the conclusion. $\square$

Lemma 8.7. *In Set-up 8.4, suppose both curves survive the birational contraction. Then C_U is smooth and (U, C_U) is dlt. The following incidence statements hold:*

- (i) *If $x \notin C$, the fence point is q . The curve Σ meets C_T at exactly one point, which is singular on T . Consequently C_U contains at least two singular points of U .*
- (ii) *If $x \in C$ and the output is a fence, then $q \notin C_U$. The morphism g is an isomorphism near C_T , and the singularities on C_U are exactly the residual boundary singularities of Lemma 8.5, with their original end-marked chains.*
- (iii) *If the output is a banana, q is one of its two transverse intersection points.*

These statements distinguish the intersection inherited from the extraction from one created by the opposite contraction.

Proof. We bound the intersections with Σ , determine the boundary incidence, and then verify smoothness and dlt singularities along C_U .

Step 1. In this step, we prove $E \cdot \Sigma > 0$ and $C_T \cdot \Sigma < 1$. First $E \cdot \Sigma > 0$. Indeed, $E^2 < 0$, whereas the nonzero image A has positive square on the rank-one surface U . If $E \cdot \Sigma = 0$, the numerical pullback of A would be E , contradicting these two signs. Intersecting (8.2) with Σ and using (8.3) gives

$$C_T \cdot \Sigma + e(E \cdot \Sigma) = -K_T \cdot \Sigma \leq 1.$$

Therefore $C_T \cdot \Sigma < 1$. Any intersection at a smooth point of T contributes a positive integer. Thus all intersections of C_T with Σ , if present, are at singular points of T .

Step 2. In this step, we determine the incidence at q according to the position of x . If $x \notin C$, then $C_T \cap E = \emptyset$. Their images can meet only at q ; the fence has exactly one intersection, so it is q . The curve C_T is smooth, and the fence makes C_U smooth at q . The restriction $C_T \rightarrow C_U$ is finite birational. It is an isomorphism near the normal point q , so its fibre there is one point. This is the singular point $C_T \cap \Sigma$ just identified. All other singular points of S on C survive isomorphically on C_U . There were at least three, hence at least two survive.

If $x \in C$, the point $C_T \cap E$ is smooth on T by Lemma 8.5. It is not on Σ , by the preceding intersection inequality. Its image is one transverse boundary intersection away from q . Since $E \cdot \Sigma > 0$, we have $q \in A$. Thus $q \in C_U$ would supply a second intersection. It occurs precisely in the banana case; in the fence case $q \notin C_U$ and g is an isomorphism near C_T . This proves (ii) and (iii).

Step 3. In this step, we prove that C_U is smooth and that (U, C_U) is dlt. The old transform C_T is smooth: off x this follows from smoothness of C , and over x it follows from the retained-end description. Away from q , smoothness transfers to C_U . At a point of C_U equal to q , it follows from the transverse fence or banana intersection. At singular points of U , lowering the reduced boundary in Theorem 8.2 preserves dlt; on the smooth locus a smooth reduced boundary is dlt. Hence (U, C_U) is dlt. $\square$

Lemma 8.8. *In Set-up 8.4, suppose the output is an off-boundary fence or a banana. Then*

$$\# \text{Sing}(U) \leq 4, \quad \# \text{Sing}(S) \leq 7.$$

The old boundary defines an anti-ample open pair on U . Its transverse intersection with A excludes the exceptional open-pair graphs.

Proof. By (8.4), $-(K_U + C_U) \equiv eA$ and $-(K_U + eA) \equiv C_U$ are ample. Lemma 8.7 makes (U, C_U) dlt. Proposition 4.10 identifies its minimal log resolution with the minimal surface resolution $\pi : V \rightarrow U$: the strict transform of C_U meets each exceptional chain at an end. Write $D = C_V + \text{Exc}(\pi)_{\text{red}}$.

Step 1. In this step, we bound $\# \text{Sing}(U)$ in the plane and low-height cases. If $U \simeq \mathbb{P}^2$, there is nothing to prove about U . If $\text{ht}(U, C_U) \leq 2$, the reduced exceptional divisor has general-fibre degree at most two, since it is contained in D . Thus $\text{ht}(U) \leq 2$, and Theorem 3.2 gives $\# \text{Sing}(U) \leq 4$.

Step 2. In this step, we exclude the exceptional open-pair graphs when $\text{ht}(U, C_U) > 2$. It remains to exclude $U \not\simeq \mathbb{P}^2$ and $\text{ht}(U, C_U) > 2$. The curve C_U is reduced and irreducible, U is $\mathbb{Q}$ -factorial, and $-(K_U + C_U)$ is ample, so Proposition 5.9 applies on this same resolution. The boundary contains a fork with centre R of square -1 and tips of squares $-2, -3, -\ell$, where $\ell \in \{3, 4, 5\}$. All other components of D are disjoint from this fork, and C_V is the only component of D not contracted by π . The entire fork cannot be contracted: eliminating its tips from the negative intersection matrix gives the central entry

$$1 - \frac{1}{2} - \frac{1}{3} - \frac{1}{\ell} < 0,$$

whereas Theorem 4.5 makes an exceptional intersection matrix negative definite. Thus C_V belongs to the fork. It cannot be a tip. In that case it would meet the middle of the exceptional three-component chain formed by the centre and the other two tips, contrary to the end incidence in Proposition 4.10.

It remains that $C_V = R$. Solving the three one-vertex pullback equations gives

$$\beta = C_U^2 = -1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{\ell} \in \left\{ \frac{1}{6}, \frac{1}{12}, \frac{1}{30} \right\}, \quad (K_U + C_U) \cdot C_U = -\beta.$$

For example, the coefficient of a tip of square $-n$ in π^*C_U is $1/n$, and its coefficient in the pullback of $K_U + C_U$ is $1 - 1/n$. Put $j = A \cdot C_U$, so $j = 1$ for the fence and $j = 2$ for the banana. We obtain

$$(K_U + eA) \cdot C_U = ej - 2\beta \geq \frac{1}{2} - \frac{1}{3} > 0.$$

This contradicts $-(K_U + eA)$ ample. The exceptional alternative is therefore impossible, and $\#\text{Sing}(U) \leq 4$.

Step 3. In this step, we transfer the four-point bound on U to the seven-point bound on S . Let $r = \#(\text{Sing}(T) \cap E)$ and $\kappa = \#(\text{Sing}(T) \cap \Sigma)$. The point q is smooth by Lemma 8.7. Since x is singular, Lemma 4.13 gives

$$\#\text{Sing}(S) = \#\text{Sing}(U) + 1 - r + \kappa \leq 4 + 1 + 2 = 7,$$

using (8.3). □

8.3. A fence with centre on the old boundary.

Theorem 8.9 (Lacini). *Let S be a rank-one klt del Pezzo surface over an algebraically closed field, and let X, Y be smooth rational curves such that $(S, X + Y)$ is a fence. Then X and Y cannot both contain exactly two singular points of S . Moreover, if X contains at most one singular point of S , then Y also contains at most one singular point of S .*

These restrictions are [Lac24, Section 4.5, Lemmas 4.24 and 4.30]. They will force the new branch to be singular as a curve.

Lemma 8.10. *In Set-up 8.4, suppose $x \in C$ and the output is a fence. Then A is singular as a curve, U with boundary A is an elliptic-boundary descendant of S , and $\#\text{Sing}(S) \leq 7$.*

The proof uses the residual boundary determinants to force $K_U + A \equiv 0$. It also shows that the singular curve A lies in the smooth locus of U .

Proof. Put $d = d_x$. We first rule out a smooth new branch, then bound the square of the old curve and identify its exceptional chains. Adjunction will then identify the descendant.

Step 1. In this step, we rule out a smooth new boundary curve A . By Lemma 8.7, C_U is smooth and its surface singularities have the residual determinants in Lemma 8.5. It is rational because C is rational. The curve A is rational as the birational image of the retained exceptional component, which is rational by Theorem 3.7. The extraction leaves at most one surface singularity on E , from its connected tail, and g can introduce only the additional point q on A . Thus A contains at most two surface singularities. If $d = 2$, the source chain is $[2]$ and has no tail, so A contains at most one.

Suppose A were smooth. Both branches would then be smooth rational curves forming a fence on the rank-one klt del Pezzo surface U . If $d = 2$, the residual determinants on C_U are $(2, 2, 2)$. The last assertion of Theorem 8.9, applied with A as the first branch, would give at most one singular point on C_U , contradicting these three points. If $d > 2$, the residual list has two entries, so C_U contains exactly two singular points. The same assertion excludes at most one singular point on A . Hence A contains exactly two, contradicting the first assertion of Theorem 8.9. Thus A is singular.

Step 2. In this step, we prove the bound $0 < C_U^2 \leq 1/d$ by adjunction on A . The curve A is rational and singular, so $p_a(A) \geq 1$. At every surface singularity on A , the reduced pair is dlt. The minimal surface resolution is therefore an isomorphism on the curve A itself: locally its smooth branch meets a chain end, and the proper birational map of curves is finite with normal target there. Intersecting the log pullback with the strict transform gives

$$(K_U + A) \cdot A = 2p_a(A) - 2 + \eta, \quad \eta = \sum_{p \in A \cap \text{Sing}(U)} \left(1 - \frac{1}{d_p}\right) \geq 0.$$

This calculation allows the curve singularities in U^{reg} ; they are retained in its arithmetic genus. Since $A \cdot C_U = 1$ and $e = 1 - 1/d$, equation (8.4) gives

$$\frac{A^2}{d} = 2p_a(A) - 1 + \eta \geq 1.$$

Picard number one and $A \cdot C_U = 1$ imply $A^2 C_U^2 = 1$. Consequently $0 < C_U^2 \leq 1/d$.

Step 3. In this step, we identify the residual chains as Du Val chains and compute C_V^2 . Let $\pi : V \rightarrow U$ be the minimal resolution, and put $s = C_V^2$. For an end-marked chain of determinant r , the self-intersection correction is the determinant of the tail divided by r . The determinant recurrence gives the following possibilities for the residual points:

r	chain	correction
2	[2]	1/2
3	[3], [2, 2]	1/3, 2/3
4	[4], [2, 2, 2]	1/4, 3/4

For the four residual lists the minimum correction sums and the upper bounds for C_U^2 are

residual determinants	minimum sum	1/d
(2, 3)	5/6	1/6
(2, 4)	3/4	1/4
(3, 3)	2/3	1/3
(2, 2, 2)	3/2	1/2

As $C_U^2 = s + \text{correction sum}$ and s is integral, each row forces $s \leq -1$. If a residual chain were non-Du-Val, the total correction would be at most 5/6, 3/4, or 1 in the first three rows. This would make $C_U^2 \leq 0$. The last row is already Du Val. Thus every residual chain is Du Val. Their correction sums are now 7/6, 5/4, 4/3, 3/2, respectively. The bounds $0 < C_U^2 \leq 1/d$ force $s = -1$ in each row.

Step 4. In this step, we identify U with boundary A as an elliptic-boundary descendant of S . Crepancy along the chains meeting C_V gives

$$K_U \cdot C_U = K_V \cdot C_V = -1.$$

Therefore $(K_U + A) \cdot C_U = 0$, and Picard number one implies $K_U + A \equiv 0$. Applying the preceding adjunction calculation again gives $0 = 2p_a(A) - 2 + \eta$. Hence $p_a(A) = 1$ and $\eta = 0$. Each surface singularity on A would contribute positively to η , so $A \subset U^{\text{reg}}$.

The minimal resolution $Y \rightarrow S$ factors through T . Exactly one component of its reduced exceptional divisor maps to E , and every other component maps to a point. Since E survives as A , the composite $Y \rightarrow U$ has reduced divisorial exceptional image A . It is an elliptic-boundary descendant in Definition 3.1. Theorem 3.6 proves $\# \text{Sing}(S) \leq 7$. $\square$

8.4. Contracting the old boundary.

Lemma 8.11. *In Set-up 8.4, suppose $C_T = \Sigma$. Then $x \in C$, its exceptional chain is $[2, 2, 2, 2, 2]$, and f retains the end adjacent to C . The complete exceptional fibre over q on the smooth resolution near Σ is $[2] - [1] - [3]$, with the strict transform of Σ in the middle. The point q is smooth, and A is rational with exactly one curve singularity, an ordinary cusp at q . Moreover,*

$$e = \frac{5}{6}, \quad K_U + \frac{5}{6}A \equiv 0, \quad \# \text{Sing}(S) = \# \text{Sing}(U) + 2.$$

The only surface singularity on A is a point $y \neq q$ of type A_4 , and the strict transform of A meets an end of its resolution chain transversely once.

The assertion concerns the complete fibre. Both the cusp and the singularity count depend on excluding any additional arms.

Proof. We identify the fibre from its intersection matrix and the scaled hunt coefficients, then contract it explicitly.

Step 1. In this step, we locate the extraction center and restrict the complete contraction fiber to three possible graphs. If $x \notin C$, then $E \cdot C_T = 0$. Changing the coefficient of E would not change the negative intersection

$$f^*(K_S + aC) \cdot C_T = (a - 1)C^2 < 0,$$

contradicting $(K_T + aC_T + bE) \cdot C_T = 0$. Thus $x \in C$. Lemma 8.5 identifies the retained end and gives $E \cdot \Sigma = 1$ in the smooth locus. The singular points of T on Σ are exactly the residual boundary points. On their minimal resolution, let M be the strict transform of Σ and write $M^2 = -j$. The complete fibre over q consists of M and their complete chains. The tail over x is disjoint from Σ and does not belong to this fibre.

For a residual arm of determinant r_i , put u_i equal to the determinant of its tail divided by r_i . The Schur complement and adjunction give

$$(8.5) \quad j > \sum_i u_i, \quad K_T \cdot \Sigma = j - 2 + \sum_i \left(1 - \frac{1}{r_i} - u_i\right) < 0.$$

For determinant two the only arm is $[2]$ and $u_i = 1/2$. For determinant three the arms are $[3]$ and $[2, 2]$, with $u_i = 1/3, 2/3$; for determinant four they are $[4]$ and $[2, 2, 2]$, with $u_i = 1/4, 3/4$. For $(2, 2, 2)$ the first inequality in (8.5) gives $j \geq 2$, and the second expression is $j - 2$, a contradiction. In any of the two-arm lists, using a longer arm forces $j \geq 2$ and makes that expression nonnegative. With one-vertex arms, negativity forces $j = 1$. Thus only

$$[2] - [1] - [3], \quad [2] - [1] - [4], \quad [3] - [1] - [3]$$

remain.

Step 2. In this step, we use the scaled coefficients to identify the complete fiber and the source chain. Write e_a for the coefficient of E in $f^*(K_S + aC)$. Projection to S gives

$$0 = (K_T + a\Sigma + bE) \cdot \Sigma = (a - 1)C^2 + (b - e_a)E \cdot \Sigma,$$

so $b > e_a$. Since the extraction has maximal coefficient for (S, aC) , the coefficient of the first vertex in every residual arm is at most e_a , hence strictly less than b . For a one-vertex arm $[n]$, that coefficient is $c_n = (n - 2 + a)/n$. Intersection with the middle (-1) -curve gives

$$0 = -1 - a + b + \sum c_n.$$

For $[2] - [1] - [4]$ this gives $b = (2 + a)/4 = c_4$; for $[3] - [1] - [3]$ it gives $b = (1 + a)/3 = c_3$. Both contradict the strict inequality. We are left with the complete fibre $[2] - [1] - [3]$. Its reduced-boundary intersection gives

$$0 = -2 + \frac{1}{2} + \frac{2}{3} + e,$$

so $e = 5/6$. The full boundary signature is consequently $(2, 3, 6)$.

An end-marked chain of determinant six is either $[6]$ or $[2, 2, 2, 2, 2]$, by the determinant recurrence. In the first case its scaled coefficient is $e_a = (4 + a)/6$. On the graph just found, the middle-curve equation gives $b = (4 + a)/6$, contradicting $b > e_a$. Thus the source chain is $[2, 2, 2, 2, 2]$. Its retained end has one residual A_4 singularity on T , disjoint from Σ . This is the asserted point y on A .

Step 3. In this step, we identify the ordinary cusp and the remaining surface singularity on A , and compute the singular-point count. Contract the middle (-1) -curve, then the image of the (-2) -arm, then the image of the (-3) -arm. At each stage the contracted curve has square -1 . Theorems 4.3 and 4.4 identify these successive contractions with the original contraction of the complete fibre; the remaining proper birational map has no exceptional curve and is an isomorphism near its normal target. In particular, q is smooth. The strict transform of E first

meets the two arm images in distinct tangent directions. After the next contraction it has contact order two with the last (-1) -curve. In coordinates for the final blowdown, its parametrization therefore has orders two and three. Since $\text{char } k \neq 2$, a formal parameter change makes the first coordinate t^2 . Subtracting the even powers from the second coordinate and dividing by a unit in $k[[t^2]]$ makes it t^3 . Thus the completed curve ring is $k[[t^2, t^3]]$.

Away from q , the map g is an isomorphism. The curve E is smooth except possibly at its end-marked A_4 point. At such an end a local equation for the surface is $uv = w^5$. In the end chart, one of the functions u, v restricts to a uniformizer on the transverse branch of the partial resolution. The image curve ring consequently contains this uniformizer and is $k[[t]]$. Hence A is regular there too. The curve A is rational because E is birational to the retained exceptional component, and the cusp is its only curve singularity.

There are no other surface singularities on A . A neighbourhood of E is obtained by retaining one end of the smooth source resolution and contracting its connected A_4 tail. The tail image is therefore the only singular point of T on E . Since g is an isomorphism away from q and q is smooth, this proves $A \cap \text{Sing}(U) = \{y\}$ before any further extraction is made.

Pushing down (8.2) gives $K_U + (5/6)A \equiv 0$. On E there is one singular point, on Σ there are exactly two, and q is smooth. Lemma 4.13 now gives $\# \text{Sing}(S) = \# \text{Sing}(U) + 1 - 1 + 2 = \# \text{Sing}(U) + 2$. $\square$

Lemma 8.12. *Let U and A be the endpoint in Lemma 8.11. There is a projective birational morphism $h : Z \rightarrow U$ from a normal projective $\mathbb{Q}$ -factorial klt surface, isomorphic over $U \setminus \{y\}$, which retains the end G adjacent to A in the A_4 chain. One has $\rho(Z) = 2$ and $K_Z = h^*K_U$ for compatible canonical divisors. Exactly one singular point of Z lies on G , of type A_3 . The strict transform A_Z lies in Z^{reg} and is isomorphic to A . It is rational with one ordinary cusp and $p_a(A_Z) = 1$. Moreover,*

$$(8.6) \quad h^*A = A_Z + \frac{4}{5}G, \quad A_Z \cdot G = 1, \quad K_Z + \frac{5}{6}A_Z + \frac{2}{3}G \equiv 0.$$

The fact that A misses every other singular point of U makes the global retained-end construction applicable.

Proof. We construct the retained-end model, check crepancy, and compute its boundary pullbacks.

Step 1. In this step, we construct Z by retaining the end adjacent to A . Let $\mu : R \rightarrow U$ be the global minimal resolution. By Lemma 8.11, $A \cap \text{Sing}(U) = \{y\}$. Thus A_R is disjoint from every exceptional fibre except the A_4 chain over y . Label that chain F_1, F_2, F_3, F_4 from the end met transversely once by A_R . All the hypotheses of Proposition 4.18 hold: U is projective, $\mathbb{Q}$ -factorial and klt, the marked fibre is a simple normal crossings chain of smooth rational curves of square -2 , and A_R misses every exceptional prime other than F_1 . It gives projective maps $R \xrightarrow{\nu} Z \xrightarrow{h} U$ with $\mu = h\nu$, retaining $G = \nu(F_1)$ and contracting every other μ -exceptional curve. In particular, $\rho(Z) = 2$, $A_Z \subset Z^{\text{reg}}$, and $A_Z \cdot G = 1$.

Step 2. In this step, we identify the residual singularity and prove $K_Z = h^*K_U$. Over y , the only contraction is the (-2) -tail $F_2 + F_3 + F_4$, so its image is an A_3 point; it is disjoint from A_Z . To check crepancy, write locally $K_R - \nu^*K_Z = \sum_{i=2}^4 c_i F_i$. Intersection with each F_j , $2 \leq j \leq 4$, gives a homogeneous system, since $K_R \cdot F_j = 0$. The tail matrix is nonsingular, so all c_i vanish. The A_4 resolution is crepant over y as well. It follows that $K_Z = h^*K_U$ there, and the equality holds away from y because h is an isomorphism there. This argument uses crepancy of μ only over y .

Step 3. In this step, we compute the pullbacks and identify the curve A_Z with A . The exceptional pullback equations give

$$\nu^*G = F_1 + \frac{3}{4}F_2 + \frac{1}{2}F_3 + \frac{1}{4}F_4,$$

whereas μ gives

$$\mu^* A = A_R + \frac{4}{5}F_1 + \frac{3}{5}F_2 + \frac{2}{5}F_3 + \frac{1}{5}F_4.$$

The strict transform A_R is disjoint from the contracted tail, so $\nu^* A_Z = A_R$. These identities prove (8.6). The restriction $A_R \rightarrow A$ is an isomorphism: it is one away from y , and over y it is a finite birational map to the regular curve germ identified in Lemma 8.11. Since ν is an isomorphism near A_R , the same holds for $A_Z \rightarrow A$. Finally, the normalization is $\mathbb{P}^1$ and $k[[t]]/k[[t^2, t^3]]$ has length one, so $p_a(A_Z) = 1$. $\square$

Lemma 8.13. *In the situation of Lemma 8.12, there is a nonzero effective divisor with one irreducible support curve N , equal to a positive rational multiple mN , such that*

$$K_Z + A_Z \equiv mN, \quad A_Z \cap N = \emptyset, \quad N^2 < 0, \quad K_Z \cdot N < 0.$$

The adjoint divisor is constructed before its support is contracted. The argument permits its support to pass through the residual A_3 point on Z .

Proof. We construct an intrinsic adjoint divisor and use Picard number two to show that its support is one curve.

Step 1. In this step, we construct a nonzero effective adjoint divisor disjoint from A_Z . Choose a smooth resolution $\tau : V \rightarrow Z$ which is an isomorphism near A_Z , and put $\sigma = h\tau$ and $B = (A_Z)_V$. Since σ is a smooth projective resolution of the rank-one klt del Pezzo surface U , Theorem 5.11 makes V rational. Thus $H^0(V, \omega_V) = H^1(V, \omega_V) = 0$. The curve B is integral Gorenstein of arithmetic genus one: its normalization is $\mathbb{P}^1$, and the quotient $k[[t]]/k[[t^2, t^3]]$ has length one. Adjunction gives an isomorphism

$$H^0(V, \mathcal{O}_V(K_V + B)) \simeq H^0(B, \omega_B).$$

The right side is one-dimensional. Thus $|K_V + B|$ consists of one effective divisor D . It is nonzero, since for $H = -K_U$ we have

$$D \cdot \sigma^* H = (K_U + A) \cdot H = \frac{1}{5}H^2 > 0.$$

Its defining section restricts to a nonzero section of the degree-zero line bundle ω_B , which has no zeros. Hence $\text{Supp}(D) \cap B = \emptyset$.

Push D to Z , obtaining $M_Z = \tau_* D$. Its further pushforward to U has positive intersection with H , so $M_Z \neq 0$. Pushing the linear equivalence $D \sim K_V + B$ gives $M_Z \sim K_Z + A_Z$. Since τ is an isomorphism near A_Z , the disjointness also descends. Indeed, over a point of A_Z the whole fibre is its single point on B , which D avoids.

Step 2. In this step, we prove that the adjoint divisor has one support curve with negative square and negative canonical degree. Write $M_Z = \sum_{i=1}^s m_i N_i$. No N_i is G , since G meets A_Z . Every $h_* N_i$ is therefore a curve. The equality $K_Z = h^* K_U$ and the projection formula give

$$K_Z \cdot N_i = K_U \cdot h_* N_i < 0.$$

Let $Q = (N_i \cdot N_j)$ and $\mathbf{m} = (m_i)$. Its off-diagonal entries are nonnegative, and

$$(Q\mathbf{m})_i = M_Z \cdot N_i = (K_Z + A_Z) \cdot N_i = K_Z \cdot N_i < 0.$$

For $P = \text{diag}(m_i)$, the symmetric matrix $-PQP$ has nonpositive off-diagonal entries and strictly positive row sums. It is positive definite: its quadratic form is bounded below by the sum of its positive row sums times the corresponding squared coordinates. Thus Q is negative definite. The classes $[N_i]$ are independent, so $s \leq 2$. If $s = 2$, they span the numerical curve space, and $A_Z \cdot N_i = 0$ would force $A_Z \cdot G = 0$, contrary to (8.6). Hence $s = 1$. Writing $M_Z = mN$ gives all the remaining assertions. $\square$

Lemma 8.14. *In the situation of Lemma 8.13, there is a projective birational morphism $p : Z \rightarrow W$, contracting exactly N to a point w and an isomorphism away from N , where W is a rank-one canonical del Pezzo surface of type D_5 with $K_W^2 = 4$. The curves $A_W = p(A_Z)$ and $G_W = p(G)$ satisfy*

$$A_W \subset W^{\text{reg}}, \quad A_W \in |-K_W|, \quad A_W \cdot G_W = 1, \quad K_W + \frac{5}{6}A_W + \frac{2}{3}G_W \equiv 0.$$

The curve A_W is rational with one ordinary cusp. In particular, $\#\text{Sing}(U) \leq 3$ and $\#\text{Sing}(S) \leq 5$.

The degree and the cusp identify the target and give a bound stronger than the one needed for the original surface.

Proof. We contract the adjoint support, determine the target, and count the singular points across the resulting link.

Step 1. In this step, we contract N projectively and prove that the target is a rank-one klt del Pezzo surface. Choose an ample Cartier divisor H on Z and put

$$\lambda_0 = -\frac{H \cdot N}{N^2} > 0, \quad L = H + \lambda_0 N.$$

Then $L \cdot N = 0$, whereas $L \cdot R \geq H \cdot R > 0$ for every irreducible curve $R \neq N$. The same inequality holds for effective cycles without N . On taking limits in the closed cone, it shows that the face $L^\perp \cap \overline{\text{NE}}(Z)$ is precisely the ray generated by N : the part without N has vanishing ample degree whenever its L -degree tends to zero. This ray is K_Z -negative. Theorem 4.17 contracts it projectively to a normal projective $\mathbb{Q}$ -factorial surface W . Only N is contracted, so the contraction is birational and is an isomorphism away from N .

Because $A_Z \cap N = \emptyset$, the map is an isomorphism near A_Z and $p^*A_W = A_Z$. Write

$$K_Z = p^*K_W + \alpha N, \quad p^*G_W = G + \lambda N.$$

Intersecting the first equality with N and using $K_Z + A_Z \equiv mN$ shows $\alpha = m > 0$. Consequently $K_W + A_W \equiv 0$. The target is klt: on any resolution above Z , the log pullback coefficients for W are those for Z minus the nonnegative coefficients of the pullback of αN . They are therefore strictly less than one. Picard number drops from two to one, and the nonzero effective class $A_W \equiv -K_W$ is ample, so W is del Pezzo.

Step 2. In this step, we identify the target as a canonical del Pezzo surface of degree four and type D_5 . The curve A_W lies in the smooth locus and has arithmetic genus one. Adjunction to A_Z and equation (8.6) give

$$(K_Z + A_Z) \cdot A_Z = 0, \quad -\frac{1}{6}A_Z^2 + \frac{2}{3} = 0.$$

Hence $A_Z^2 = A_W^2 = 4$ and $K_W^2 = 4$. The pullback of $K_W + (5/6)A_W + (2/3)G_W$ differs from the numerically trivial left side of (8.6) by $((2/3)\lambda - m)N$. Intersection with N makes this coefficient zero. The displayed numerical relation on W follows, and $A_W \cdot G_W = A_Z \cdot G = 1$.

To obtain linear equivalence and canonicity, take a smooth rational resolution $V' \rightarrow W$ that is an isomorphism near A_W . The same adjunction exact sequence used above gives an effective divisor in $|K_{V'} + (A_W)_{V'}|$. Its pushforward is effective and linearly equivalent to $K_W + A_W$. It is numerically trivial, hence zero by ampleness of $-K_W$. Thus $K_W + A_W \sim 0$. Since A_W is Cartier, K_W is Cartier. The discrepancies are integral and greater than -1 , so they are nonnegative. Hence W is canonical. The curve A_W is integral, rational, contained in W^{reg} and anticanonical, and its only singularity is the ordinary cusp. Since $K_W^2 = 4$, Theorem 5.13 gives type D_5 . Thus $\#\text{Sing}(W) = 1$.

Step 3. In this step, we count singular points through the adjoint link and obtain the bounds on U and S . In the link $U \xleftarrow{h} Z \xrightarrow{p} W$, the extracted end G contains exactly one surface

singularity, the residual A_3 point. By Proposition 4.12, at most two singular points of Z lie on N . Lemma 4.13 therefore gives

$$\# \text{Sing}(U) = \# \text{Sing}(W) + 1 - 1 + \#(\text{Sing}(Z) \cap N) - \mathbf{1}_{w \in \text{Sing}(W)} \leq 3.$$

Lemma 8.11 gives $\# \text{Sing}(S) = \# \text{Sing}(U) + 2 \leq 5$. $\square$

8.5. The direct surface and the earlier tiger link.

Corollary 8.15. *Let S be a rank-one klt del Pezzo surface over k , and let C be a reduced irreducible rational curve such that (S, C) is plt and $K_S + C \equiv 0$. Then C contains at least three singular points of S , and $\# \text{Sing}(S) \leq 7$.*

Proof. Lemma 8.3(i) makes C smooth and supplies at least three boundary singularities. In particular, S is singular. Theorem 4.16 makes S $\mathbb{Q}$ -factorial. Run a pair hunt near coefficient one, with the notation of Set-up 8.4.

Theorem 8.2 gives five possibilities. For a net, Lemma 8.6 gives $\# \text{Sing}(S) \leq 7$. If C_T is contracted, Lemmas 8.11–8.14 give $\# \text{Sing}(S) \leq 5$. For an off-boundary fence or a banana, Lemma 8.8 gives $\# \text{Sing}(U) \leq 4$ and $\# \text{Sing}(S) \leq 7$. The remaining possibility is a fence with centre on C , which is covered by Lemma 8.10. These cases exhaust the outputs and prove the bound. $\square$

Theorem 8.16. *Let S be a singular rank-one klt del Pezzo surface over k , and let C be a reduced irreducible curve with (S, C) plt and $K_S + C \equiv 0$. Suppose that S has no exceptional tiger on its minimal resolution. Then $\# \text{Sing}(S) \leq 7$.*

Proof. The absence of an exceptional tiger gives the second alternative of Lemma 8.3. Hence C is smooth and rational. Corollary 8.15 now gives $\# \text{Sing}(S) \leq 7$. $\square$

The direct hypothesis is used to rule out a smooth elliptic boundary. For an endpoint of a prior extraction, the rationality of the extracted curve supplies that step instead. The singularity estimate must then be carried back to the original surface.

Lemma 8.17. *Let $X \xleftarrow{f_0} T_0 \xrightarrow{g_0} S$ be a link satisfying the hypotheses of Lemma 4.13, with distinct exceptional curves E_0, Σ_0 , and put $C = g_0(E_0)$. Assume S is $\mathbb{Q}$ -Gorenstein, $K_{T_0} \cdot \Sigma_0 < 0$, and T_0 is $\mathbb{Q}$ -factorial. If C contains at least three singular points of S and $\# \text{Sing}(S) \leq 7$, then $\# \text{Sing}(X) \leq 7$.*

It is the number of singularities away from C that pays for the earlier link.

Proof. Proposition 4.12 gives $\#(\text{Sing}(T_0) \cap \Sigma_0) \leq 2$. Writing $m = \#(C \cap \text{Sing}(S))$, the second assertion of Lemma 4.13 gives

$$\# \text{Sing}(X) \leq \# \text{Sing}(S) + 3 - m = \#(\text{Sing}(S) \setminus C) + 3 \leq 4 + 3 = 7.$$

$\square$

Theorem 8.18. *Let X be a rank-one klt del Pezzo surface over k . Suppose its exceptional-tiger reduction is output (iii) of Theorem 6.5, with endpoint (S, C) satisfying $K_S + C \equiv 0$. Then $\# \text{Sing}(X) \leq 7$.*

Proof. We bound the endpoint using the rational origin of its boundary and then transfer the estimate through the original link.

Step 1. In this step, we bound $\# \text{Sing}(S)$ and the number of singular points on its boundary C . Write that link as $X \xleftarrow{f_0} T_0 \xrightarrow{g_0} S$, with exceptional curves E_0 and Σ_0 . The curve E_0 survives as $C = g_0(E_0)$, so $E_0 \neq \Sigma_0$. On the minimal resolution of X , the component retained as E_0 is rational by Theorem 3.7. Its birational image C is therefore rational. Corollary 8.15 gives

$$\# \text{Sing}(S) \leq 7, \quad \#(C \cap \text{Sing}(S)) \geq 3.$$

Step 2. We check the hypotheses of Lemma 8.17 for transferring this estimate. The point $f_0(E_0)$ is singular on X , since the extraction retains a component of its minimal resolution. Each morphism contracts one curve. Normality and the connected fibres of Theorem 4.8 give the one-curve fibres and the isomorphisms away from the centres required in Lemma 4.13. For compatible canonical divisors write

$$f_0^*K_X = K_{T_0} + e_0E_0, \quad 0 \leq e_0 < 1,$$

where the coefficient bounds follow from Theorem 3.7. This is the crepant pullback of the klt pair $(X, 0)$ with effective boundary, so Theorem 4.16 makes T_0 $\mathbb{Q}$ -factorial. Since $f_0(\Sigma_0)$ is a curve and $-K_X$ is ample,

$$K_{T_0} \cdot \Sigma_0 = K_X \cdot (f_0)_*\Sigma_0 - e_0(E_0 \cdot \Sigma_0) < 0.$$

The surface S is klt and hence $\mathbb{Q}$ -Gorenstein. All the hypotheses of Lemma 8.17 are satisfied, and it gives $\#\text{Sing}(X) \leq 7$. $\square$

Corollary 8.19. *Let X be a rank-one klt del Pezzo surface over k with a tiger on its minimal resolution. Then $\#\text{Sing}(X) \leq 7$.*

All tiger reductions are accounted for by the negative and numerically trivial boundary cases.

Proof. Apply Theorem 6.5. Its net and singular anticanonical outputs are covered by Proposition 7.2. For an anti-nef endpoint (S, C) , Picard number one implies that either $-(K_S + C)$ is ample or $K_S + C \equiv 0$. In the exceptional-tiger realization these alternatives are covered by Proposition 7.7 and Theorem 8.18, respectively. If there is no exceptional tiger, the boundary lies on X itself. Proposition 7.5 covers the anti-ample case, and Theorem 8.16 covers the numerically trivial case when X is singular. A smooth X has no singular points. $\square$

9. FIRST NETS AND SINGULAR FIRST BOUNDARIES

Corollary 8.19 settles surfaces with a tiger on their minimal resolution. We now consider the complementary case, using the hunt of Definition 6.2, including its convention for equal maximal coefficients. Let $p : Y \rightarrow S_0$ be the minimal resolution, and put $D = \text{Exc}(p)_{\text{red}}$. The geometric arguments below produce a $\mathbb{P}^1$ -fibration on Y whose general fibre meets D in degree at most four. Corollary 3.17 then bounds the number of singular points of S_0 . All klt surfaces occurring below are $\mathbb{Q}$ -factorial by Theorem 4.16.

We first record the small-coefficient conclusion from the opening and Case 1 of the proof of [Lac24, Lemma 4.2], before the use of Riemann–Hurwitz. Its local inputs are [KM97, Proposition 10.1(1) and Lemma 10.4]; its validity in characteristic three is explained in [Nag25a, Section 3.1].

Theorem 9.1 (Lacini–Nagaoka). *Let S_0 be a rank-one klt del Pezzo surface over the algebraically closed field k of characteristic 3, without a tiger on its minimal resolution. Run the first hunt of Definition 6.2 from $(S_0, 0)$, with extraction $f_1 : T_1 \rightarrow S_0$, extracted curve E_1 , and net output $h : T_1 \rightarrow C$. Define e_0 by*

$$K_{T_1} + e_0E_1 = f_1^*K_{S_0},$$

and assume $e_0 < 2/5$. Then the following hold:

- (i) $e_0 \geq 1/3$.
- (ii) Every non-Du Val singularity of S_0 has resolution graph [3].
- (iii) $E_1 \subset T_1^{\text{reg}}$.
- (iv) There are a closed point $c \in C$ and an integral curve $G \subset T_1$ such that $h^*(c) = 3G$ as Weil divisors.

The smooth-locus condition allows us to restrict the multiple fibre to E_1 and recover its degree over the base.

Lemma 9.2. *Let S_0 be a rank-one klt del Pezzo surface over k without a tiger on its minimal resolution $p : Y \rightarrow S_0$. Suppose that the first hunt is a net $h : T_1 \rightarrow C$. Write*

$$K_{T_1} + e_0 E_1 = f_1^* K_{S_0}.$$

Assume that either $e_0 \geq 2/5$, or that $e_0 \geq 1/3$, $E_1 \subset T_1^{\text{reg}}$, and some fibre of h is $3G$ for an integral curve G . Then the induced fibration on Y has general fibre F with $F \cdot D \leq 4$. In the second alternative, $F \cdot D = 3$.

The multiple fibre controls the horizontal degree even in characteristic three.

Proof. Since f_1 extracts a component of the minimal resolution, there is a morphism $\tau : Y \rightarrow T_1$ with $p = f_1 \tau$. Exactly one component of D maps to E_1 ; every other component maps to a point. The curve E_1 is horizontal for h , since its numerical class spans the edge different from the fibre edge. If F_T is a general fibre and $d = F_T \cdot E_1$, then d is a positive integer and

$$F \cdot D = d, \quad -2 + e_0 d = f_1^* K_{S_0} \cdot F_T < 0.$$

Here F_T avoids the finitely many singular points of T_1 , and $f_1(F_T)$ is a curve. Thus $e_0 < 2/d$. If $e_0 \geq 2/5$, this gives $d < 5$.

In the other alternative, restrict h to the normalization of E_1 . This is a finite map of degree d . Near E_1 , the surface is smooth and the curve G is Cartier. The fibre divisor restricts to three times an integral divisor on this normalization. Its degree is d , so $3 \mid d$. The inequalities $1/3 \leq e_0 < 2/d$ give $d < 6$; hence $d = 3$. This divisor calculation applies also when the map from E_1 to C is inseparable. $\square$

Proposition 9.3. *Let S_0 be a rank-one klt del Pezzo surface over k without a tiger on its minimal resolution. If its first hunt step has a net output, then $\# \text{Sing}(S_0) \leq 7$.*

Proof. Write $K_{T_1} + e_0 E_1 = f_1^* K_{S_0}$. If $e_0 \geq 2/5$, the first alternative of Lemma 9.2 applies. If $e_0 < 2/5$, Theorem 9.1 supplies its second alternative. In both cases the minimal resolution of S_0 has a fibration of exceptional degree at most four, and Corollary 3.17 gives $\# \text{Sing}(S_0) \leq 7$. $\square$

The higher-genus birational output is classified by [Nag25a, Proposition 3.1 and Example 3.6]. Of the three families there, two require characteristics two and five, respectively. The remaining family gives the following statement.

Proposition 9.4. *Let S_0 be a rank-one klt del Pezzo surface over k without a tiger on its minimal resolution. If its first hunt step is birational and $p_a(A_1) > 1$, then its singularity type is $[2, 3, 2, 2] + [3, 2, 2, 2, 2]$. In particular $\# \text{Sing}(S_0) = 2$.*

The two connected summands count the two singular points of S_0 .

We next prepare the genus-one case. Its two uses of the old double point are different: it prevents the old curve from being a fibre, and it determines which boundary has negative adjoint divisor after the second contraction.

The local contraction results we use come from the arbitrary-characteristic analysis in [Lac24, Section 3.4]. The smoothness assertion is stated there immediately after Notation 3.19.

Theorem 9.5 (Keel–McKernan–Lacini). *Work over an algebraically closed field k of arbitrary characteristic. Let T be a projective klt surface, and let $g : T \rightarrow S$ be a proper birational contraction of a K_T -negative extremal ray onto a normal surface. Let Σ be its irreducible exceptional curve and put $q = g(\Sigma)$. Let $W \subset T$ be a reduced curve with smooth components meeting transversely and at most two components in the étale local configuration along Σ . Assume that $g|_W$ is finite. Let Δ_W be a rational boundary with support W , with every component coefficient strictly between zero and one. Suppose that $K_T + \Delta_W$ is $\mathbb{Q}$ -Cartier, g -numerically trivial, and flush. Put $D = g(W)_{\text{red}}$. If $\text{mult}_q D = 2$, then $q \in S^{\text{reg}}$.*

The hypothesis that the log divisor is trivial on the contracted ray will follow from the scaling in the hunt. To state the exceptional local case, we use the configurations of [Lac24, Definitions 3.17–3.18] and [KM97, Sections 11.0–11.2]. The induced morphism between minimal resolutions exists by Lemma 4.9.

Definition 9.6. Let $g : T \rightarrow S$, W , D , Σ , and q be as in Theorem 9.5, with $\text{mult}_q D = 2$. Let $\tilde{T} \rightarrow T$ and $\tilde{S} \rightarrow S$ be the minimal surface resolutions, and let $h : \tilde{T} \rightarrow \tilde{S}$ be the induced morphism. The following configurations describe the sequence of blowups of h over q .

- (i) A *node of order m* has two smooth branches meeting with intersection multiplicity $m \geq 1$. The $m - 1$ successive blowups at their common point that make their strict transforms transverse form *configuration 0*. Blowing up their intersection once more gives *configuration I*.
- (ii) A *cusp of order m* is a unibranch double point whose strict transform becomes smooth after $m \geq 1$ successive blowups along the branch. These m blowups form *configuration I*.

We say that g has a specified configuration when the blowups of h over q are exactly the indicated sequence.

For an ordinary node or cusp, configuration I consists of one blowup. The resulting smooth strict transform of a cusp is still tangent to the exceptional curve. The first hunt with configuration I is classified in [Nag25a, Lemma 3.2 and Example 3.3].

Theorem 9.7 (Nagaoka). *Let S_0 be a rank-one klt del Pezzo surface over an algebraically closed field k of arbitrary characteristic, without a tiger on its minimal resolution. Run the first hunt from $(S_0, 0)$ and suppose its opposite contraction $g_1 : T_1 \rightarrow S_1$ is birational. Let $A_1 = g_1(E_1)$ and $q_1 = g_1(\Sigma_1)$. Assume that A_1 has a double point at q_1 , $p_a(A_1) \geq 1$, and g_1 has configuration I of Definition 9.6. Then $\text{char } k = 3$, $p_a(A_1) = 1$, and the weighted dual graph of the exceptional divisor of the minimal resolution of S_0 is one of*

$$[2, 3] + 2[3] + E_6, \quad [2, 3] + 2[3] + 3[2, 2].$$

In particular, $\# \text{Sing}(S_0)$ is respectively four or six.

This case already satisfies the required bound. The other ordinary double-point configurations give the following coefficient estimates.

Theorem 9.8 (Keel–McKernan–Lacini). *Let $g : T \rightarrow S$, W , Δ_W , D , Σ , and q satisfy all the hypotheses of Theorem 9.5, over an algebraically closed field of arbitrary characteristic. Assume that D has an ordinary node or an ordinary cusp at q , and that g is not in configuration I. Then the following hold:*

- (i) *If D has an ordinary cusp, W has one component in the local configuration, and that component has coefficient c in Δ_W , then*

$$c \in \left\{ \frac{2}{3}, \frac{3}{4}, \frac{5}{7}, \frac{7}{9}, \frac{11}{14}, \frac{10}{13}, \frac{15}{19} \right\}.$$

In particular, $2/3 \leq c < 4/5$.

- (ii) *If D has an ordinary node, the two local branches have the same coefficient a in Δ_W , and configuration 0 also does not occur, then $a \geq 2/3$.*

The cusp coefficients are the order-one entries of [Lac24, Lemma 3.21]. The node assertion uses the alternatives of [Lac24, Lemma 3.20] and the local coefficient comparison in [KM97, Lemmas 11.0.3, 11.1.1 and the proof of Lemma 17.2(6)]. Thus only the local contraction hypotheses in Theorem 9.5 enter this numerical assertion.

Finally, flushness at a surface singularity gives the reduced boundary structure needed for the next extraction. We use the one-boundary form of [Lac24, Lemma 3.7(1)], whose local proof is [KM97, Lemma 8.3.5(1)].

Theorem 9.9 (Keel–McKernan–Lacini). *Work over an algebraically closed field k of arbitrary characteristic. Let p be a singular point of a klt surface S , let $C \subset S$ be a reduced irreducible curve through p , and let $a \in \mathbb{Q}$ satisfy $0 < a < 1$. Assume that $K_S + aC$ is $\mathbb{Q}$ -Cartier near p . Let $\pi : \tilde{S} \rightarrow S$ be the minimal resolution over p . If*

$$e(F; S, aC) < a \quad \text{for every } \pi\text{-exceptional prime divisor } F \text{ over } p,$$

then (S, C) is dlt at p .

At such a point, Proposition 4.10 then identifies the exceptional chain and its attachment to the boundary.

Lemma 9.10. *Let S_0 be a rank-one klt del Pezzo surface over k , let $p : Y \rightarrow S_0$ be its minimal resolution, and put $D = \text{Exc}(p)_{\text{red}}$. Suppose its first hunt is birational and a second hunt step is defined. Then Y admits morphisms $\tau : Y \rightarrow T_1$ and $\mu : Y \rightarrow T_2$ satisfying*

$$p = f_1\tau, \quad f_2\mu = g_1\tau.$$

Every component of D maps under μ to a point, to the strict transform A_T of A_1 , or to E_2 . If the second hunt is birational and both boundaries survive, the map $g_2\mu$ factors through the minimal resolution $\pi : V \rightarrow S_2$. Under the resulting morphism $Y \rightarrow V$, every component of D maps to a point or to a component of $A_V + B_V + \text{Exc}(\pi)_{\text{red}}$.

These are morphisms from the original resolution, so fibrations on the later models pull back without further blowups of Y .

Proof. The first extraction is a partial resolution of S_0 , so it is dominated by Y and retains exactly the selected component of D . The morphism $g_1\tau : Y \rightarrow S_1$ factors through the minimal resolution $R \rightarrow S_1$ by Lemma 4.9. The second extraction is a partial resolution of S_1 , giving $R \rightarrow T_2$ and hence the morphism μ . The first assertion about components is Proposition 4.15.

In the birational case the same factorization lemma gives $Y \rightarrow V \rightarrow S_2$. A component whose image on S_2 is a curve must map to A_2 or B_2 , so its image on V is the corresponding strict transform. A component whose image on S_2 is a point maps on V either to a point or to an exceptional curve. $\square$

Lemma 9.11. *Let S_0 be a rank-one klt del Pezzo surface over k without a tiger on its minimal resolution. If its first hunt is birational, then $0 < a_1 < 1$.*

This bound does not require any assumption on the singularities of A_1 .

Proof. Let $p : Y \rightarrow S_0$ be the minimal resolution. The first extraction is dominated by Y , so $Y \rightarrow T_1 \rightarrow S_1$ is also a birational model of S_1 . Define e_0 by

$$K_{T_1} + e_0 E_1 = f_1^* K_{S_0}.$$

The coefficient e_0 is a canonical pullback coefficient on Y ; Theorem 3.7 gives $e_0 \geq 0$. By Definition 6.2 and Theorem 6.3, $a_1 = \lambda(e_0 + \varepsilon) > 0$.

The same theorem gives $\rho(S_1) = 1$, $-(K_{S_1} + a_1 A_1)$ ample, and no tiger for $(S_1, a_1 A_1)$ on this fixed model Y . The curve A_1 is $\mathbb{Q}$ -Cartier by Theorem 4.16. If $a_1 \geq 1$, rank one gives a rational number $t > 0$ with

$$K_{S_1} + a_1 A_1 + t A_1 \equiv 0.$$

The effective addition $t A_1$ to the weighted pair makes the coefficient of its strict transform on Y equal to $a_1 + t \geq 1$. This is a tiger on Y , a contradiction. $\square$

Lemma 9.12. *Let M be a normal projective $\mathbb{Q}$ -factorial surface over k , and let $C \subset M$ be an integral singular curve. Then*

$$(K_M + C) \cdot C \geq 0.$$

If a morphism $h : M \rightarrow Z$ to a smooth projective curve is a K_M -negative extremal contraction, such a curve C is horizontal.

The conductor of the singular curve supplies at least the degree needed to cancel the canonical degree of a rational normalization.

Proof. We use the conductor to bound the adjoint intersection and then exclude a vertical singular curve.

Step 1. In this step, we prove $(K_M + C) \cdot C \geq 0$ by adjunction on the normalization. Let $\nu : C^\nu \rightarrow C$ be the normalization, with conductor divisor J . At a singular point with at least two branches the conductor has positive order on every branch: order zero on one branch would put a nontrivial idempotent in the completed local ring. At a unibranch singular point its order is at least two. Indeed, order zero or one would put $k + tk[[t]] = k[[t]]$ in that local ring, making the point smooth. Thus $\deg J \geq 2$.

For a node the completed normalization is $k[[u]] \oplus k[[v]]$, and the conductor ideal is $(u) \oplus (v)$. For a cusp the inclusion is $k[[t^2, t^3]] \subset k[[t]]$, and its conductor is $t^2k[[t]]$. Thus the conductor is the sum of the two points over the node, or twice the point over the cusp. Its degree is two. In each case the normalization quotient has length one.

Theorem 5.1, namely Keel's adjunction formula [Kee99, Adjunction Formula 5.3], gives an effective divisor Θ on C^ν such that

$$(K_M + C) \cdot C = 2p_a(C^\nu) - 2 + \deg J + \deg \Theta \geq 0.$$

Step 2. In this step, we prove that C is horizontal for a K_M -negative extremal contraction to a curve. If C were vertical, then $C^2 \leq 0$. Indeed, intersecting C with the full fibre containing it gives zero, and all intersections with the other fibre components are nonnegative. On the contracted ray $K_M \cdot C < 0$, so $(K_M + C) \cdot C < 0$, a contradiction. $\square$

Lemma 9.13. *Let S_0 be a rank-one klt del Pezzo surface over k without a tiger on its minimal resolution. Suppose its first hunt is birational, $p_a(A_1) = 1$, and A_1 has an ordinary node or cusp at $q_1 \in S_1^{\text{reg}}$. Assume $a_1 \geq 2/3$ and that (S_1, A_1) is dlt near $A_1 \cap \text{Sing}(S_1)$. Then*

$$2/3 \leq a_1 < 1, \quad e_1 \geq 1/3.$$

Moreover A_1 is rational, q_1 is its only singular point, and $A_1 \cap \text{Sing}(S_1)$ is nonempty.

The lower bound concerns the actual maximal extraction, rather than a prescribed component next to the boundary.

Proof. We determine the geometry of A_1 and then compare the maximal extraction coefficient with a boundary-adjacent coefficient.

Step 1. In this step, we prove that A_1 is rational with one curve singularity and meets $\text{Sing}(S_1)$. The node or cusp has normalization quotient of length one. Since $p_a(A_1) = 1$, the normalization is $\mathbb{P}^1$ and there are no other singular points of A_1 .

Lemma 9.11 gives $a_1 < 1$, and Theorem 6.3 preserves the absence of a tiger for $(S_1, a_1 A_1)$ on Y . If $A_1 \subset S_1^{\text{reg}}$, then A_1 is Cartier and adjunction gives $(K_{S_1} + A_1) \cdot A_1 = 0$. Rank one implies $K_{S_1} + A_1 \equiv 0$. Adding $(1 - a_1)A_1$ to the weighted pair again gives a tiger. Therefore A_1 meets a surface singularity.

Step 2. In this step, we bound the actual second extraction coefficient by computing an end coefficient over p . Choose $p \in A_1 \cap \text{Sing}(S_1)$. By Proposition 4.10, applied near p , the exceptional divisor is a chain $[b_1, \dots, b_s]$, with $b_i \geq 2$, whose first component meets the strict transform of A_1 once. Let M be its negative intersection matrix, $r = \det M$, and γ_i its canonical pullback coefficients. These are nonnegative by Theorem 3.7. If η_i are the coefficients for the reduced pair, their intersection equations give

$$M\eta = (b_i - 2)_{i=1}^s + (1, 0, \dots, 0)^t, \quad M(\mathbf{1} - \eta) = (0, \dots, 0, 1)^t.$$

The corner cofactor of this tridiagonal matrix is one. Hence $\eta_1 = 1 - 1/r$, with $r \geq 2$, including when $s = 1$. The first component consequently has coefficient

$$e(P_1; S_1, a_1 A_1) = a_1 \left(1 - \frac{1}{r}\right) + (1 - a_1)\gamma_1 \geq \frac{a_1}{2} \geq \frac{1}{3}.$$

The coefficient e_1 of the actual second extraction is maximal among all exceptional components of the minimal resolution. It is at least this coefficient, whether or not P_1 is selected. $\square$

Lemma 9.14. *Let $\psi : Y \rightarrow \mathbb{P}^2$ be a birational morphism from a smooth projective surface, and suppose that ψ contracts a curve to $q \in \mathbb{P}^2$. Let A, B be distinct integral plane curves, and let D be a reduced divisor on Y whose components map to points, A , or B . Then the pencil of lines through q induces a $\mathbb{P}^1$ -fibration on Y with general fibre F satisfying*

$$F \cdot D \leq \deg A + \deg B - \text{mult}_q A - \text{mult}_q B + 1.$$

The point q need not belong to either boundary curve.

Proof. By Theorem 4.2, ψ is a sequence of point blowups. A curve is contracted over q , so the sequence contains the blowup of q . The earlier centres lie over other points and commute with this blowup. Thus ψ factors through $V = \text{Bl}_q \mathbb{P}^2$. Write L for the exceptional curve of $V \rightarrow \mathbb{P}^2$ and H for the pullback of a line. The pencil has fibre class $H - L$. On V , every component of D maps to a point or to a component of $A_V + B_V + L$. Further exceptional curves are vertical for the pulled-back pencil. Therefore

$$F \cdot D \leq (H - L) \cdot (A_V + B_V + L) = \deg A - \text{mult}_q A + \deg B - \text{mult}_q B + 1.$$

$\square$

Lemma 9.15. *Let S be a rank-one klt del Pezzo surface over k , and let A, B be distinct integral curves with $A \cdot B \geq 1$ and (S, B) dlt. Suppose that $-(K_S + cA + dB)$ is ample for rational numbers $c > 1/2$ and $d \geq 0$. Let $\pi : V \rightarrow S$ be the minimal resolution. Then $B_V + \text{Exc}(\pi)_{\text{red}}$ has no connected component consisting of four smooth rational curves R, L_1, L_2, L_3 , with*

$$R^2 = -1, \quad L_1^2 = -2, \quad L_2^2 = -3, \quad L_3^2 = -\ell, \quad \ell \in \{3, 4, 5\},$$

where R meets each of the other three curves transversely once and the three tips are pairwise disjoint.

Only one curve can be retained on S ; every choice is incompatible with either dlt incidence or the weighted adjoint inequality.

Proof. We locate the retained boundary component in the fork and then compare its intersections with the weighted adjoint divisor.

Step 1. In this step, we show that B_V must be the central curve of the fork. Put

$$\beta = -1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{\ell} = \begin{cases} 1/6 & \ell = 3, \\ 1/12 & \ell = 4, \\ 1/30 & \ell = 5. \end{cases}$$

The square of $R + L_1/2 + L_2/3 + L_3/\ell$ is $\beta > 0$. Thus the whole fork cannot be exceptional by Theorem 4.5. Since B_V is the unique nonexceptional component of the reduced divisor, it belongs to this fork.

If B_V is a tip, the other three fork components form an exceptional chain and B_V meets its middle component. This contradicts the end incidence in Proposition 4.10. Hence $B_V = R$.

Step 2. In this step, we compute the intersections with B and contradict the weighted anti-ameness. The pullback of B is

$$\pi^* B = R + \frac{1}{2} L_1 + \frac{1}{3} L_2 + \frac{1}{\ell} L_3.$$

It has no coefficients on the exceptional components disjoint from the fork, so $B^2 = \beta$. The canonical coefficients on the three tips are $0, 1/3, 1 - 2/\ell$. As $K_V \cdot R = -1$, adjunction gives

$$(K_S + B) \cdot B = -2 + \left(1 - \frac{1}{2}\right) + \left(1 - \frac{1}{3}\right) + \left(1 - \frac{1}{\ell}\right) = -\beta.$$

It follows that

$$(K_S + cA) \cdot B = cA \cdot B - 2\beta > 0.$$

On the other hand, anti-ampleness and $dB^2 \geq 0$ imply $(K_S + cA) \cdot B < 0$, a contradiction. The argument considers each possible position of B_V in the fork. $\square$

Lemma 9.16. *Let S_0 be a rank-one klt del Pezzo surface over k with minimal resolution $p : Y \rightarrow S_0$ and $D = \text{Exc}(p)_{\text{red}}$. Let S be another rank-one klt del Pezzo surface, with distinct integral curves A, B forming a fence, banana, or tacnode configuration. Suppose (S, B) is dlt and there are rational numbers a, b such that*

$$a > 2/3, \quad b \geq 0, \quad a + b > 1, \quad -(K_S + aA + bB) \text{ and } -(K_S + B) \text{ are ample.}$$

Suppose there is a birational morphism $\psi : Y \rightarrow S$ whose images of components of D are points, A , or B , and which contracts a curve to a point q . In the tacnode case assume that q is the tangency point. Then Y admits a $\mathbb{P}^1$ -fibration with general fibre F satisfying $F \cdot D \leq 4$, and $\#\text{Sing}(S_0) \leq 7$.

The degree bound is measured on the original resolution of S_0 .

Proof. We first use a pencil when S is the plane. Otherwise the open-pair theorem gives either a ruling of small degree or an exceptional fork; we lift the ruling and exclude the fork.

Step 1. In this step, we obtain the required exceptional-degree bound from a plane pencil. Suppose $S = \mathbb{P}^2$, and put $u = \deg A$ and $v = \deg B$. For a fence, Bézout gives $uv = 1$, hence $u + v = 2$. For a banana it gives $uv = 2$, hence $u + v = 3$. In either case Lemma 9.14 gives degree at most four for the pencil through q . No incidence condition at q is needed in these two cases.

In the tacnode case both curves pass through q . The inequality $-(K_S + B) > 0$ gives $v \leq 2$, while the weighted anti-ampleness gives $au + bv < 3$. Since $b \geq 0$, $v \geq 1$, and $b > 1 - a$, we obtain $au < 2 + a$. Thus $u < 1 + 2/a < 4$ and $u \leq 3$. The pencil estimate is now

$$F \cdot D \leq u - 1 + v - 1 + 1 \leq 4.$$

Step 2. In this step, we lift a ruling of the nonplane open pair and bound its degree on the original exceptional divisor. Suppose $S \not\cong \mathbb{P}^2$, and let $\pi : V \rightarrow S$ be its minimal surface resolution. By Proposition 4.10, B is smooth and B_V meets the exceptional chains only at their ends. The exceptional divisor is an snc forest by Theorem 3.7. Consequently π is also the minimal log resolution of (S, B) . Put $E = \text{Exc}(\pi)_{\text{red}}$. By Lemma 4.9, ψ factors through a birational morphism $\rho : Y \rightarrow V$.

If $\text{ht}(S, B) \leq 2$, choose a fibration on V with general fibre G such that $G \cdot (B_V + E) \leq 2$. On this minimal resolution write

$$\pi^*(K_S + aA + bB) = K_V + aA_V + bB_V + \sum_i c_i P_i,$$

where the P_i are the exceptional components. The canonical coefficients are nonnegative by Theorem 3.7. The coefficients of the pullbacks of A and B are nonnegative as well. To see this, put $Q_{ij} = -P_i \cdot P_j$. For either curve the exceptional coefficient vector z solves $Qz = v$ with $v \geq 0$. If $z = z_+ - z_-$ has a nonzero negative part, negative definiteness of the exceptional intersection matrix and $Q_{ij} \leq 0$ for $i \neq j$ give

$$0 \leq z_-^t v = z_-^t Q z_+ - z_-^t Q z_- < 0,$$

a contradiction. Hence all $c_i \geq 0$. Intersecting with G gives

$$-2 + aG \cdot A_V + bG \cdot B_V + \sum_i c_i G \cdot P_i < 0.$$

In particular $G \cdot A_V < 2/a < 3$, so $G \cdot A_V \leq 2$. Every component of D maps on V to a point or to $A_V + B_V + E$. Curves contracted by ρ are vertical for the pulled-back fibration, and the remaining components map birationally to distinct curves. It follows that

$$F \cdot D \leq G \cdot (A_V + B_V + E) \leq 4.$$

Step 3. In this step, we exclude the exceptional open-pair alternative using the weighted boundary. Theorem 5.2 describes the other alternative on the same resolved pair $(V, B_V + E)$. If $\text{ht}(S, B) > 2$, Proposition 5.9 gives one of the forks in Lemma 9.15. The intersections in a fence, banana, or tacnode configuration lie in S^{reg} ; they give $A \cdot B \geq 1$. The lemma, with $c = a$ and $d = b$, excludes each of them. Its argument does not identify B_V with the strict transform of the cuspidal curve on the other surface in the open-pair theorem.

Thus a fibration of degree at most four exists in every case, and Corollary 3.17 gives the assertion. $\square$

Lemma 9.17. *Let S_0 be a rank-one klt del Pezzo surface over k without a tiger on its minimal resolution. Suppose its first hunt is birational, $p_a(A_1) = 1$, and A_1 has an ordinary node or cusp at $q_1 \in S_1^{\text{reg}}$. Assume $a_1 \geq 2/3$ and (S_1, A_1) dlt near $A_1 \cap \text{Sing}(S_1)$. Then $\#\text{Sing}(S_0) \leq 7$.*

These local hypotheses suffice for every possible second hunt output.

Proof. Lemma 9.13 gives $2/3 \leq a_1 < 1$ and $e_1 \geq 1/3$. We exclude contraction of the old curve, bound the degree of a second net, and treat the two surviving boundaries by the preceding lemma.

Step 1. In this step, we exclude contraction of the old boundary A_T . If g_2 contracted A_T , alternative (ii) of Theorem 6.6 would make (S_1, A_1) dlt everywhere. Theorem 4.6 would make A_1 smooth at q_1 , a contradiction.

Step 2. In this step, we bound the original exceptional degree by four when the second output is a net. The extraction f_2 is centred at a singular point of S_1 , so it is an isomorphism near q_1 . Thus A_T has the same double point at a smooth point of T_2 . Suppose g_2 is a net with general fibre F_T . By Lemma 9.12, A_T is horizontal. Set

$$d_A = F_T \cdot A_T \geq 1, \quad d_E = F_T \cdot E_2 \geq 0.$$

The unscaled pullback is

$$K_{T_2} + a_1 A_T + e_1 E_2 = f_2^*(K_{S_1} + a_1 A_1).$$

Intersecting with a general fibre and using anti-ampleness on S_1 gives $-2 + a_1 d_A + e_1 d_E < 0$. Therefore

$$2d_A + d_E < 6, \quad 2d_A + d_E \leq 5, \quad d_A + d_E \leq 4.$$

Lemma 9.10 gives a ruling on Y for which $F \cdot D \leq d_A + d_E$. The conclusion follows from Corollary 3.17.

Step 3. In this step, we prove that $-(K_{S_2} + B_2)$ is ample and that (S_2, B_2) is dlt when both boundaries survive. It remains that g_2 is birational and both boundaries survive. By Definition 6.2 and Theorem 6.3,

$$a_2 = \lambda a_1 > 2/3, \quad b_2 = \lambda(e_1 + \varepsilon) > 1/3, \quad a_2 + b_2 > 1,$$

and $-(K_{S_2} + a_2 A_2 + b_2 B_2)$ is ample. Theorem 6.6 now applies to the two-component endpoint. In particular, Σ_2 avoids the old double point. The map to S_2 is an isomorphism near it. Denote its image by r_2 ; then

$$r_2 \in S_2^{\text{reg}}, \quad r_2 \neq q_2,$$

and A_2 still has the old node or cusp at r_2 . Lemma 9.12 gives $(K_{S_2} + A_2) \cdot A_2 \geq 0$. Since the alternatives theorem says at least one of $-(K_{S_2} + A_2)$ and $-(K_{S_2} + B_2)$ is ample, it is the latter.

In the banana and tacnode cases (S_2, B_2) is plt by that theorem. For a fence, $(S_2, A_2 + B_2)$ is dlt at the surface singularities, so (S_2, B_2) is dlt there and B_2 is smooth there. If B_2 had a singular point in S_2^{reg} , Lemma 9.12 would contradict the anti-amenability of $K_{S_2} + B_2$. Thus B_2 is smooth, and the pair is dlt also over S_2^{reg} . In every case (S_2, B_2) is dlt.

Step 4. In this step, we apply the two-boundary ruling bound to the original minimal resolution. By Lemma 9.10, the morphism $\psi = g_2\mu : Y \rightarrow S_2$ sends the components of D to points or to A_2, B_2 . The strict transform of Σ_2 is contracted to q_2 . In the tacnode case the alternatives theorem identifies q_2 as the tangency point. All the hypotheses of Lemma 9.16 are now satisfied, which finishes the proof. $\square$

Theorem 9.18. *Let S_0 be a rank-one klt del Pezzo surface over k without a tiger on its minimal resolution. Suppose the first hunt step is birational and A_1 has arithmetic genus one, with an ordinary node or ordinary cusp at q_1 . Then $\#\text{Sing}(S_0) \leq 7$.*

The local classification treats the one-blowup case directly. In every other case it supplies the hypotheses of the second-hunt argument above.

Proof. We first check the local contraction hypotheses, then separate configuration I and derive the coefficient and dlt conditions for Lemma 9.17.

Step 1. In this step, we verify the local flush contraction hypotheses and prove that q_1 is smooth. Lemma 9.11 gives $0 < a_1 < 1$ independently of the local shape of A_1 . Theorem 6.6 gives (T_1, E_1) dlt and $(T_1, a_1 E_1)$ flush. By Theorem 4.6, E_1 is a smooth curve. The opposite contraction g_1 is K_{T_1} -negative by Theorem 6.3, and its exceptional curve Σ_1 is distinct from E_1 . Thus the proper restriction $g_1|_{E_1} : E_1 \rightarrow A_1$ is nonconstant and finite. It is birational, so the smooth curve E_1 is the normalization of A_1 . The first scaled boundary is $a_1 E_1$, so $K_{T_1} + a_1 E_1$ is $\mathbb{Q}$ -Cartier and g_1 -numerically trivial by Definition 6.2. In the *étale local configuration*, the preimages of the branches of the ordinary double point are disjoint smooth components, at most two, each with coefficient a_1 . In particular they satisfy the transversality condition. All hypotheses of Theorem 9.5 therefore hold with $W = E_1$ and $\Delta_W = a_1 E_1$. It follows that $q_1 \in S_1^{\text{reg}}$.

Step 2. In this step, we treat configuration I and exclude configuration 0 for an ordinary node. Use the configurations of Definition 9.6 for g_1 . If configuration I occurs, Theorem 9.7 gives $\#\text{Sing}(S_0) = 4$ or 6, and proves the assertion. For an ordinary node, configuration 0 would leave the two branches meeting on the source boundary, contrary to smoothness of E_1 . We may hence exclude it as well.

Step 3. In this step, we obtain the coefficient and dlt hypotheses needed for the genus-one reduction. Theorem 9.8 now applies. For a cusp it gives the displayed seven-element list; for a node the two local coefficients are both a_1 . In either case $a_1 \geq 2/3$.

The endpoint $(S_1, a_1 A_1)$ is flush by Theorem 6.6. For each $z \in A_1 \cap \text{Sing}(S_1)$ and every exceptional prime P of the minimal resolution over z , this means

$$e(P; S_1, a_1 A_1) < a_1.$$

Since $0 < a_1 < 1$ and S_1 is klt and $\mathbb{Q}$ -factorial, Theorem 9.9 gives (S_1, A_1) dlt near z . We have obtained all three additional hypotheses of Lemma 9.17: the old double point lies in S_1^{reg} , the coefficient is at least $2/3$, and the reduced boundary is dlt along the surface singularities. That lemma handles every second output and gives $\#\text{Sing}(S_0) \leq 7$. $\square$

10. SMOOTH FIRST BOUNDARIES AND SECOND FIBRATIONS

We use the hunt of Definition 6.2, including its convention for equal maximal coefficients. We treat the case where the first boundary is smooth rational and the second hunt produces a

fibration. The ruling lifts to the original minimal resolution, where its general fibre meets the reduced exceptional divisor in degree at most three.

We use the following coefficient estimates from [Lac24, Lemma 4.33]. Their validity in arbitrary characteristic is established in [Nag25a, Remark 3.16(1)].

Theorem 10.1 (Lacini–Nagaoka). *Let S_0 be a rank-one klt del Pezzo surface over k without a tiger on its minimal resolution. Suppose its first hunt step is birational and A_1 is smooth. For the coefficients in Definition 6.2, one has $a_1 > 2/3$ and $e_1 > 1/2$.*

Here e_1 is the coefficient in the unscaled second pullback, as in Definition 6.2. These strict bounds turn an inequality for a weighted boundary into a bound for its reduced support. We isolate that numerical argument.

Lemma 10.2. *Let $h : T \rightarrow \Lambda$ be a $\mathbb{P}^1$ -fibration from a normal projective surface to a smooth projective curve over an algebraically closed field of arbitrary characteristic. Let C_1, C_2 be distinct reduced irreducible curves on T , and put $d_i = F \cdot C_i$ for a general fibre F and $i = 1, 2$. Let $a, e \in \mathbb{Q}$ satisfy*

$$a > \frac{2}{3}, \quad e > \frac{1}{2}, \quad ad_1 + ed_2 < 2.$$

Let $\phi : V \rightarrow T$ be a projective birational morphism with V smooth, and let D_V be a reduced effective divisor on V . Assume every component of D_V that is horizontal for $h\phi$ is the strict transform of C_1 or C_2 . Then $h\phi : V \rightarrow \Lambda$ is a $\mathbb{P}^1$ -fibration whose general fibre G satisfies $G \cdot D_V \leq 3$.

Only horizontal components contribute to the intersection with a general fibre. The lemma therefore applies even when D_V contains additional curves contracted by ϕ .

Proof. We bound the sum of the two fiber degrees and then transfer that bound to D_V .

Step 1. In this step, we prove $d_1 + d_2 \leq 3$ from the coefficient inequalities. The integers d_1, d_2 are nonnegative. For a horizontal curve, the intersection with F is the degree of the map from its normalization to Λ ; for a vertical curve, it is zero. The coefficient inequalities give

$$\frac{2}{3}d_1 + \frac{1}{2}d_2 \leq ad_1 + ed_2 < 2, \quad 3(d_1 + d_2) \leq 4d_1 + 3d_2 < 12.$$

It follows that $d_1 + d_2 \leq 3$.

Step 2. In this step, we verify that the composite is a $\mathbb{P}^1$ -fibration and bound its intersection with D_V . The morphism ϕ is an isomorphism away from finitely many points of the normal surface T . Choose F to avoid these points and the images of the vertical components of D_V . Its inverse image G is then isomorphic to F . By Theorem 4.8, applied to ϕ ,

$$(h\phi)_*\mathcal{O}_V = h_*\mathcal{O}_T = \mathcal{O}_\Lambda,$$

so $h\phi$ has connected fibres and is a $\mathbb{P}^1$ -fibration. Every vertical component of D_V is disjoint from G . Since D_V is reduced, each of the two possible horizontal strict transforms occurs at most once. Thus

$$G \cdot D_V \leq F \cdot C_1 + F \cdot C_2 = d_1 + d_2 \leq 3.$$

□

Theorem 10.3. *Let S_0 be a rank-one klt del Pezzo surface over k without a tiger on its minimal resolution. Suppose the first hunt step is birational, A_1 is smooth rational with (S_1, A_1) dlt, and the second output is a net. Then $\#\text{Sing}(S_0) \leq 7$.*

The second fibration controls the singularities of S_0 through the exceptional divisor on its own minimal resolution.

Proof. We construct the comparison with the second extraction, use the anti-ample pullback to bound the fibre degrees, and transfer that bound to the original exceptional divisor.

Step 1. In this step, we construct the comparison morphisms from the original minimal resolution to the second extraction. Let $p : Y \rightarrow S_0$ be the minimal resolution and put $D = \text{Exc}(p)_{\text{red}}$. The first extraction retains one exceptional curve of p , say E_Y . By the extraction construction in Definition 6.2, there is a projective birational morphism $\tau : Y \rightarrow T_1$ with $p = f_1\tau$. It maps E_Y birationally onto E_1 and contracts every other component of D . The opposite contraction maps E_1 birationally onto A_1 . By Theorem 6.3, S_1 is a rank-one log del Pezzo surface and $-(K_{S_1} + a_1A_1)$ is ample.

Let $\theta : R_1 \rightarrow S_1$ be the minimal resolution. The composite $\eta = g_1\tau : Y \rightarrow S_1$ is a resolution, so it factors through θ . More explicitly, contract η -exceptional (-1) -curves successively. Theorem 4.3 permits each contraction over S_1 , leaving a smooth projective resolution. Each contraction lowers the Picard number, so the process ends at the unique minimal resolution. It gives a projective birational morphism $\alpha : Y \rightarrow R_1$ with $\eta = \theta\alpha$.

The second hunt extraction retains one exceptional divisor of θ . Hence there is a projective birational morphism $\rho_2 : R_1 \rightarrow T_2$ with $\theta = f_2\rho_2$. Define

$$\mu = \rho_2\alpha : Y \rightarrow T_2.$$

The comparison is expressed by the commutative diagram

$$\begin{array}{ccccc} Y & \xrightarrow{\alpha} & R_1 & \xrightarrow{\rho_2} & T_2 \\ \tau \downarrow & & \theta \downarrow & & \downarrow f_2 \\ T_1 & \xrightarrow{g_1} & S_1 & = & S_1. \end{array}$$

In particular, $p = f_1\tau$ and $f_2\mu = g_1\tau$.

Step 2. In this step, we derive the strict weighted degree inequality on a general fiber of the second net. Apply Theorem 6.3 to the second step. Its nonbirational assertion is [Nag25a, Lemma 2.7(1),(6)], which applies in characteristic three. The second output is therefore a $\mathbb{P}^1$ -fibration $h : T_2 \rightarrow \Lambda$ over a smooth projective curve. Let A_T be the strict transform of A_1 on T_2 . Since f_2 extracts just E_2 over a point, it is an isomorphism at the generic point of A_1 . The unscaled second log pullback is

$$K_{T_2} + a_1A_T + e_1E_2 = f_2^*(K_{S_1} + a_1A_1).$$

Choose a general fibre F of h inside T_2^{reg} . Its class lies on the ray contracted by h , whereas E_2 spans the other ray, contracted by f_2 . Thus F is not contracted by f_2 , and $(f_2)_*F$ is a nonzero effective curve. Put $d_A = F \cdot A_T$ and $d_E = F \cdot E_2$. Since $F^2 = 0$ and $F \simeq \mathbb{P}^1$, adjunction gives $K_{T_2} \cdot F = -2$. The projection formula now yields

$$\begin{aligned} -2 + a_1d_A + e_1d_E &= (K_{T_2} + a_1A_T + e_1E_2) \cdot F \\ &= (K_{S_1} + a_1A_1) \cdot (f_2)_*F < 0. \end{aligned}$$

The last inequality follows from the ampleness of $-(K_{S_1} + a_1A_1)$ and the nonvanishing of $(f_2)_*F$.

Step 3. In this step, we bound the reduced exceptional degree on the original resolution by three. The comparison morphisms satisfy all the hypotheses of Proposition 4.15. Every component of D therefore maps under μ to a point, A_T , or E_2 . Consequently, the only possible horizontal components of D for $h\mu$ are the strict transforms of A_T and E_2 . Theorem 10.1 gives $a_1 > 2/3$ and $e_1 > 1/2$. Applying Lemma 10.2 with $\phi = \mu$, $C_1 = A_T$, $C_2 = E_2$, and $D_V = D$, we obtain a $\mathbb{P}^1$ -fibration $h\mu : Y \rightarrow \Lambda$ whose general fibre G satisfies

$$G \cdot D \leq d_A + d_E \leq 3.$$

Corollary 3.17 applies to this fibration on the minimal resolution of S_0 and gives $\# \text{Sing}(S_0) \leq 7$. $\square$

11. CONTRACTING THE SMOOTH FIRST BOUNDARY

We retain the notation of the two hunt steps in Definition 6.2. We consider the case in which the second opposite contraction removes the smooth first boundary. We first identify its entire fibre on the minimal resolution. The new boundary is then a rational cuspidal curve of arithmetic genus one. Retaining the exceptional end which meets this curve leads to an adjoint contraction and a count on a canonical del Pezzo surface.

Theorem 11.1. *Let S_0 be a rank-one klt del Pezzo surface over k without a tiger on its minimal resolution. Suppose its first two hunt steps are birational, A_1 is smooth rational with (S_1, A_1) dlt, and g_2 contracts $(A_1)_{T_2}$. Then $\# \text{Sing}(S_0) \leq 7$.*

The first boundary will contain three surface singularities. Two of them are absorbed by the second opposite contraction; the remaining one determines the singular point on the new boundary.

Lemma 11.2. *Let S be a projective klt surface, let C be a reduced irreducible curve, and assume (S, C) is dlt along C . Let $p \in C \cap \text{Sing}(S)$, and order the exceptional chain of the minimal resolution $\pi : Y \rightarrow S$ over p as $F_1, \dots, F_n$, starting at the end met by C_Y . Fix $0 < a \leq 1$ and write, near this fibre,*

$$\pi^*(K_S + aC) = K_Y + aC_Y + \sum_{i=1}^n c_i F_i.$$

If $c_1 < a$, then $c_1 > c_2 > \dots > c_n$. An extraction from this chain which retains a component of maximal coefficient therefore retains F_1 . If its target is klt and $\mathbb{Q}$ -factorial, the image of F_1 is isomorphic to $\mathbb{P}^1$, and its intersection with the strict transform of C is a smooth transverse point of the target.

This identifies both the extracted divisor and the position of the remaining singularity on it.

Proof. We use the chain recurrence to locate the maximal coefficient and then check smoothness of the retained curve.

Step 1. In this step, we prove that the coefficients strictly decrease from the boundary-adjacent end. Put $b_i = -F_i^2 \geq 2$, $c_0 = a$, $c_{n+1} = 0$, and $u_i = 1 - c_i$. Adjunction and orthogonality of a pullback to the exceptional curves give

$$b_i c_i = b_i - 2 + c_{i-1} + c_{i+1}, \quad b_i u_i = u_{i-1} + u_{i+1}.$$

Here $u_0 \geq 0$ and $u_1 - u_0 > 0$. The identity

$$u_{i+1} - u_i = (u_i - u_{i-1}) + (b_i - 2)u_i$$

shows inductively that $u_0 < u_1 < \dots < u_n$. Thus F_1 is the unique component of maximal coefficient.

Step 2. In this step, we prove that the retained curve is smooth and meets the boundary transversely in the smooth locus. The induced morphism from Y to the extraction contracts precisely the tail $F_2 + \dots + F_n$ near this fibre. It is an isomorphism at $C_Y \cap F_1$, since this point is disjoint from the tail. It remains to check regularity of the retained curve at the tail image. For a nonempty tail $H_1, \dots, H_l$, put $\beta_j = -H_j^2 \geq 2$ and write the pullback of the canonical divisor plus the retained curve with coefficients d_j on the H_j . For $v_j = 1 - d_j$, the equations are

$$\beta_j v_j = v_{j-1} + v_{j+1}, \quad v_0 = 0, \quad v_{l+1} = 1.$$

The numbers $s_1 = \beta_1$ and $s_j = \beta_j - 1/s_{j-1}$ satisfy $s_j > 1$. Elimination gives $v_{j+1} = s_j v_j$ and $s_l v_l = 1$, so every $v_j > 0$. Consequently every exceptional coefficient d_j is less than one. The resolution has simple normal crossing support, so the retained pair is plt. Theorem 4.6 makes its boundary smooth. The proper birational map from $F_1 \simeq \mathbb{P}^1$ to this smooth curve is an isomorphism. $\square$

Lemma 11.3. *Let S_0 be a rank-one klt del Pezzo surface without a tiger on its minimal resolution. Suppose its first two hunt steps are birational, A_1 is smooth rational, (S_1, A_1) is dlt, and $\Sigma_2 = (A_1)_{T_2}$. Let $\pi : Y \rightarrow S_1$ be the minimal resolution, let $\rho : Y \rightarrow T_2$ be the induced morphism, and write $\Delta_2 = b_2 B_2$. Then the following hold.*

- (i) A_1 contains exactly three singular points of S_1 , one of which is x_2 . The extraction retains the boundary-adjacent end of the chain over x_2 .
- (ii) The full reduced divisorial fibre of $g_2 \rho$ over q_2 is a chain with self-intersections $-2, -1, -3$. Its middle curve is A_Y , and the strict transform of B_2 meets that middle curve transversely away from the two arms. Moreover $4/5 < b_2 \leq 5/6$.
- (iii) The normalization of B_2 is $\mathbb{P}^1$, its only curve singularity is an ordinary cusp at $q_2 \in S_2^{\text{reg}}$, and $p_a(B_2) = 1$.
- (iv) $B_2 \cap \text{Sing}(S_2) = \{y\}$ for one point $y \neq q_2$. On the minimal resolution $r : X \rightarrow S_2$, the fibre over y is a nonempty chain met transversely at one end by B_X . The curve B_X misses all other exceptional curves, and $r|_{B_X} : B_X \rightarrow B_2$ is an isomorphism.

The statement concerns the full minimal resolutions. Exceptional fibres away from the boundaries remain present throughout the construction.

Proof. We locate the second extraction, determine the complete contraction fibre, and then follow the new boundary through its contraction. Theorem 6.3 gives rank-one klt del Pezzo endpoints and the anti-ampleness of $K_{S_i} + \Delta_i$. These endpoints are $\mathbb{Q}$ -factorial by Theorem 4.16. Theorem 6.6 gives $0 < a_1 \leq 1$, flushness of $(S_1, a_1 A_1)$, and the smooth point q_2 at which B_2 is singular and unibranch.

The minimal resolution of S_0 maps to T_1 and hence to S_1 . By Lemma 4.9, it factors through the minimal resolution Y of S_1 , which in turn maps to T_2 . Applying Theorem 6.3 twice shows that the induced pairs on S_1 and S_2 have no tiger on this model. In particular, neither strict boundary transform can supply a tiger.

Step 1. In this step, we identify the second extraction center and prove that A_1 contains exactly three surface singularities. Put $P = A_1 \cap \text{Sing}(S_1)$. The curve A_1 is $\mathbb{Q}$ -Cartier and $A_1^2 > 0$, since it is a nonzero effective curve on a rank-one $\mathbb{Q}$ -factorial projective surface. We claim that $(K_{S_1} + A_1) \cdot A_1 > 0$. Otherwise

$$t = -\frac{K_{S_1} \cdot A_1}{A_1^2} \geq 1, \quad K_{S_1} + tA_1 \equiv 0.$$

The effective divisor $(t - a_1)A_1$ would give a tiger on the strict transform of A_1 . This contradicts the preceding no-tiger assertion. Proposition 4.10 now gives

$$(11.1) \quad 0 < (K_{S_1} + A_1) \cdot A_1 = -2 + \sum_{p \in P} \left(1 - \frac{1}{d_p}\right).$$

Each summand is less than one, so $\#P \geq 3$.

For each $p \in P$, write the exceptional chain as $F_{p,1}, \dots, F_{p,n_p}$, starting at the end met by A_Y . Since $q_2 \in B_2$, the curves E_2 and Σ_2 meet. Their intersection maps to $x_2 \in A_1$. An extraction from the minimal resolution has a singular centre, so $x_2 \in P$. Flushness and Lemma 11.2 show that ρ retains $F_{x_2,1}$ and contracts the tail. In particular $E_2 \cap \Sigma_2$ is one smooth transverse point.

Every point of $P \setminus \{x_2\}$ persists as a singular point of T_2 on Σ_2 . The opposite ray is K_{T_2} -negative, so Proposition 4.12 bounds their number by two. It follows that

$$P = \{x_2, p_L, p_R\}, \quad \#(\text{Sing}(T_2) \cap \Sigma_2) = 2.$$

Step 2. In this step, we determine the complete fiber over q_2 and derive its determinant identity. The fibres over p_L and p_R are the only ρ -exceptional fibres whose images lie on Σ_2 . Indeed, every exceptional fibre over a point outside P maps to a point of S_1 outside A_1 . The tail over x_2 meets the retained end away from $A_Y \cap F_{x_2,1}$; its image is therefore a different point of E_2 . The local

isomorphism at $A_Y \cap F_{x_2,1}$ also rules out an identification of these two points. Thus the full fibre over q_2 is

$$(11.2) \quad \sum_{i=1}^{n_{p_L}} F_{p_L,i} + A_Y + \sum_{j=1}^{n_{p_R}} F_{p_R,j}.$$

All other exceptional fibres are disjoint from this divisor.

Every ρ -exceptional curve is exceptional on the minimal resolution of S_1 and has square at most -2 , by Theorem 3.7 and minimality. Consequently $Y \rightarrow T_2$ is minimal. Over the smooth point q_2 , the morphism $g_2\rho$ factors into point blowups by Theorem 4.2. Its exceptional fibre contains a (-1) -curve. In (11.2), only A_Y can be such a curve. Hence $A_Y^2 = -1$.

Write the two arms as $L_1, \dots, L_m$ and $R_1, \dots, R_n$, ordered from A_Y . Let d_L, d_R be their determinants and let e_L, e_R be their determinants after deleting the first component, with value one for an empty tail. Then $1 \leq e_L < d_L$ and $1 \leq e_R < d_R$. The complete fibre of a sequence of point blowups has determinant one: in the basis of total transforms its negative intersection matrix is the identity, and the change to strict transforms is integral unimodular. Expanding the determinant at A_Y therefore gives

$$(11.3) \quad d_L d_R - e_L d_R - e_R d_L = 1.$$

Step 3. In this step, we determine the two arms and bound the second boundary coefficient. Let α_2 be the coefficient of A_Y in the pullback of $K_{S_2} + b_2 B_2$, and let u_i, v_j be the coefficients on the arms. By Theorem 6.6, this endpoint is not level and Σ_2 is its only exceptional divisor where flushness fails. The scaled hunt divisor is a boundary. Thus

$$0 < b_2 < \alpha_2 \leq 1, \quad u_i < b_2, \quad v_j < b_2.$$

Put $\zeta = 1 - \alpha_2 \geq 0$. For a chain of determinant d and tail determinant e , the first diagonal and corner entries of the inverse of its positive intersection matrix are e/d and $1/d$. Solving the arm equations consequently gives

$$u_1 = 1 - \frac{1 + e_L \zeta}{d_L}, \quad v_1 = 1 - \frac{1 + e_R \zeta}{d_R}.$$

For completeness, the coefficient vector of an arm with weights a_i satisfies $B\mathbf{u} = (a_i - 2)_i + \alpha_2 \mathbf{e}_1$. Since $B\mathbf{1} = (a_i - 2)_i + \mathbf{e}_1 + \mathbf{e}_m$, the displayed formula follows by Cramer's rule, also when $m = 1$. Intersection with A_Y gives $-1 + b_2 - \alpha_2 + u_1 + v_1 = 0$. Using (11.3), we obtain

$$(11.4) \quad b_2 = \frac{d_L + d_R - \zeta}{d_L d_R}.$$

Interchange the arms so that $d_L \leq d_R$. The inequalities $b_2 < \alpha_2$ and $v_1 < b_2$ imply

$$(11.5) \quad \zeta(d_L d_R - 1) < d_L d_R - d_L - d_R, \quad d_L d_R - 2d_L - d_R < \zeta(e_R d_L - 1).$$

If $d_L = 2$, then $e_L = 1$ and (11.3) gives $d_R = 2e_R + 1$. For $d_R \geq 5$, (11.5) would give

$$\frac{d_R - 4}{d_R - 2} < \zeta < \frac{d_R - 2}{2d_R - 1}.$$

The left endpoint is at least the right endpoint, since cross multiplication leaves $d_R(d_R - 5) \geq 0$. Thus $d_R = 3$ in this case.

Suppose instead that $d_L \geq 3$. Equation (11.3) excludes $d_L = d_R$, so $d_R > d_L$. Both denominators below are positive, and

$$\frac{d_L d_R - 2d_L - d_R}{e_R d_L - 1} \geq \frac{d_L d_R - d_L - d_R}{d_L d_R - 1}.$$

Here is the integer calculation. Put $N = d_L d_R - d_L - d_R$ and write $e_L d_R = \ell d_L - 1$. Then $\ell \geq e_L + 1$, and the difference of the cross products is

$$d_R(e_L N - (d_L^2 - d_L + 1)) = d_L d_R(\ell(d_L - 1) - e_L - d_L).$$

The last factor is at least $d_L - 3$ when $e_L = 1$, and at least $e_L(d_L - 2) - 1$ when $e_L \geq 2$. It is therefore nonnegative. This contradicts the two strict bounds for ζ in (11.5). We conclude that $(d_L, d_R) = (2, 3)$. Equation (11.3) then gives $e_L = e_R = 1$. A nonempty chain of weights at least two has determinant at least two, so each arm has only one component. They have weights two and three. Finally

$$b_2 = \frac{4 + \alpha_2}{6}, \quad \frac{4}{5} < b_2 \leq \frac{5}{6}.$$

Step 4. In this step, we compute the cusp of B_2 and identify its normalization and arithmetic genus. On Y the strict transform of B_2 is $F_{x_2,1}$. It meets A_Y transversely at a point of A_Y distinct from the two arm attachments. Using Theorems 4.3 and 4.4, contract successively A_Y , the image of the (-2) -arm, and the image of the (-3) -arm over S_2 . After the first contraction, the two arms and the boundary have three distinct tangent directions at one point. After the second, the last exceptional curve has square -1 and the boundary is smooth with contact order two with it. Choose completed coordinates u, v at that contact point with the last exceptional curve given by $u = 0$. Its blowdown has coordinates $x = u, z = uv$. On the boundary, u has order two and v has order one. Since $\text{char}(k) = 3 \neq 2$, a change of parameter makes $x = t^2$. Subtracting a power series in x from z removes its even powers of t ; the result is t^3 times a unit in $k[[x]]$. Dividing by this unit gives

$$\hat{\mathcal{O}}_{B_2, q_2} \simeq k[[t^2, t^3]].$$

These three contractions factor $g_2\rho$ over a neighbourhood of q_2 , since they remove its complete fibre.

Lemma 11.2 gives $E_2 \simeq \mathbb{P}^1$. The morphism $E_2 \rightarrow B_2$ is proper and birational, and is an isomorphism away from q_2 , because g_2 is an isomorphism off Σ_2 . It is the normalization. Thus B_2 is regular away from its ordinary cusp. The normalization quotient $k[[t]]/k[[t^2, t^3]]$ has dimension one, and hence $p_a(B_2) = 1$.

Step 5. In this step, we identify the unique surface singularity on B_2 and its incidence on the global minimal resolution. Only the tail over x_2 can give a singular point of T_2 on E_2 . Indeed, the other ρ -exceptional fibres map under f_2 to points different from x_2 , whereas every point of E_2 maps to x_2 . Since q_2 is smooth, this proves $\#(B_2 \cap \text{Sing}(S_2)) \leq 1$. If this intersection were empty, B_2 would be Cartier and adjunction would give $(K_{S_2} + B_2) \cdot B_2 = 0$. Picard number one would give $K_{S_2} + B_2 \equiv 0$. Adding $(1 - b_2)B_2$ would then give a tiger, contrary to no-tiger preservation. Thus there is exactly one such point y , and the tail over x_2 is nonempty.

Globally, the three smooth contractions just described give a smooth surface X and a morphism $r : X \rightarrow S_2$. All other exceptional fibres are unchanged: they are disjoint from (11.2). Their components still have square at most -2 , so r is the minimal resolution. The fibre over y is precisely the nonempty x_2 -tail, and B_X meets its first component transversely once. It misses all other exceptional curves. The tail coefficient calculation in Lemma 11.2 proves that (S_2, B_2) is plt near y ; in particular B_2 is regular there. The restriction $B_X \rightarrow B_2$ is an isomorphism near y , since it is a proper birational map of smooth curves there, and away from y the surface resolution is already an isomorphism along B_2 . $\square$

Lemma 11.4. *Let S be a rank-one $\mathbb{Q}$ -factorial klt del Pezzo surface. Let A be an integral $\mathbb{Q}$ -Cartier curve with $p_a(A) = 1$ and $A \cap \text{Sing}(S) = \{y\}$. Let $r : X \rightarrow S$ be the minimal resolution, and assume $r|_{A_X} : A_X \rightarrow A$ is an isomorphism. Suppose A_X meets exactly one exceptional prime E , transversely once, and the normalization of E is $\mathbb{P}^1$. Suppose that $r = h\tau$ with $\tau : X \rightarrow Z$ and $h : Z \rightarrow S$ projective birational, where Z is $\mathbb{Q}$ -factorial and klt, τ contracts every r -exceptional prime except E , τ is an isomorphism near A_X , and h is an isomorphism off y . Put $A_Z = \tau(A_X)$, $G = \tau(E)$, and $K_Z = \tau_*K_X$. Then the following hold.*

- (i) $\text{Exc}(h) = G$, $\rho(Z) = 2$, $A_Z \subset Z^{\text{reg}}$, and $A_Z \cdot G = 1$.

- (ii) The system $|K_X + A_X|$ has a unique effective member D_X . Both D_X and $D_Z = \tau_* D_X$ are nonzero, and their supports are disjoint from A_X and A_Z , respectively. There are a positive integer m and an irreducible curve $M \neq G$ such that

$$D_Z = mM \sim K_Z + A_Z, \quad M^2 < 0, \quad K_Z \cdot M = mM^2 < 0.$$

- (iii) There is a projective birational morphism $p : Z \rightarrow W$ to a rank-one canonical del Pezzo surface, contracting exactly M to a point q , with $p^{-1}(q) = M$ as sets and p an isomorphism off q . For $A_W = p(A_Z)$ with its reduced structure,

$$A_W \subset W^{\text{reg}}, \quad p^* A_W = A_Z, \quad p_a(A_W) = 1, \quad K_W + A_W \sim 0, \quad A_Z^2 = A_W^2 \geq 1.$$

Here A_W and K_W are Cartier, $A_W \in |-K_W|$, and compatible canonical divisors satisfy $K_Z = p^* K_W + mM$.

- (iv) $\#(\text{Sing}(Z) \cap M) \leq 2$ and $0 < -K_Z \cdot M \leq 1$. The curve G survives on W ; its image $G_W = p(G)$ has normalization $\mathbb{P}^1$ and satisfies $(-K_W) \cdot G_W = A_W \cdot G_W = 1$.

The adjoint divisor is supported away from the genus-one curve. Its contraction preserves that curve and turns it into a Cartier anticanonical divisor.

Proof. We construct the adjoint divisor, prove that its image has one negative component, and contract that component.

Step 1. In this step, we construct the unique effective adjoint divisor and prove that its nonzero image avoids A_Z . Only E survives among the r -exceptional primes, so $\text{Exc}(h) = G$ and $\rho(Z) = \rho(S) + 1 = 2$. The local isomorphism near A_X gives $A_Z \subset Z^{\text{reg}}$ and $A_Z \cdot G = 1$. Theorem 5.11 makes X rational. Consequently $H^0(X, \mathcal{O}_X(K_X)) = 0$ and $H^1(X, \mathcal{O}_X) = 0$; Serre duality gives $H^1(X, \mathcal{O}_X(K_X)) = 0$ as well. Adjunction on the smooth surface X gives

$$0 \longrightarrow \mathcal{O}_X(K_X) \longrightarrow \mathcal{O}_X(K_X + A_X) \longrightarrow \omega_{A_X} \longrightarrow 0.$$

These vanishings identify the global sections of the middle sheaf with $H^0(A_X, \omega_{A_X})$, which has dimension $p_a(A_X) = 1$. Thus $|K_X + A_X|$ consists of one effective divisor D_X . The integral Cartier curve A_X is Gorenstein, and its dualizing sheaf has degree zero. A nonzero section of that invertible sheaf has no zeros. The section defining D_X restricts to such a section, so $\text{Supp}(D_X) \cap A_X = \emptyset$.

Pushforward of a principal divisor gives $D_Z = \tau_* D_X \sim K_Z + A_Z$. It is effective and $\mathbb{Q}$ -Cartier, and its support misses A_Z . We show that $D_Z \neq 0$, and hence that $D_X \neq 0$. If $D_Z = 0$, then D_X is effective and τ -exceptional. Every component F_i of its support satisfies

$$D_X \cdot F_i = (K_X + A_X) \cdot F_i = K_X \cdot F_i \geq 0$$

because A_X is disjoint from the τ -exceptional curves and $K_X \cdot F_i = -2 - F_i^2 \geq 0$ for the smooth rational minimal exceptional curves. Negative definiteness would contradict $D_X^2 \geq 0$ unless $D_X = 0$. But $D_X = 0$ would give

$$0 = (K_X + A_X) \cdot E = K_X \cdot E + 1 \geq 1.$$

Hence $D_Z \neq 0$. Since G meets A_Z , it is not a component of D_Z .

Step 2. In this step, we prove that D_Z is supported on one curve with negative square and negative canonical degree. Write $D_Z = \sum_{i=1}^s m_i M_i$ with distinct irreducible curves and positive integers m_i . By Theorem 3.7,

$$K_Z = h^* K_S - c_E G, \quad c_E \geq 0.$$

Every $h_* M_i$ is a curve, because $M_i \neq G$. Thus

$$K_Z \cdot M_i = K_S \cdot h_* M_i - c_E G \cdot M_i < 0.$$

Here $-K_S$ is ample, and distinct effective $\mathbb{Q}$ -Cartier curves have nonnegative intersection. Since $A_Z \cdot M_i = 0$, we obtain $D_Z \cdot M_i = K_Z \cdot M_i < 0$.

Let $Q = (M_i \cdot M_j)$. For a real vector with coordinates $x_i = m_i y_i$, expansion gives

$$\mathbf{x}^t Q \mathbf{x} = \sum_i m_i (D_Z \cdot M_i) y_i^2 - \sum_{i < j} m_i m_j (M_i \cdot M_j) (y_i - y_j)^2.$$

This is strictly negative for every nonzero vector. Hence Q is negative definite, and the classes of the M_i are linearly independent. The intersection form on Z has signature $(1, 1)$, by the Hodge index theorem and $\rho(Z) = 2$. Thus $s = 1$. Write $D_Z = mM$. Then

$$mM^2 = D_Z \cdot M = K_Z \cdot M < 0.$$

Step 3. In this step, we construct the projective contraction of M and prove that its target is klt. Choose an ample Cartier divisor H on Z . The divisor

$$L = H - \frac{H \cdot M}{M^2} M$$

is nef, has $L \cdot M = 0$, and satisfies $L \cdot C \geq H \cdot C > 0$ for every irreducible curve $C \neq M$. It supports the extremal ray generated by M . Theorem 4.17, applied to $(Z, 0)$ over $\text{Spec}(k)$, contracts this K_Z -negative ray. Only M has numerical class on the ray, so the resulting morphism $p : Z \rightarrow W$ is birational, has exceptional locus M , and has $\rho(W) = 1$. The target W is normal, projective, and $\mathbb{Q}$ -factorial. Put $q = p(M)$. Connectedness of the fibres, from Theorem 4.8, gives $p^{-1}(q) = M$ as sets, and p is an isomorphism off q .

Choose compatible canonical divisors. Since K_W is $\mathbb{Q}$ -Cartier, we can write $K_Z = p^* K_W + cM$. Intersecting with M gives $c = (K_Z \cdot M)/M^2 = m > 0$. This also proves that W is klt. Indeed, on a resolution $\sigma : V \rightarrow Z$ resolving M , write $K_V = \sigma^* K_Z + \sum_F a_F F$. Every $a_F > -1$, and $\sigma^* M$ is effective. In

$$K_V = (p\sigma)^* K_W + m\sigma^* M + \sum_F a_F F,$$

the discrepancy of the strict transform of M is $m > 0$, and the discrepancy of each σ -exceptional prime F is $a_F + m \text{mult}_F(\sigma^* M) > -1$. Thus all discrepancies over W are greater than -1 .

Step 4. In this step, we identify the canonical anticanonical model and prove the remaining intersection and singular-point bounds. The morphism p is an isomorphism near A_Z , because M is disjoint from A_Z . Thus $A_W = p(A_Z)$ is contained in W^{reg} , is Cartier, and satisfies $p^* A_W = A_Z$. Its arithmetic genus is one. Push forward $K_Z + A_Z \sim mM$ to obtain

$$K_W + A_W \sim 0.$$

In particular K_W is Cartier. A klt surface with Cartier canonical divisor is canonical: its discrepancies are integers greater than -1 , hence nonnegative.

The nonzero effective Cartier divisor A_W has positive numerical class on a surface of Picard number one. It is ample by the Nakai–Moishezon criterion. Therefore $-K_W$ is ample, $A_W \in |-K_W|$, and A_W^2 is a positive integer. The projection formula gives $A_Z^2 = A_W^2 \geq 1$.

Proposition 4.12 applied to p gives $\#(\text{Sing}(Z) \cap M) \leq 2$ and $0 < -K_Z \cdot M \leq 1$. Finally G is not contracted, since $A_Z \cdot G = 1$ whereas $A_Z \cdot M = 0$. If $G_W = p(G)$, then

$$(-K_W) \cdot G_W = A_W \cdot G_W = p^* A_W \cdot G = 1.$$

The maps $E \rightarrow G \rightarrow G_W$ are birational on these curves, so their normalizations agree and are $\mathbb{P}^1$. $\square$

Lemma 11.5. *Let S, Z be normal projective $\mathbb{Q}$ -factorial klt surfaces, let $y \in \text{Sing}(S)$, and let $h : Z \rightarrow S$ be a projective birational morphism with exceptional locus $G \simeq \mathbb{P}^1$, with $h^{-1}(y) = G$ as sets, and which is an isomorphism off y . Let A be an integral $\mathbb{Q}$ -Cartier curve on S with strict transform $A_Z \subset Z^{\text{reg}}$ and $A_Z \cdot G = 1$. Let $p : Z \rightarrow W$ be a projective birational morphism to a rank-one canonical del Pezzo surface, with exceptional locus an irreducible curve M satisfying $M^2 < 0$, $K_Z \cdot M < 0$, and $M \cap A_Z = \emptyset$. Suppose that $p^{-1}(q) = M$ as sets for $q = p(M)$, and*

that p is an isomorphism off q . Put $A_W = p(A_Z)$ and $G_W = p(G)$, both with reduced structure. Assume

$$A_W \subset W^{\text{reg}}, \quad A_W \sim -K_W, \quad p^*A_W = A_Z, \quad (-K_W) \cdot G_W = 1,$$

and assume A_W is rational with an ordinary cusp as its only singularity. If

$$\Theta = h^* \left(K_S + \frac{2}{3}A \right) = K_Z + \frac{2}{3}A_Z + \eta G, \quad \eta \geq \frac{1}{3},$$

and $-(K_S + \frac{2}{3}A)$ is ample, then $\# \text{Sing}(S) \leq 5$. If $\# \text{Sing}(W) \geq 3$, then $\# \text{Sing}(S) \leq 4$.

The cases with three or four singular points require the positions of those points on G_W . The actual coefficient η rules out the two exceptional incidence patterns in degree one.

Proof. Theorem 5.13 applies to (W, A_W) . Thus either $\# \text{Sing}(W) \leq 2$, or W has type $3A_2$ and $K_W^2 = 3$, or W has type $4A_2$ and $K_W^2 = 1$. Put

$$r_G = \#(\text{Sing}(Z) \cap G), \quad \kappa = \#(\text{Sing}(Z) \cap M), \quad \delta = \begin{cases} 1 & q \in \text{Sing}(W), \\ 0 & q \notin \text{Sing}(W). \end{cases}$$

Lemma 4.13 and Proposition 4.12 give

$$(11.6) \quad \# \text{Sing}(S) = \# \text{Sing}(W) + 1 - r_G + \kappa - \delta, \quad \kappa \leq 2.$$

If $\# \text{Sing}(W) \leq 2$, this already gives $\# \text{Sing}(S) \leq 5$.

Step 1. In this step, we compute the possible markings of G_W and settle the target of type $3A_2$. Suppose W has type nA_2 with $n = 3$ or 4 , and put $H = -K_W$. On the minimal resolution $\mu : \widetilde{W} \rightarrow W$, write each exceptional chain as $E_{i,1} + E_{i,2}$. Let C be the strict transform of G_W and put

$$x_i = C \cdot E_{i,1}, \quad y_i = C \cdot E_{i,2}, \quad Q_i = x_i^2 + x_i y_i + y_i^2.$$

Since W is canonical, this resolution is crepant. Hence $K_{\widetilde{W}} \cdot C = -1$ and $C^2 = 2p_a(C) - 1$. The local correction in the numerical pullback of G_W is

$$\frac{2x_i + y_i}{3} E_{i,1} + \frac{x_i + 2y_i}{3} E_{i,2}.$$

It contributes $2Q_i/3$ to the difference $G_W^2 - C^2$. Since $\rho(W) = 1$ and $H \cdot G_W = 1$, we obtain

$$(11.7) \quad \frac{1}{H^2} = G_W^2 = 2p_a(C) - 1 + \frac{2}{3} \sum_i Q_i.$$

When $n = 3$, we have $H^2 = 3$ and $\sum_i Q_i = 2 - 3p_a(C)$. All terms are nonnegative integers, so $p_a(C) = 0$ and the sum is two. The form $x^2 + xy + y^2$ takes value one exactly at $(1, 0)$, $(0, 1)$ and never takes value two on nonnegative integers. Thus exactly two singular points lie on G_W , each with a transverse end intersection on the resolution. A marked point different from q lifts to a singular point on G , and a marked point equal to q contributes to δ . Consequently $r_G + \delta \geq 2$, and (11.6) gives $\# \text{Sing}(S) \leq 4$.

Step 2. In this step, we use the two-thirds pullback on the target of type $4A_2$ to show that p identifies G with G_W . When $n = 4$, we have $H^2 = 1$ and $\sum_i Q_i = 3(1 - p_a(C))$. There are precisely three possibilities:

- (i) $p_a(C) = 1$ and all intersections x_i, y_i vanish;
- (ii) $p_a(C) = 0$ and three pairs (x_i, y_i) are $(1, 0)$ or $(0, 1)$;
- (iii) $p_a(C) = 0$ and one pair is $(1, 1)$, with all the others zero.

Indeed, the only positive values of $x^2 + xy + y^2$ at most three are one and three, and the latter occurs only at $(1, 1)$.

The curves M and G are distinct, because their intersections with A_Z are zero and one. Since h contracts only G , ampleness on S gives

$$\Theta \cdot M = \left(K_S + \frac{2}{3}A \right) \cdot h_*M < 0.$$

Here $G_W \equiv H$ and $A_W \sim H$, so $p_*\Theta \equiv (\eta - 1/3)H$. The difference of a divisor and the pullback of its pushforward is supported on M . Hence, for a real number t ,

$$\Theta \equiv p^*((\eta - 1/3)H) + tM, \quad t = \frac{\Theta \cdot M}{M^2} > 0.$$

Intersecting with G gives

$$(11.8) \quad 0 = \Theta \cdot G = (\eta - 1/3) + t(M \cdot G).$$

Since both terms on the right are nonnegative, $M \cdot G = 0$. Thus M and G are disjoint. If q belonged to G_W , properness would make G meet $p^{-1}(q) = M$. It follows that $q \notin G_W$, and p identifies G with G_W . In particular G_W is smooth and rational.

Step 3. In this step, we exclude possibilities (i) and (iii) and count the remaining marked singular points. Possibility (i) is now excluded: G_W avoids $\text{Sing}(W)$, so it would be isomorphic to C and have arithmetic genus one. To exclude (iii), we use the elementary local form of an A_2 point. A regular curve germ in

$$k[[u, v, w]]/(uv - w^3)$$

has $(u, v, w)\mathcal{O}_C = (t)$. Blowing up this principal ideal on the curve leaves it unchanged, so its strict transform has one point over the singular point. On the w -chart of the surface blowup,

$$u = \xi^2 v, \quad v = \xi v^2, \quad w = \xi v.$$

At the crossing of the two exceptional components, all three functions would lie in (t^2) , which is impossible. At a smooth point of the exceptional divisor, these functions generate its local parameter ideal; their images generate (t) only for transverse contact. The u - and v -charts give the same conclusion at the two ends. Thus a regular curve has intersection pattern $(1, 0)$ or $(0, 1)$, and cannot have pattern $(1, 1)$. This excludes (iii).

Only (ii) remains, giving three marked singular points on G_W . As before, $r_G + \delta \geq 3$. Equation (11.6) yields $\#\text{Sing}(S) \leq 4 + 1 + 2 - 3 = 4$. Together with the first case, this proves the asserted bounds. $\square$

Proof of Theorem 11.1. We apply the adjoint construction to the cuspidal second boundary, check its two-thirds pullback coefficient, and return from the endpoint bound to the original surface by counting the two hunt links.

Step 1. In this step, we construct the retained-end model and bound the two-thirds pullback coefficient. Apply Lemma 11.3, and let $r : X \rightarrow S_2$ be the minimal resolution supplied there. Write the fibre over the unique point $y \in B_2 \cap \text{Sing}(S_2)$ as $H_1, \dots, H_l$, starting at the end met by B_X . The surface S_2 is projective, $\mathbb{Q}$ -factorial and klt, and this nonempty fibre is a simple normal crossings chain of smooth rational curves of square at most -2 . The reduced irreducible curve B_X is not exceptional; it meets H_1 transversely once and is disjoint from every other r -exceptional prime, including those over points away from B_2 . Thus Proposition 4.18 gives a projective $\mathbb{Q}$ -factorial klt surface Z and

$$X \xrightarrow{\tau} Z \xrightarrow{h} S_2, \quad G = \tau(H_1) \simeq \mathbb{P}^1, \quad B_Z = \tau(B_X) \subset Z^{\text{reg}}, \quad B_Z \cdot G = 1.$$

Here $r = h\tau$, the morphism τ contracts every r -exceptional prime other than H_1 and is an isomorphism near B_X , and h is an isomorphism off y . In particular the other exceptional fibres give the same singular points on Z as on S_2 away from y . Theorem 4.8 gives $h^{-1}(y) = G$ as sets. For $\lambda \in \mathbb{Q}$, let $c(\lambda)$ be the coefficient of H_1 in $r^*(K_{S_2} + \lambda B_2) - K_X - \lambda B_X$. It depends affinely on λ . The canonical coefficient is nonnegative by Theorem 3.7. To compute $c(1)$ locally at y , let Q be the negative intersection matrix of $H_1, \dots, H_l$ and put $d = \det Q \geq 2$. For the

coefficients c_i of these curves in the pullback of $K_{S_2} + B_2$, set $v_i = 1 - c_i$. The chain equations in Lemma 11.2 give $Q\mathbf{v} = \mathbf{e}_l$, where $\mathbf{e}_l$ is the last standard basis vector. The corner entry of Q^{-1} is $1/d$, so $v_1 = 1/d$. Consequently

$$c(0) \geq 0, \quad c(1) = 1 - \frac{1}{d} \geq \frac{1}{2}.$$

With $K_Z = \tau_* K_X$, the actual pullback is therefore

$$(11.9) \quad h^* \left(K_{S_2} + \frac{2}{3} B_2 \right) = K_Z + \frac{2}{3} B_Z + \eta G, \quad \eta = c(2/3) = \frac{1}{3} c(0) + \frac{2}{3} c(1) \geq \frac{1}{3}.$$

Moreover $b_2 > 4/5 > 2/3$ and $-(K_{S_2} + b_2 B_2)$ is ample. The nonzero effective curve B_2 is ample on the rank-one surface S_2 , so

$$-\left(K_{S_2} + \frac{2}{3} B_2 \right) = -(K_{S_2} + b_2 B_2) + (b_2 - 2/3) B_2$$

is ample as well.

Step 2. In this step, we construct the adjoint endpoint and deduce $\# \text{Sing}(S_2) \leq 5$. The cusp lemma gives $p_a(B_2) = 1$ and an isomorphism $B_X \rightarrow B_2$. Together with the factorization just constructed, these facts check the hypotheses of Lemma 11.4 for $(S, A) = (S_2, B_2)$. It produces $p : Z \rightarrow W$ with exceptional curve M . Its curve $B_W = p(B_Z)$ is isomorphic to B_2 , lies in W^{reg} , and belongs to $|-K_W|$. In particular it is rational with one ordinary cusp. Moreover $p^* B_W = B_Z$, $M \cap B_Z = \emptyset$, and $M^2 < 0$, $K_Z \cdot M < 0$. The image $G_W = p(G)$ has $(-K_W) \cdot G_W = 1$. Thus Lemma 11.5, with the coefficient and anti-ameness established above, gives

$$\# \text{Sing}(S_2) \leq 5.$$

Step 3. In this step, we count singular points through the two original hunt links. The extraction centres are singular because the hunt extracts divisors from minimal resolutions. The opposite contractions are K -negative, so Proposition 4.12 bounds each of their singular-point counts by two. Since $\#(A_1 \cap \text{Sing}(S_1)) = 3$, the first-link inequality in Lemma 4.13 gives

$$\# \text{Sing}(S_0) \leq \# \text{Sing}(S_1) + 3 - 3 = \# \text{Sing}(S_1).$$

For the second link, there is exactly one singular point on E_2 and exactly two on Σ_2 , while q_2 is smooth. The exact identity in that lemma gives

$$\# \text{Sing}(S_1) = \# \text{Sing}(S_2) + 1 - 1 + 2 = \# \text{Sing}(S_2) + 2 \leq 7.$$

Combining the two counts gives $\# \text{Sing}(S_0) \leq 7$. $\square$

12. TWO SURVIVING BOUNDARIES

We use the hunt and its prescribed choices from Definition 6.2. When both curves survive the second contraction, we use a reduced boundary on S_2 to count its singular points. The first link does not increase this count when we return to S_0 ; the second link costs at most three points. The choice of reduced boundary will depend on whether the two curves form a fence or a banana.

We first record three local results about the surviving curves. The tacnode exclusion is [Lac24, Lemma 4.34], with the characteristic-free corrections in [Nag25a, Remark 3.16(1),(2)].

Theorem 12.1 (Lacini–Nagaoka). *Let S_0 be a rank-one klt del Pezzo surface over k without a tiger on its minimal resolution. Suppose its first two hunt steps are birational, A_1 is smooth rational with (S_1, A_1) dlt, and both $(A_1)_{T_2}$ and E_2 survive g_2 . Then $(S_2, A_2 + B_2)$ is not a tacnode configuration.*

The local geometry of a fence is given by the preliminary assertions in [Nag25a, proof of Proposition 3.18] and [Lac24, Lemma 4.39(1)–(3) and its proof].

Theorem 12.2 (Local geometry of a fence). *Let S_0 be a rank-one klt del Pezzo surface over k without a tiger on its minimal resolution. Suppose its first two hunt steps are birational, A_1 is smooth rational with (S_1, A_1) dlt, both $(A_1)_{T_2}$ and E_2 survive g_2 , and $(S_2, A_2 + B_2)$ is a fence. Then A_2 is smooth at every point of $A_2 \cap S_2^{\text{reg}}$, whereas B_2 is singular as a curve and $\#(B_2 \cap \text{Sing}(S_2)) \leq 1$.*

For a banana, we use the bound on the surface singular points lying on the new boundary from [Nag25a, Section 3.6].

Theorem 12.3 (Singularities on the new boundary). *Let S_0 be a rank-one klt del Pezzo surface over k without a tiger on its minimal resolution. Suppose its first two hunt steps are birational, A_1 is smooth rational with (S_1, A_1) dlt, both $(A_1)_{T_2}$ and E_2 survive g_2 , and $(S_2, A_2 + B_2)$ is a banana. Then $\#(B_2 \cap \text{Sing}(S_2)) \leq 1$.*

Lemma 12.4. *Let S_0 be a rank-one klt del Pezzo surface over k without a tiger on its minimal resolution. Suppose its first two hunt steps are birational and A_1 is smooth rational with (S_1, A_1) dlt. Then*

$$\#(A_1 \cap \text{Sing}(S_1)) \geq 3, \quad \# \text{Sing}(S_0) \leq \# \text{Sing}(S_1) \leq \# \text{Sing}(S_2) + 3.$$

In particular, if $\text{ht}(S_2) \leq 2$, then $\# \text{Sing}(S_0) \leq 7$.

The first boundary contains at least three singular points, which absorb the possible loss in the first birational comparison.

Proof. We first count the points on A_1 , and then apply the link formula. Let $p : Y \rightarrow S_0$ be the minimal resolution. By construction, p factors through f_1 , so the first link gives a morphism $Y \rightarrow S_1$. Theorem 6.3 says that S_1 is rank-one log del Pezzo and that $(S_1, a_1 A_1)$ has no tiger on Y .

Step 1. In this step, we use adjunction and the absence of a tiger to count singular points on A_1 . The curve A_1 is $\mathbb{Q}$ -Cartier by Theorem 4.16. Its numerical class is a positive multiple of an ample class, since $\rho(S_1) = 1$. In particular $A_1^2 > 0$. Put

$$t = -\frac{K_{S_1} \cdot A_1}{A_1^2}.$$

Then $K_{S_1} + tA_1 \equiv 0$, and the anti-ampleness of $K_{S_1} + a_1 A_1$ gives $t > a_1$. If $t \geq 1$, the effective divisor $(t - a_1)A_1$ produces a tiger: its addition to $a_1 A_1$ gives coefficient t on the strict transform of A_1 on Y . Thus $t < 1$ and

$$(K_{S_1} + A_1) \cdot A_1 = (1 - t)A_1^2 > 0.$$

Proposition 4.10 now gives

$$0 < -2 + \sum_{z \in A_1 \cap \text{Sing}(S_1)} \left(1 - \frac{1}{d_z}\right).$$

Every summand is less than one. There must therefore be at least three summands.

Step 2. In this step, we bound the total change in the singular-point count through the two links. Theorem 6.3 makes each g_i K_{T_i} -negative. Each T_i is a partial minimal resolution of the klt surface S_{i-1} , hence is klt and is $\mathbb{Q}$ -factorial by Theorem 4.16. Thus Proposition 4.12 gives

$$\kappa_i := \#(\Sigma_i \cap \text{Sing}(T_i)) \leq 2 \quad (i = 1, 2).$$

The second assertion of Lemma 4.13, applied to the first link with $C = A_1$, yields

$$\# \text{Sing}(S_0) \leq \# \text{Sing}(S_1) + 3 - \#(A_1 \cap \text{Sing}(S_1)) \leq \# \text{Sing}(S_1).$$

For the second link put $r_2 = \#(E_2 \cap \text{Sing}(T_2))$, and let δ_2 be one or zero according as q_2 is singular or smooth. The same lemma gives

$$\# \text{Sing}(S_1) = \# \text{Sing}(S_2) + 1 - r_2 + \kappa_2 - \delta_2 \leq \# \text{Sing}(S_2) + 3.$$

Finally, Theorem 3.2 gives $\# \text{Sing}(S_2) \leq 4$ when $\text{ht}(S_2) \leq 2$, since the characteristic is different from two. $\square$

Lemma 12.5. *Let S be a normal projective surface over k , and let $B \subset S$ be a reduced irreducible singular rational curve with $K_S + B$ $\mathbb{Q}$ -Cartier. Then $(K_S + B) \cdot B \geq 0$.*

The conductor records the failure of B to be normal. A singular point contributes at least two to its degree.

Proof. Let $\nu : B^\nu \rightarrow B$ be the normalization, and let J be the conductor divisor, so

$$\mathcal{O}_{B^\nu}(-J) = \text{Ann}_{\mathcal{O}_B}(\nu_* \mathcal{O}_{B^\nu} / \mathcal{O}_B) \mathcal{O}_{B^\nu}.$$

Theorem 5.1 supplies an effective $\mathbb{Q}$ -divisor Θ on B^ν with

$$\nu^*((K_S + B)|_B) \sim_{\mathbb{Q}} K_{B^\nu} + J + \Theta.$$

It suffices to prove $\deg J \geq 2$. Choose a singular point p of B . In the completed normalization over p , the conductor has the form $\prod_{i=1}^r t_i^{c_i} k[[t_i]]$. If $r \geq 2$, each c_i is positive. Otherwise the conductor, and hence the completed local ring of B , would contain a nontrivial idempotent supported on a single branch. This is impossible in a local ring. Thus $\sum_i c_i \geq 2$.

If $r = 1$, exponent zero would make the completed local ring equal to its normalization. Exponent one has the same consequence: the local ring would contain both k and $tk[[t]]$, hence all of $k[[t]]$. Both contradict singularity at p . Therefore $c_1 \geq 2$ in this case as well. It follows that $\deg J \geq 2$. Since $B^\nu \simeq \mathbb{P}^1$, adjunction gives

$$(K_S + B) \cdot B = -2 + \deg J + \deg \Theta \geq 0.$$

$\square$

Lemma 12.6. *Let S_0 be a rank-one klt del Pezzo surface over k without a tiger on its minimal resolution. Suppose its first two hunt steps are birational, A_1 is smooth rational with (S_1, A_1) dlt, and both $(A_1)_{T_2}$ and E_2 survive g_2 . Assume $(S_2, A_2 + B_2)$ is a fence or a banana. In the fence case set $C = A_2$, and in the banana case set $C = B_2$. In either case, $-(K_{S_2} + C)$ is ample, (S_2, C) is dlt, and the minimal surface resolution of S_2 is the minimal log resolution of (S_2, C) . Moreover,*

	$A_2 \cdot B_2$	$\#(C \cap \text{Sing}(S_2))$
fence	1	≤ 3
banana	2	≤ 1

Thus a fence uses its old smooth side, whereas a banana uses its new plt side.

Proof. We establish the boundary choice in each case, then check its resolution and intersection numbers. Theorems 10.1 and 6.3 give $a_2 + b_2 > 1$. Theorem 6.6 therefore supplies both the alternative between the two reduced adjoint negatives and, in the banana case, the plt property of (S_2, B_2) . The surface S_2 is rank-one klt del Pezzo and is $\mathbb{Q}$ -factorial by Theorem 4.16.

Step 1. In this step, we select A_2 in the fence case and bound its surface singularities. Theorem 12.2 says that B_2 is singular. It is rational by Definition 6.4. Lemma 12.5 gives $(K_{S_2} + B_2) \cdot B_2 \geq 0$, so $-(K_{S_2} + B_2)$ is not ample. The alternative above therefore selects $-(K_{S_2} + A_2)$. At a singular point of S_2 , the pair $(S_2, A_2 + B_2)$ is dlt by the definition of a fence. Lowering the boundary gives that (S_2, A_2) is dlt there. At a smooth point of S_2 , Theorem 12.2 makes A_2 smooth and gives the same conclusion. Proposition 4.10 then makes A_2 smooth everywhere and gives

$$0 > (K_{S_2} + A_2) \cdot A_2 = -2 + \sum_{p \in A_2 \cap \text{Sing}(S_2)} \left(1 - \frac{1}{d_p}\right).$$

Since every summand is at least $1/2$, there are at most three of them.

Step 2. In this step, we select B_2 in the banana case and prove that its adjoint divisor is anti-ample. Proposition 4.10 applies to the plt pair (S_2, B_2) . The rational curve B_2 is smooth, and Theorem 12.3 bounds its singular support by one point. Thus

$$(K_{S_2} + B_2) \cdot B_2 = -2 + \sum_{p \in B_2 \cap \text{Sing}(S_2)} \left(1 - \frac{1}{d_p}\right) < 0.$$

There is at most one summand, and it is less than one. Since $\rho(S_2) = 1$, this strict inequality implies that $-(K_{S_2} + B_2)$ is ample.

Step 3. In this step, we identify the minimal log resolution of the selected pair and compute the boundary intersections. Let $\pi : V \rightarrow S_2$ be the minimal surface resolution and put $E = \text{Exc}(\pi)_{\text{red}}$. Over each point of $C \cap \text{Sing}(S_2)$, Proposition 4.10 says that C_V meets an end of the exceptional chain transversely once. Away from C , the exceptional divisor is simple normal crossings by Theorem 3.7. Consequently $C_V + E$ is simple normal crossings on V . Every log resolution factors through V by Lemma 4.9; since V already resolves the pair, it is its minimal log resolution.

Finally, Definition 6.4 places all intersections of A_2 and B_2 in S_2^{reg} and makes them transverse. Each contributes one to $A_2 \cdot B_2$, and there are respectively one and two such points. $\square$

Lemma 12.7. *Let S be a rank-one klt del Pezzo surface over k , and let C, H be distinct reduced irreducible curves with (S, C) dlt. Let $\pi : V \rightarrow S$ be the minimal resolution and put $D_C = C_V + \text{Exc}(\pi)_{\text{red}}$. Assume that, for some $\ell \in \{3, 4, 5\}$, one connected component of D_C is a four-component fork of smooth rational curves, and that every other component of D_C is a curve of square -2 . Write R for the central curve of the fork and L_1, L_2, L_3 for its tips, with*

$$R^2 = -1, \quad L_1^2 = -2, \quad L_2^2 = -3, \quad L_3^2 = -\ell.$$

Assume there are rational numbers $c > 1/2$ and $d \geq 0$ such that $H \cdot C \geq 1$ and $-(K_S + cH + dC)$ is ample. Then these hypotheses are incompatible.

The proof considers every possible position of the unique component of D_C which survives on S .

Proof. Put $n_1 = 2, n_2 = 3, n_3 = \ell$, and set

$$\beta = -1 + \sum_{j=1}^3 \frac{1}{n_j} = \begin{cases} 1/6, & \ell = 3, \\ 1/12, & \ell = 4, \\ 1/30, & \ell = 5. \end{cases}$$

We first locate C_V in the fork, then exclude a tip and the centre separately.

Step 1. In this step, we prove that the retained curve C_V belongs to the fork. If C_V did not belong to the fork, all four fork components would be exceptional. But the divisor

$$Z = R + \sum_{j=1}^3 \frac{1}{n_j} L_j$$

has $Z^2 = \beta > 0$. This contradicts Theorem 4.5.

Step 2. In this step, we exclude each tip as the retained curve by dlt end incidence. Suppose $C_V = L_i$, and let L_j, L_k be the other two tips. The connected exceptional component meeting C_V is exactly $L_j + R + L_k$: all other components of D_C are exceptional and the remaining connected components are disjoint from this fork. It contracts to a point p on C , which is singular since the minimal resolution is an isomorphism over S^{reg} . It is the full exceptional fibre over p by Theorem 4.8. The curve C_V meets the middle component R , not an end of this chain. This contradicts Proposition 4.10 for the dlt pair (S, C) at p . This argument applies to each of the three tips.

Step 3. In this step, we exclude the central curve by computing its intersection with the scaled adjoint divisor. It remains to suppose $C_V = R$. Each L_j is exceptional, and all other exceptional curves are disjoint from the displayed fork. Write

$$\pi^*C = R + \sum_{j=1}^3 x_j L_j + M,$$

where M is supported on exceptional curves outside the fork. The equations $\pi^*C \cdot L_j = 0$ give $x_j = 1/n_j$. Hence

$$C^2 = -1 + \sum_{j=1}^3 \frac{1}{n_j} = \beta.$$

Similarly, write

$$\pi^*(K_S + C) = K_V + R + \sum_{j=1}^3 u_j L_j + N$$

with N supported outside the fork. Adjunction on L_j gives $K_V \cdot L_j = n_j - 2$, so the pullback equations give $u_j = 1 - 1/n_j$. Adjunction on R then gives

$$(K_S + C) \cdot C = -2 + \sum_{j=1}^3 \left(1 - \frac{1}{n_j}\right) = -\beta.$$

Consequently

$$(K_S + cH) \cdot C = c(H \cdot C) - 2\beta > \frac{1}{2} - \frac{1}{3} > 0.$$

On the other hand, anti-ampleness and $d \geq 0$ imply

$$(K_S + cH) \cdot C = (K_S + cH + dC) \cdot C - dC^2 < 0.$$

This is the required contradiction. In particular, the argument does not require an upper bound on d . $\square$

Lemma 12.8. *Let S_0 be a rank-one klt del Pezzo surface over k without a tiger on its minimal resolution. Suppose its first two hunt steps are birational, A_1 is smooth rational with (S_1, A_1) dlt, both $(A_1)_{T_2}$ and E_2 survive g_2 , and $(S_2, A_2 + B_2)$ is a banana with $S_2 \simeq \mathbb{P}^2$. Then $\# \text{Sing}(S_0) \leq 3$.*

The smoothness of the plane leaves only the points lost in the two birational comparisons to count.

Proof. The two transverse intersections of the banana give, by Bézout,

$$\deg A_2 \deg B_2 = 2, \quad \{\deg A_2, \deg B_2\} = \{1, 2\}.$$

Thus the two curves are a line and a smooth conic. Since the surface S_2 is smooth, Lemma 12.4 gives

$$\# \text{Sing}(S_0) \leq \# \text{Sing}(S_2) + 3 = 3.$$

$\square$

Theorem 12.9. *Let S_0 be a rank-one klt del Pezzo surface over k without a tiger on its minimal resolution. Suppose its first two hunt steps are birational, A_1 is smooth rational with (S_1, A_1) dlt, and both $(A_1)_{T_2}$ and E_2 survive g_2 . Then $\# \text{Sing}(S_0) \leq 7$.*

The proof separates the plane cases from the open-pair alternatives. An ordinary ruling gives the count, and the exceptional graphs exclude the case of height greater than two.

Proof. We first choose a reduced boundary. We then apply the singularity count when its resolved height is at most two and exclude the remaining graphs. Theorems 10.1 and 6.3 give

$$a_2 = \lambda a_1 > \frac{2}{3}, \quad b_2 = \lambda(e_1 + \varepsilon) > \frac{1}{2}, \quad a_2 + b_2 > 1,$$

as well as the anti-ampleness of $K_{S_2} + a_2 A_2 + b_2 B_2$. The surface S_2 is rank-one log del Pezzo. Theorem 6.6 now makes $(S_2, A_2 + B_2)$ a fence, banana, or tacnode configuration. Theorem 12.1 excludes the last case.

Step 1. In this step, we exclude a plane fence and prove the bound for a plane banana. If a fence lies in $\mathbb{P}^2$, its unique transverse intersection gives $\deg A_2 \deg B_2 = 1$. Both curves are lines, contradicting Theorem 12.2, which makes B_2 singular. A plane banana satisfies the required bound by Lemma 12.8.

Step 2. In this step, we select a reduced boundary on a nonplane endpoint and verify the open-pair hypotheses. Assume $S_2 \not\subset \mathbb{P}^2$. Set

$$(C, H, c, d) = \begin{cases} (A_2, B_2, b_2, a_2), & \text{for a fence,} \\ (B_2, A_2, a_2, b_2), & \text{for a banana.} \end{cases}$$

Lemma 12.6 gives that $-(K_{S_2} + C)$ is ample and (S_2, C) is dlt. It also identifies the minimal surface resolution $\pi : V \rightarrow S_2$ with the minimal log resolution of this pair. Put $D_C = C_V + \text{Exc}(\pi)_{\text{red}}$. We have $c > 1/2$, $d \geq 0$, $H \cdot C \geq 1$, and $-(K_{S_2} + cH + dC)$ ample. Every component of D_C is smooth rational: this holds for C_V by Lemma 12.6, and for the exceptional curves by Theorem 3.7.

All the hypotheses of Theorem 5.2 hold: S_2 is normal projective of Picard number one, it is not the plane, C is nonzero and reduced, and its reduced adjoint negative is ample.

Step 3. In this step, we prove the bound when the selected pair has height at most two. Suppose first that $\text{ht}(S_2, C) \leq 2$. Then V admits a $\mathbb{P}^1$ -fibration whose general fibre F satisfies $F \cdot D_C \leq 2$. Since V is the minimal surface resolution and C_V is effective,

$$F \cdot \text{Exc}(\pi)_{\text{red}} \leq F \cdot D_C \leq 2.$$

Thus $\text{ht}(S_2) \leq 2$. Lemma 12.4 gives $\# \text{Sing}(S_0) \leq 7$.

Step 4. In this step, we exclude height greater than two using the exceptional open-pair graphs. Assume instead that $\text{ht}(S_2, C) > 2$. The selected curve C is irreducible, and S_2 is $\mathbb{Q}$ -factorial by Theorem 4.16. Theorem 5.3 supplies a canonical surface W , its cuspidal curve T , the integer $r \in \{6, 7, 8\}$, and a morphism $\tau : V \rightarrow W$, with

$$D_C = T_V + \text{Exc}(\tau)_{\text{red}}.$$

We now have all the hypotheses of Proposition 5.9, including the strict height inequality and its specified common resolution. The proposition describes the graph of this same divisor D_C . Its marked tip is T_V ; it need not be C_V , the curve retained by π . Regardless of which component is C_V , Lemma 12.7, with $\ell = r - 3$, excludes this graph using the dlt and scaled anti-ampleness properties established above. The ordinary case gives at most four singularities on S_2 , and Lemma 12.4 adds at most three on returning to S_0 . Together with the plane cases, this proves the assertion. $\square$

13. PROOF OF THE MAIN THEOREM

We retain the hunt notation of Definition 6.2. We now combine the preceding bounds, first according to the existence of a tiger and then according to the geometry of the first boundary and the second contraction.

Theorem 13.1. *Let k be an algebraically closed field of characteristic 3, and let X be a normal projective klt del Pezzo surface over k with $\rho(X) = 1$. Then X has at most seven singular points.*

The bounds in the preceding sections apply to the singularities of the original surface, even when the hunt passes to a different surface.

Proof. We first dispose of the tiger case. In its absence we consider the first hunt, and use a second hunt when the first boundary is smooth rational. If X is smooth, then $\# \text{Sing}(X) = 0$. We may therefore assume that X is singular. Put $S_0 = X$, and let $p : Y \rightarrow S_0$ be its minimal resolution. The hypotheses say precisely that S_0 is a rank-one log del Pezzo surface over k .

Step 1. In this step, we prove the bound when the minimal resolution carries a tiger. If S_0 has a tiger on Y , Corollary 8.19 gives $\# \text{Sing}(S_0) \leq 7$. We henceforth assume that S_0 has no tiger on Y .

Step 2. In this step, we treat the first hunt and reduce to a smooth rational first boundary. Start the hunt from $(S_0, 0)$, using the choices in Definition 6.2 at each step, including its convention for tied maximal coefficients. The initial adjoint divisor is K_{S_0} , whose negative is ample. The hypotheses of Theorem 6.6 [Nag25a, Proposition 2.10] are therefore satisfied. If the first output is a net, Proposition 9.3 gives $\# \text{Sing}(S_0) \leq 7$.

Suppose that the first output is birational, and write the first link as

$$S_0 \xleftarrow{f_1} T_1 \xrightarrow{g_1} S_1, \quad A_1 = g_1(E_1), \quad \Delta_1 = a_1 A_1.$$

By Theorem 6.3 [Nag25a, Lemma 2.7(7)], the surface S_1 is again rank-one log del Pezzo and $-(K_{S_1} + a_1 A_1)$ is ample. Thus the effective hunt boundary Δ_1 satisfies the adjoint ampleness hypothesis for the second hunt.

Theorem 6.6 gives four possibilities for A_1 . If $p_a(A_1) > 1$, Proposition 9.4 gives $\# \text{Sing}(S_0) = 2$. If A_1 has arithmetic genus one and an ordinary node or an ordinary cusp at q_1 , Theorem 9.18 gives $\# \text{Sing}(S_0) \leq 7$. In the remaining possibility, A_1 is rational and (S_1, A_1) is dlt. The dlt condition includes that $K_{S_1} + A_1$ is $\mathbb{Q}$ -Cartier. Since S_1 is klt and A_1 is reduced and irreducible, Proposition 4.10 applies and shows that A_1 is smooth. Hence the remaining case has a smooth rational first boundary with (S_1, A_1) dlt.

Step 3. In this step, we treat every second hunt output with a smooth rational first boundary. Continue this chosen hunt for $(S_1, a_1 A_1)$, and write $f_2 : T_2 \rightarrow S_1$ for its extraction and E_2 for the extracted curve. Theorem 6.6 gives three possible outputs. If the output is a net, Theorem 10.3 gives $\# \text{Sing}(S_0) \leq 7$.

Otherwise the second opposite contraction $g_2 : T_2 \rightarrow S_2$ is birational, so both hunt steps are birational. If g_2 contracts the strict transform $(A_1)_{T_2}$, all the hypotheses of Theorem 11.1 hold, and again $\# \text{Sing}(S_0) \leq 7$. In the remaining alternative, both curves $(A_1)_{T_2}$ and E_2 survive g_2 , with images A_2 and B_2 . Theorem 12.9 then gives $\# \text{Sing}(S_0) \leq 7$.

These cases exhaust the chosen hunt, so $\# \text{Sing}(X) = \# \text{Sing}(S_0) \leq 7$. $\square$

APPENDIX A. DETAILS OF THE USE OF GENERATIVE AI

The results of this paper were obtained using Danus [Liu⁺26], an automated mathematical reasoning system built on Rethlas [Ju⁺26]. The retained records describe an earlier research project and a separate manuscript-writing run. We distinguish their settings below (§A.1), reproduce the retained problem formulations (§A.2), and give the recorded guidance with its available attribution (§A.3). Historical proposals and reports quoted here are records of the process; they are not additional premises of the proofs in the body.

Private service identities, deployment paths, and project handles have been anonymized; the mathematical content, model settings, and non-identifying provenance are retained.

A.1. How the system was run. The retained configuration of the earlier research project specifies `gpt-5.5` and seven research workers: three at reasoning effort `high` and four at `xhigh`. This is a configuration snapshot, not a complete history of the models used by every actor. The available records do not establish the original main-agent or verifier settings, the research launch

time, or the elapsed time from launch to acceptance of the final target. Guidance was recorded on 13 July 2026. The archived target’s filesystem timestamp is not an authenticated acceptance time, so we do not use it to infer a research duration. No comparison-run outcome is established by these records.

The separate writing view was created on 20 September 2026 at 04:29:58 UTC. Its main session started at 04:32:20 UTC. This run used `gpt-6-astra` through a configured private API service, with reasoning effort `ultra` for the main session and `max` for the native helpers, section writers, and verifiers. It used six helper homes, eleven body-section writer jobs, eleven separate section-verifier homes, and a whole-paper verifier. The main session prepared the introduction and notation. No research workers were launched in this writing run.

The mathematical source was a frozen graph of 1,193 active records. The target’s complete literal citation closure contained 422 records; their statements and proofs were read for this manuscript. Section writers produced the body from that source and the shared outline. All eleven body sections passed independent native review before the first whole-paper pass, recorded at 06:32 UTC on 20 September 2026, approximately two hours after the writing main session started. This interval measures manuscript production and review, not discovery of the theorem. Introduction, front matter, and prose polish followed that first pass. These reviews are model-based mathematical assessments; they are not formal proof-assistant certification. The retained records do not identify a named human checker or support a complete account of individual human contributions.

A.2. The problem as posed. An authenticated transcript of the original human launch question is not available in the retained records. The first quotation below reproduces the consolidated historical problem-and-repair directive, including its later corrections. The second reproduces the problem file supplied to the September writing run. They have different roles and are kept separate.

Throughout the quotations in this appendix, internal record identifiers and project paths are replaced by `[internal identifier]` and `[project path]`, respectively. These substitutions keep technical provenance tokens out of the article; no mathematical wording, instruction, attribution, or stored transcription defect is otherwise changed. The complete, unmodified records and a precise substitution map are retained in the private provenance archive. The quotations are therefore not byte-for-byte transcriptions of those originals.

Retained consolidated research directive. The following reproduces the retained historical problem-and-repair directive with the technical substitutions specified above. It includes later corrections and invalidations. Its author and revision dates are not identified in the retained document; it is not presented as the original human launch prompt. Its historical mathematical assertions are reproduced as a record, not adopted as proof premises.

```
# Seven singular points in characteristic three: clean mathematical directive
```

```
## The theorem
```

```
Let (k) be an algebraically closed field of characteristic (3). Let (X) be a
normal projective klt surface such that  $(-K_X)$  is ample and  $(\rho(X)=1)$ . Prove
```

```
\[
  \#\operatorname{Sing}(X)\leq 7.
\]
```

```
If  $(\pi:Y\rightarrow X)$  is the minimal resolution and
 $(D=\operatorname{Exc}(\pi)_{\mathrm{red}})$ , then the desired inequality is
equivalent to
```

```
\[
```

$\#\pi_0(D) \leq 7$.
 $\backslash]$

The argument must be a proof of the upper bound. Searching for an eight-point construction, enumerating candidate surfaces, or inferring the theorem from failure to find a counterexample is irrelevant.

Invalidated characteristic-transfer claims

The following six claims are unavailable. Each imported a result from the characteristic-zero Keel--McKernan memoir into characteristic (3) without reproducing the needed proof.

- ``[internal identifier]``: the full one-singularity KM 12.4 genus-one transformation package;
- ``[internal identifier]``: the KM 12.5 bound of two surface singularities on a fixed component;
- ``[internal identifier]``: the KM 12.0 restriction isomorphism, KM 12.1(6) disjointness, and the KM 12.4 divisor identity;
- ``[internal identifier]``: the KM 12.1(7) fixed-part component bound;
- ``[internal identifier]``: the KM 12.4(1) Picard-rank formula;
- ``[internal identifier]``: the old-boundary local profile obtained from KM 23.5 and KM 11.2.

The later source audit quarantined the following additional roots and every active successor. They are also unavailable:

- ``[internal identifier]`` and duplicate ``[internal identifier]``: the ordinary-cusp coefficient table was cited only from KM Lemma 11.2.1;
- ``[internal identifier]``: the ordinary-node coefficient lower bound was imported from KM Lemmas 11.1.1 and 17.2;
- ``[internal identifier]`` and ``[internal identifier]``: actual characteristic-three hunt links were put into the KM Section 11.4 list without an on-page proof;
- ``[internal identifier]``: boundary-adjacent retention was imported from KM Lemma 8.0.7 without the resolution-chain recurrence;
- ``[internal identifier]``: Palka--Peřka Theorem 1.2(a) was used backwards to type an arbitrary common-resolution witness as $(3A_2/4A_2)$;
- ``[internal identifier]`` and ``[internal identifier]``: coarse net alternatives were silently upgraded to full minimal-resolution comparison diagrams;
- ``[internal identifier]``: the smooth $(A_1=\Sigma_2)$ profile came from KM Lemma 18.7/Nagaoka Lemma 3.21 outside their proved characteristic range.

Two terminal assembly overclaims, ``[internal identifier]`` and ``[internal identifier]``, are likewise unavailable. A true statement with an incomplete characteristic transfer must be reproved as a new fact; the old content-addressed node and its descendants do not become valid retroactively.

No statement that used one of these claims may be used. In particular, the old target-theorem facts and their all-leaf assembly consequences are unavailable. A phrase such as "the local part of the KM proof is characteristic-free" is not a proof. Any needed assertion from KM Sections 11, 12, or 23 must be derived on the page from intersection theory, adjunction, formal local algebra, and surface contraction results valid over (k) , or else replaced by a different argument.

This invalidation does not affect facts whose proofs are independent of the six claims. Especially useful surviving inputs include:

- ``[internal identifier]`` and ``[internal identifier]``: a $(\mathbb{P}^1)$ -fibration on (Y) with $(F \cdot D \leq 4)$, equivalently height at most four, implies at most seven singular points;
- ``[internal identifier]``: the exact singular-support identity for a finite-centre birational diagram;
- ``[internal identifier]``: the intrinsic fixed-component $((-1))$ -curve calculation on a smooth resolution;
- ``[internal identifier]``: an adjoint bridge that explicitly avoids KM Section 12;
- ``[internal identifier]``: the numerical anti-ampleness calculation at the $(5/6)$ -boundary endpoint;
- ``[internal identifier]``: the degree-one $(4A_2)$ lattice trichotomy;
- ``[internal identifier]``: the $(2/3)$ adjunction correction for an (A_2) end curvette;
- ``[internal identifier]``: the explicit blow-up of $(xy=z^3)$ in characteristic (3) and the implication "regular curve germ $(\rightarrow)$ transverse end contact";
- ``[internal identifier]``: the fifteen coarse tiger/no-tiger source alternatives, subject to checking every external hypothesis used in its proof;
- ``[internal identifier]``: the purely conditional reduction from closure of those fifteen alternatives to the target theorem.

The last two facts are reductions, not a proof of the theorem.

Global proof architecture

Assume for contradiction that $(n = \#\operatorname{Sing}(X) \geq 8)$. The characteristic-three height-four theorem and the Palka--Peřka height/descendant results reduce to

```
\[
  \operatorname{ht}(X) \geq 5
  \quad \text{and} \quad
  X \text{ has no elliptic-boundary descendant}.
\]
```

Run the tiger/no-tiger hunt only through statements whose hypotheses and characteristic range have been proved. The surviving coarse reduction has fifteen source alternatives. Each alternative must end in one of the following three contradictions:

1. a morphism $(Y \rightarrow B)$ with general fibre (F) satisfying $(F \cdot D \leq 4)$;
2. a genuine elliptic-boundary descendant in the precise Palka--Peřka sense;
3. a direct support calculation giving $(n \leq 7)$.

$$\backslash [M_Z = \sum_{i=1}^s m_i N_i, \quad \text{quad } m_i > 0,$$

\]

for the image of the non-exceptional adjoint fixed divisor. Derive, rather than assume,

$$\begin{aligned} & \text{A_Z}\cap\operatorname{operatorname{Supp}}\text{M_Z}=\varnothing, \\ & \text{K_Z}+\text{A_Z}\equiv \text{M_Z}. \end{aligned}$$

In the actual one-centre diagram, construct an effective (h)-exceptional boundary (J) for which $(-(K_Z + J))$ is ample, or give another direct proof that

$$\backslash[\quad K_Z \cdot N_i < 0 \quad \backslash \text{quad} \backslash \text{text}\{\text{for every } \} i. \quad \backslash]$$

The equality above then gives $(M_Z \cdot N_i < 0)$. If $(Q = (N_i \cdot N_j))$, then (Q) is symmetric, its off-diagonal entries are nonnegative, and

$$\begin{aligned} & \backslash[\\ & \quad Q(m_1, \ldots, m_s)^t < 0. \\ & \backslash] \end{aligned}$$

Prove directly that these conditions make (Q) negative definite. One convenient proof rescales by the positive vector (m) and observes that the negative matrix is a symmetric nonsingular (M)-matrix with positive row sums. Contract the whole support by characteristic-free surface contraction. Check after each contraction that the remaining curves are still (K)-negative:

$$\begin{aligned} & \backslash[\\ & K_Z = p_1^* K_{Z_1} + c N_1 \\ & \quad \quad \quad \Longrightarrow \\ & K_{Z_1} \cdot N_i' = K_Z \cdot N_i - c(N_1 \cdot N_i) < 0 . \\ & \backslash] \end{aligned}$$

When $(\rho(Z)=2)$, the (s) contracted numerical curve classes are independent, so

$$\rho(W) = 2 - s \geq 1.$$

Together with $(M_Z \neq 0)$, this gives $(s=1)$. This is the intended replacement for the unsupported KM component and rank formulas; it needs no injection of fixed components into marked ends of resolution chains.

G3. Geometry of the target

For the contraction $(p:Z \rightarrow W)$, prove only the conclusions actually needed later:

- (p) is an isomorphism in a neighbourhood of (A_Z) ;
- $(\rho(W)=1)$;
- the push-forward of $(K_Z+A_Z \equiv M_Z)$ gives $(K_W+A_W \equiv 0)$, where $(A_W=p(A_Z))$;

- the relevant singularities of (W) are klt or canonical, with discrepancies computed from the displayed contraction rather than borrowed from KM;
- $(-K_W)$ is ample, using $(\rho(W)=1)$, $(-K_W \cdot A_W > 0)$, or the surviving numerical lemma ``[internal identifier]`` as appropriate;
- every use of "Gorenstein", "Du Val", or $(A_W \sim K_W)$ is justified at the level of linear equivalence and Cartier indices, not merely numerical equivalence.

Use the exact support identity ``[internal identifier]`` for the diagram $(U \leftarrow Z \rightarrow W)$. A generic claim that a fixed component contains at most two singularities is not available. Either prove the required local bound from adjunction on the actual component, or compute the integers (m, r, k, δ) directly in the actual resolution graph.

0. Old-boundary contraction in characteristic three

Consider the numerical-zero plt endpoint $((S, C))$, its first pair-hunt link

$\backslash$
 $S \xleftarrow{f} T \xrightarrow{g} U$,
 $\backslash$

and the boundary-contracted case $(C_T = \Sigma = \text{operatorname{Exc}}(g))$, with $(E = \text{operatorname{Exc}}(f))$ and $(A = g(E))$. The following local assertions must be proved from a fixed complete resolution diagram:

1. the full exceptional locus over each marked point of (T) is specified on one chosen minimal dlt or minimal resolution;
2. no extra exceptional component is attached to either displayed arm;
3. the component retained by the hunt is identified with a definite component of that resolution;
4. the total graph contracted by (g) is

$\backslash$
 $[2] - [1] - [3]$,
 $\backslash$

where the middle curve is the strict transform of (Σ) ;

5. $(q = g(\Sigma))$ is smooth on (U) , (E) meets (Σ) at a smooth transverse point away from the two arms, and (A) has an ordinary cusp at (q) .

The determinant calculation must use the complete graph. For a chain $([r] - [1] - [s])$, the determinant is $(rs - r - s)$; a smooth contraction forces $((r-1)(s-1)=2)$, hence $(\{r, s\} = \{2, 3\})$. This observation is useful only after smoothness or the corresponding determinant-one statement has been proved independently. The ordinary-cusp conclusion should be checked by an explicit sequence of blow-downs or completed-local parameters, not by naming "configuration III".

After the local diagram is proved, combine it with the actual numerical relation

$\backslash$
 $K_U + \frac{56}{A} \equiv 0$
 $\backslash$

and the intrinsic genus-one lemmas $G1--G3$. The desired endpoint conclusion is a direct height-four exit, an elliptic descendant, or a support bound strong enough to give $(\#\operatorname{Sing}(S)) \leq 7$. It is not enough to construct an auxiliary surface whose relation to the original hunt is conditional.

A. The characteristic-three (A_2) curvette package

Work in the completed local ring

```
\[
  R=k[[x,y,z]]/(xy-z^3).
\]
```

The Jacobian singular subscheme is nonreduced in characteristic (3), but the underlying singular point is isolated: in every prime containing (x,y) , the equation forces (z) into the prime. The explicit blow-up calculation in ``[internal identifier]`` is therefore the correct starting point.

Prove the converse missing from that fact. If a regular strict transform meets one end component transversely at a smooth point of the (A_2) exceptional chain, then its image curve germ is regular. On the (x) -chart

```
\[
  y=x^2Z^3, \text{ } \sqcup \text{ } z=xZ,
\]
```

transversality makes (x) a uniformizer; on the (z) -chart away from the node, one of (X,Y) is a unit and $(z=XY)$ is a uniformizer. Treat all charts and state the completion/strict-transform compatibility used to return to the algebraic germ.

Also prove that a Cartier curve through the (A_2) point cannot be regular. If $(C=(f))$, then either $(f \in \mathfrak{m}^2)$, leaving embedding dimension at least two, or (f) has a nonzero linear term; after eliminating that variable, the remaining plane-curve equation has multiplicity at least two. Thus $(R/(f))$ is not a regular one-dimensional local ring.

Apply these two local statements to the ($4A_2$) lattice trichotomy ``[internal identifier]``. In the end-marked case the curve is regular. In the $((1,1))$ case its local class is zero, hence it is Cartier and therefore nonregular. In the actual birational diagram, if (p) is an isomorphism away from (q) and the strict transform is regular, the unique $((1,1))$ -marked (A_2) point must be (q) . Every arrow in

```
\[
  (1,1) \rightarrow [G]=0 \rightarrow G \text{ } \{ \text{Cartier} \}
  \rightarrow G \text{ } \{ \text{nonregular at the } A_2 \text{ point} \}
  \rightarrow q \text{ } \{ \text{is that point} \}
\]
```

must be written explicitly.

S. Smooth ($A_1=\Sigma_2$) branch

The citation to Nagaoka Lemma 3.21 is unavailable because its stated characteristic range is at least five. Prove the characteristic-three branch directly.

There are two viable mathematical routes.

1. **Half-boundary height route.** Starting from the actual relation $\backslash(-K_{S_2}+\tfrac{1}{2}B_2)\backslash$ ample, derive the second-hunt coefficient from the scaling equations. The coefficient (b_2) of the actual boundary is **not** $(2/3)$; $(2/3)$ is only an auxiliary scaling coefficient in one later pullback. The local calculation should prove the true interval $\backslash[\tfrac{4}{5}<b_2<\tfrac{5}{6}, \quad e_1>\tfrac{1}{2}, \backslash]$ if that route is used. Then run a $\backslash((K_{S_2}+\tfrac{2}{3}B_2)\backslash)$ -negative contraction or construct the pencil dictated by the complete cusp resolution. Pull the resulting ruling to the original minimal resolution and prove that every horizontal component of the original exceptional boundary is among (B_2) and at most two residual second-link components. The coefficient inequality must exclude total horizontal degree five, including inseparable degree-three packets. Conclude $(F\cdot D)\leq 4$.
2. **Intrinsic genus-one route.** Prove all hypotheses of G1--G3 for the actual pair $((S_2, B_2))$. Let $(m=\#(B_2\cap\operatorname{Sing}S_2))$. For $(m=1)$, the one-centre Picard argument gives a single fixed component. For $(m\geq 2)$, do not quote the KM formula: use the exact finite-centre support identity and the independent numerical classes of the contracted curves to prove either the required support saving or a new extremal contraction yielding height at most four. The residual $(m=2)$ case must be resolved by an actual graph or support computation, not a rank-two endpoint classification imported from characteristic zero.

Either route must finish with $(\#\operatorname{Sing}(S_0)\leq 7)$ in the original two-hunt diagram.

Source scope

Whenever an external theorem is used, its stated characteristic range and all hypotheses must match the application. Exact theorem or lemma numbers are required. Results stated only in characteristic zero or in characteristic different from $(2,3)$ are unavailable at $(p=3)$ unless the needed assertion is proved independently here. This warning applies in particular to KM Sections 11, 12, 18, and 23; Lacini Proposition 5.1; and Nagaoka Lemma 3.21.

A published theorem whose stated scope includes characteristic $\backslash(3\backslash)$ may be cited as a theorem; this proof is not required to re-audit the cited author's proof. In particular, Nagaoka Proposition 2.10(7) legitimately supplies its stated smooth-point, unibranch, and unique-non-flush conclusions in characteristic $\backslash(3\backslash)$, and Lacini's results explicitly stated in the arbitrary-characteristic part of Section 3 may be cited within that stated scope. This does not extend the characteristic range of results whose own statements exclude characteristic $\backslash(3\backslash)$.

The required endpoint is a complete implication from the hypotheses of the theorem to the seven-point bound, with no conditional local profile, no unproved "actual endpoint" identification, and no use of any invalidated fact.

The separate September 2026 writing assignment. The following reproduces the operational problem file for the new manuscript-writing view created on 20 September 2026, with the same

technical substitutions. It is a writing assignment based on an older result, not a transcript of the initial research question or authenticated direct human speech.

```
# Fixed writing-only assignment
```

```
Read [project path] completely and execute the installed native paper workflow now.
```

```
This project is a new manuscript-writing view of an already verified theorem, not a research launch.
```

```
The theorem is: let  $k$  be algebraically closed of characteristic 3, and let  $X$  be a normal projective klt del Pezzo surface over  $k$  with  $\rho(X)=1$ . Then  $\#Sing(X)\leq 7$ . Its frozen verified root is [internal identifier]; the complete literal citation closure has 422 active nodes.
```

```
Use the independent frozen fact_graph as mathematical authority. [project path] and [project path] are original historical records. The historical graph and memory must remain unchanged. No new research workers, research Goal, source graph edits, global configuration edits, or external publication are authorized. The empty workers directory is only native scope scaffolding.
```

```
Produce a fresh readable manuscript through the remote native v3.1 paper skill and style, native section writers and independent verifiers, global verification, polish, and a checked build. See [project path] for the exact scope, provider, preservation, and delivery requirements.
```

```
Read [project path] before any native CLI command.
```

A.3. Guidance given during the run. The six dated historical guidance records identify their record authors as `main_agent` or `codex_dialogue`. They do not identify a human speaker or delimit a direct human utterance. Their timestamps are recording times. We reproduce the claim and evidence fields with the technical substitutions specified above and with their recorded attributions, and describe only the effects reported in the records. The absence of an authenticated direct quotation does not establish that no human guidance was given.

The retained archive also contains eighteen mathematical consult notes and nine operational directives. They do not carry authenticated human speaker boundaries. One correction directive is reproduced in the evidence field of historical record 6 below. We do not convert these mathematical documents into purported human dialogue.

For the September run, the instruction index and delegated writing request are reproduced below with the same substitutions, followed by the dated historical records and the two short operational startup instructions. The writing request led to the fresh outline, native section writing and review, whole-paper review, and the present disclosure. It authorized manuscript production from the frozen source, not a new research launch.

Provenance of the writing-run instructions. The current instruction register explicitly distinguishes the delegated writing directive from a historical research-user quotation. Its text is reproduced here with the technical substitutions specified above.

```
# Writing-run instruction provenance
```

```
The 2026-09-20 authorized writing directive is preserved verbatim as a delegated operational directive in [project path]. It is not represented as a direct historical research-user quotation. Historical research guidance remains in [project path] and [project path]. Native generated startup messages are not human utterances.
```

Delegated writing directive, 20 September 2026. The following operational directive is the document referenced by the current instruction register. The date of authorization is recorded

in the directive. It is reproduced as a delegated formulation, not attributed to a named human speaker. Internal identifiers and project paths have the substitutions specified above; the unmodified original remains in the private archive.

Authorized native manuscript rewrite

This is a fresh **writing-only** assignment, authorized on 2026-09-20. Read this entire file before doing work. It overrides generic research-startup behavior in the main-agent contract. Enter the installed native Danus v3.1 paper workflow immediately; do not create or resume any research worker, mathematical research Goal, or exploratory campaign.

Fixed theorem and mathematical authority

Write a new, readable mathematical paper proving:

> Let k be an algebraically closed field of characteristic 3. Let X be a normal projective klt del Pezzo surface over k with Picard number $\rho(X)=1$. Then X has at most seven singular points.

The Picard-number-one hypothesis is indispensable. The exact verified source root is ``[internal identifier]``. Its declared predecessor closure has 421 nodes; including all literal fact references gives 422 nodes. Every node is present and active, with no revoked predecessor or content-ID mismatch. All 1193 active source files match the final historical graph by SHA-256 and recompute under the installed native v3.1 parser. See ``[project path]`` for manifests and closure records.

The mathematical authority is the frozen fact graph in this writing view. Read the actual statements and proofs, including the full dependency closure. Do not replace fact proofs with the old manuscript. Do not use revoked facts as positive premises. Preserve every load-bearing hypothesis and every necessary argument. If native review exposes a real load-bearing gap in the source, report the exact missing implication and supporting source paths; never hide it, invent a proof, weaken the theorem silently, or restart mathematical research to repair it.

The old v2 source project is ``[project path]``. It is historical and must not be modified. This view contains an independent copy of its graph, with write permission removed from source files. Do not change permissions, graph files, revocation records, historical memory, ``[project path]``, or other frozen source materials. The main's MCP role is ``reader`` by design; do not rewrite it to a fact-writing role.

Exposition to produce

Reorganize the accepted mathematics into an intelligible proof, led by the overall proof strategy and the principal geometric mechanisms. Establish consistent definitions and notation. Organize necessary details into mathematically meaningful lemmas, with their actual proofs. The reader must be able to follow why each geometric reduction applies and how it contributes to the seven-point bound.

Do not mechanically expand fact IDs, worker reports, chronological research logs, branch acronyms, or a checklist of accepted nodes into the article. Internal fact IDs belong only in private provenance comments/maps. Keep necessary technical detail in the body or a clearly cited appendix. There is no page, byte, section-count, compression, or deadline target.

Historical manuscripts under `[project path]` may be consulted only to identify prior readability problems and bibliographic leads. Do not copy their article body as the basis of this fresh manuscript. Historical assertions of completion, objections, and superseded drafts must be interpreted according to their dates and provenance; they do not supersede the frozen final graph.

Required native workflow

Use the installed remote `[project path]`, its current native companion skills, and the native style/template resolved by `[project path]`. Do not import or impose the local historical Jihao writing profile during this remote generation phase. A separate local danus-transcribe pass will follow after delivery.

The main organizes proof architecture and `[project path]`, using the native planning and outline skills. Then use native `danus paper assign` and `danus paper start` for section writers and their independent native verifiers. Continue through section passes, independent whole-paper verification, polish, introduction/front matter, and a checked build with resolved references and citations. Read and address actual verifier findings rather than treating a status string as mathematical proof. No section may be silently omitted to make the build pass.

Use the command-local project root consistently:

...

```
DANUS_AGENTS_ROOT=[project path]
project=[project-01]
```

All outputs belong in this new view, whose absolute path is `[project path]`. Always verify that `danus paper` points to its `[project path]` directory. Do not run any research-level `danus start`, `danus assign`, `danus verify up`, or Goal command. An empty `[project path]` directory exists only for native scope recognition; no research roster has been created.

All model requests, including main, helpers, section writers, section verifiers, whole-paper verifier, and any later continued writing sessions, must use the existing [private provider] provider at `[private API endpoint]` with configured model `gpt-6-astra`. Preserve native reasoning tiers: main ultra, native section/verifier settings max. Native shared configuration and credentials are already installed; retain those links. Do not change global provider configuration, switch to an account-backed provider, expose credentials, or print keys. Before starting a new actor, confirm its private home links to the shared [private provider] configuration; report only provider/base URL, model, effort, and credential-present booleans.

History, privacy, and delivery

This is a new manuscript-writing run based on an older research result, not a new proof-discovery run. Preserve that distinction in the AI disclosure. Original problem, project metadata, and guidance are preserved as attributed historical material; do not fabricate an original launch, researcher utterance, model history, or elapsed time. Generated startup instructions are not human speech. The later local transcription will redact only direct human utterances, preserving AI-generated formulation and narrative.

Keep `[project path]` current at meaningful milestones: source reading, architecture, outline, section jobs, verifier findings, final review, and build. Start substantive native planning promptly. Continue the complete native pipeline without waiting for another operator confirmation.

Deliver a complete compile-ready source tree, final PDF, any genuine supplementary dependencies, precise bibliography, and the native verification/build receipts. Write `[project path]` with the final source/PDF paths and SHA-256 values, section and global verdict paths, build command/result, and any unresolved issue. Leave no unresolved issue hidden behind a success label. No external publication or Overleaf upload is authorized for this new problem in this task.

Historical guidance record 1. Recorded UTC timestamp:
2026-07-13T04:59:04.923030+00:00.

Recorded author label: `main_agent`. This is the record's timestamp and recorder label, not an identified human speaker or the time of a recovered human utterance. The decoded `claim` and `evidence` fields follow with the technical substitutions specified above. Their reported results and proposed arguments are historical record content, not independent verification by this appendix.

This earlier formulation differs from the consolidated problem file above. Stored line breaks within four occurrences of the letters for rho are retained without repair.

Recorded claim field.

Seven singular points in characteristic three: clean mathematical directive

The theorem

Let (k) be an algebraically closed field of characteristic (3) . Let (X) be a normal projective klt surface such that $(-K_X)$ is ample and $(h^0(X)=1)$. Prove

$$\begin{aligned} & \backslash[\\ & \quad \backslash\#\backslash\operatornamename{Sing}\}(X)\backslash\leq 7. \\ & \backslash] \end{aligned}$$

If $(\pi:Y\rightarrow X)$ is the minimal resolution and $(D=\backslash\operatornamename{Exc}\}(\backslash\pi)_*\backslash\mathrm{red}\})$, then the desired inequality is equivalent to

$$\begin{aligned} & \backslash[\\ & \quad \backslash\#\backslash\pi_0(D)\backslash\leq 7. \\ & \backslash] \end{aligned}$$

The argument must be a proof of the upper bound. Searching for an eight-point construction, enumerating candidate surfaces, or inferring the theorem from failure to find a counterexample is irrelevant.

Invalidated characteristic-transfer claims

The following six claims are unavailable. Each imported a result from the characteristic-zero Keel--McKernan memoir into characteristic (3) without reproducing the needed proof.

- `[internal identifier]`: the full one-singularity KM 12.4 genus-one transformation package;

- ``[internal identifier]``: the KM 12.5 bound of two surface singularities on a fixed component;
- ``[internal identifier]``: the KM 12.0 restriction isomorphism, KM 12.1(6) disjointness, and the KM 12.4 divisor identity;
- ``[internal identifier]``: the KM 12.1(7) fixed-part component bound;
- ``[internal identifier]``: the KM 12.4(1) Picard-rank formula;
- ``[internal identifier]``: the old-boundary local profile obtained from KM 23.5 and KM 11.2.

No statement that used one of these claims may be used. In particular, the old target-theorem facts and their all-leaf assembly consequences are unavailable. A phrase such as "the local part of the KM proof is characteristic-free" is not a proof. Any needed assertion from KM Sections 11, 12, or 23 must be derived on the page from intersection theory, adjunction, formal local algebra, and surface contraction results valid over (k) , or else replaced by a different argument.

This invalidation does not affect facts whose proofs are independent of the six claims. Especially useful surviving inputs include:

- ``[internal identifier]`` and ``[internal identifier]``: a $(\mathbb{P}^1)$ -fibration on (Y) with $(F \cdot D) \leq 4$, equivalently height at most four, implies at most seven singular points;
- ``[internal identifier]``: the exact singular-support identity for a finite-centre birational diagram;
- ``[internal identifier]``: the intrinsic fixed-component $((-1))$ -curve calculation on a smooth resolution;
- ``[internal identifier]``: an adjoint bridge that explicitly avoids KM Section 12;
- ``[internal identifier]``: the numerical anti-ampleness calculation at the $(5/6)$ -boundary endpoint;
- ``[internal identifier]``: the degree-one $(4A_2)$ lattice trichotomy;
- ``[internal identifier]``: the $(2/3)$ adjunction correction for an (A_2) end curvette;
- ``[internal identifier]``: the explicit blow-up of $(xy=z^3)$ in characteristic (3) and the implication "regular curve germ $(\rightarrow)$ transverse end contact";
- ``[internal identifier]``: the fifteen coarse tiger/no-tiger source alternatives, subject to checking every external hypothesis used in its proof;
- ``[internal identifier]``: the purely conditional reduction from closure of those fifteen alternatives to the target theorem.

The last two facts are reductions, not a proof of the theorem.

Global proof architecture

Assume for contradiction that $(n = \#\operatorname{Sing}(X) \geq 8)$. The characteristic-three height-four theorem and the Palka--Peřka height/descendant results reduce to

```
\[
  \operatorname{ht}(X) \geq 5
  \quad \text{and} \quad
  X \text{ has no elliptic-boundary descendant}.
\]
```

Run the tiger/no-tiger hunt only through statements whose hypotheses and characteristic range have been proved. The surviving coarse reduction has fifteen source alternatives. Each alternative must end in one of the following three contradictions:

1. a morphism $(Y \rightarrow B)$ with general fibre (F) satisfying $(F \cdot D \leq 4)$;
2. a genuine elliptic-boundary descendant in the precise Palka--Peřka sense;
3. a direct support calculation giving $(n \leq 7)$.

Most alternatives already have independent closures. The load-bearing missing mathematics is concentrated in the genus-one transformation used at two actual endpoints, the old-boundary local contraction, and the characteristic-three (A_2) curvette application. These must be repaired without reinstating any of the six invalidated claims.

G. Intrinsic genus-one adjoint transformation

The strongest useful replacement should be proved in small, explicit lemmas.

G1. The adjoint divisor on the smooth resolution

Let $(r: \widetilde{U} \rightarrow U)$ be a smooth projective resolution, and let $(\widetilde{A})$ be the strict transform of an integral Gorenstein curve (A) of arithmetic genus one. Establish the exact hypotheses under which

$$\begin{aligned} & \left[\begin{array}{l} 0 \rightarrow \omega_{\widetilde{U}} \\ \rightarrow \omega_{\widetilde{U}}(\widetilde{A}) \\ \rightarrow \omega_{\widetilde{A}} \rightarrow 0 \end{array} \right] \end{aligned}$$

gives a one-dimensional space $(H^0(\widetilde{U}, K_{\widetilde{U}} + \widetilde{A}))$. It is enough in the actual applications to prove $(H^0(K_{\widetilde{U}} + \widetilde{A}) = H^1(K_{\widetilde{U}}) = 0)$, for example from rationality or the required cohomology vanishing on the particular resolution. Since an integral Gorenstein genus-one curve has trivial dualizing sheaf, the nonzero restricted section has no zero. Deduce that the unique effective adjoint divisor

$$\left[\begin{array}{l} \widetilde{M} + \widetilde{E} \in |K_{\widetilde{U}} + \widetilde{A}| \end{array} \right]$$

does not contain $(\widetilde{A})$ and is disjoint from it. Do not cite KM 12.0 or 12.1 for this conclusion.

Prove separately that the non-exceptional part is nonzero in each actual endpoint. In the $(5/6)$ -endpoint this should follow from

$$\left[\begin{array}{l} K_U + \frac{5}{6}A \equiv 0, \\ \quad \quad \quad K_U + A \equiv \frac{1}{6}A > 0 \end{array} \right]$$

and the push-forward of the adjoint divisor. State whatever additional numerical relation is used at the smooth $(A_1 = \Sigma_2)$ endpoint rather than silently importing the KM setup.

G2. Negative definiteness and one-component support

Let $(h:Z \rightarrow U)$ be the partial resolution actually used in the application, let (A, Z) be the strict transform of (A) , and write

$$\backslash[\begin{array}{l} M_Z=\sum_{i=1}^{\infty} m_i N_i, \\ \quad\quad\quad m_i>0, \end{array}\backslash]$$

for the image of the non-exceptional adjoint fixed divisor. Derive, rather than assume.

$$\begin{aligned} & A_Z \cap \operatorname{operatorname{Supp}} M_Z = \varnothing, \\ & \quad K_Z + A_Z \equiv M_Z. \end{aligned}$$

In the actual one-centre diagram, construct an effective (h)-exceptional boundary (J) for which $(-(K_Z+J))$ is ample, or give another direct proof that

$$\backslash[\quad K_Z \cdot N_i < 0 \quad \quad \quad \backslash \text{quad} \backslash \text{text}\{\text{for every } \} i. \quad \backslash]$$

The equality above then gives $(M_Z \cdot N_i < 0)$. If $(Q = (N_i \cdot N_j))$, then (Q) is symmetric, its off-diagonal entries are nonnegative, and

$$\begin{aligned} & \backslash[\\ & \quad Q(m_1, \ldots, m_s)^t < 0. \\ & \backslash] \end{aligned}$$

Prove directly that these conditions make (Q) negative definite. One convenient proof rescales by the positive vector (m) and observes that the negative matrix is a symmetric nonsingular (M)-matrix with positive row sums. Contract the whole support by characteristic-free surface contraction. Check after each contraction that the remaining curves are still (K)-negative:

$$\begin{aligned} & \backslash[\\ & K_Z = p_1^*K_{\{Z_1\}} + cN_1 \\ & \quad \Longleftrightarrow \\ & K_{\{Z_1\}} \cdot N_i' = K_Z \cdot N_i - c(N_1 \cdot N_i) < 0 . \\ & \backslash] \end{aligned}$$

When $(\text{ho}(Z)=2)$, the (s) contracted numerical curve classes are independent, so

$$\rho(W) = 2 - s \geq 1.$$

Together with $(M_Z \neq 0)$, this gives $(s=1)$. This is the intended replacement for the unsupported KM component and rank formulas; it needs no injection of fixed components into marked ends of resolution chains.

G3. Geometry of the target

For the contraction $(p:Z \rightarrow W)$, prove only the conclusions actually needed later:

- (p) is an isomorphism in a neighbourhood of (A_Z) ;
- $($
 $h_0(W)=1)$;
- the push-forward of $(K_Z+A_Z \equiv M_Z)$ gives $(K_W+A_W \equiv 0)$, where $(A_W=p_*(A_Z))$;
- the relevant singularities of (W) are klt or canonical, with discrepancies computed from the displayed contraction rather than borrowed from KM;
- $(-K_W)$ is ample, using $($
 $h_0(W)=1)$, $(-K_W \cdot A_W > 0)$, or the surviving numerical lemma "[internal identifier]" as appropriate;
- every use of "Gorenstein", "Du Val", or " $(A_W \sim K_W)$ " is justified at the level of linear equivalence and Cartier indices, not merely numerical equivalence.

Use the exact support identity "[internal identifier]" for the diagram $(U \rightarrow Z \rightarrow W)$. A generic claim that a fixed component contains at most two singularities is not available. Either prove the required local bound from adjunction on the actual component, or compute the integers (m, r, k, δ) directly in the actual resolution graph.

0. Old-boundary contraction in characteristic three

Consider the numerical-zero plt endpoint $((S, C))$, its first pair-hunt link

$$\begin{array}{c} \backslash [\\ S \xrightarrow{f} T \xrightarrow{g} U, \\ \backslash] \end{array}$$

and the boundary-contracted case $(C_T = \Sigma = \text{operatorname{Exc}}(g))$, with $(E = \text{operatorname{Exc}}(f))$ and $(A = g(E))$. The following local assertions must be proved from a fixed complete resolution diagram:

1. the full exceptional locus over each marked point of (T) is specified on one chosen minimal dlt or minimal resolution;
2. no extra exceptional component is attached to either displayed arm;
3. the component retained by the hunt is identified with a definite component of that resolution;
4. the total graph contracted by (g) is

$$\begin{array}{c} \backslash [\\ [2] - [1] - [3], \\ \backslash] \end{array}$$
 where the middle curve is the strict transform of (Σ) ;
5. $(q = g(\Sigma))$ is smooth on (U) , (E) meets (Σ) at a smooth transverse point away from the two arms, and (A) has an ordinary cusp at (q) .

The determinant calculation must use the complete graph. For a chain $([r]-[1]-[s])$, the determinant is $(rs-r-s)$; a smooth contraction forces $((r-1)(s-1)=2)$, hence $(\{r,s\}=\{2,3\})$. This observation is useful only after smoothness or the corresponding determinant-one statement has been proved independently. The ordinary-cusp conclusion should be checked by an explicit sequence of blow-downs or completed-local parameters, not by naming "configuration III".

After the local diagram is proved, combine it with the actual numerical relation

$$\begin{aligned} & \backslash[\\ & \quad K_U + \frac{56A}{\text{equiv}0} \\ & \backslash] \end{aligned}$$

and the intrinsic genus-one lemmas $G1--G3$. The desired endpoint conclusion is a direct height-four exit, an elliptic descendant, or a support bound strong enough to give $(\#\operatorname{Sing}(S) \leq 7)$. It is not enough to construct an auxiliary surface whose relation to the original hunt is conditional.

A. The characteristic-three (A_2) curvette package

Work in the completed local ring

$$\begin{aligned} & \backslash[\\ & \quad R = k[[x,y,z]]/(xy-z^3). \\ & \backslash] \end{aligned}$$

The Jacobian singular subscheme is nonreduced in characteristic (3), but the underlying singular point is isolated: in every prime containing (x,y) , the equation forces (z) into the prime. The explicit blow-up calculation in ``[internal identifier]`` is therefore the correct starting point.

Prove the converse missing from that fact. If a regular strict transform meets one end component transversely at a smooth point of the (A_2) exceptional chain, then its image curve germ is regular. On the (x) -chart

$$\begin{aligned} & \backslash[\\ & \quad y = x^2Z^3, \text{qqquad } z = xZ, \\ & \backslash] \end{aligned}$$

transversality makes (x) a uniformizer; on the (z) -chart away from the node, one of (X,Y) is a unit and $(z=XY)$ is a uniformizer. Treat all charts and state the completion/strict-transform compatibility used to return to the algebraic germ.

Also prove that a Cartier curve through the (A_2) point cannot be regular. If $(C=(f))$, then either $(f \in \mathfrak{m}^2)$, leaving embedding dimension at least two, or (f) has a nonzero linear term; after eliminating that variable, the remaining plane-curve equation has multiplicity at least two. Thus $(R/(f))$ is not a regular one-dimensional local ring.

Apply these two local statements to the $(4A_2)$ lattice trichotomy ``[internal identifier]``. In the end-marked case the curve is regular. In the $((1,1))$ case its local class is zero, hence it is Cartier and therefore nonregular. In the actual birational diagram, if (p) is an isomorphism away from (q) and the strict transform is regular, the unique $((1,1))$ -marked (A_2) point must be (q) . Every arrow in

```

\[
  (1,1)\rightarrow [G]=0\rightarrow G\text{ Cartier}
  \rightarrow G\text{ nonregular at the }A_2\text{ point}
  \rightarrow q\text{ is that point}
\]

```

must be written explicitly.

S. Smooth ($A_1=\Sigma_2$) branch

The citation to Nagaoka Lemma 3.21 is unavailable because its stated characteristic range is at least five. Prove the characteristic-three branch directly.

There are two viable mathematical routes.

1. **Half-boundary height route.** Starting from the actual relation $-(K_{S_2}+\frac{1}{2}B_2)$ ample and the proved second-hunt coefficients $(b_2=2/3)$, $(e_1>1/2)$, run a $(K_{S_2}+\frac{2}{3}B_2)$ -negative contraction or construct the pencil dictated by the complete cusp resolution. Pull the resulting ruling to the original minimal resolution and prove that every horizontal component of the original exceptional boundary is among (B_2) and at most two residual second-link components. The coefficient inequality must exclude total horizontal degree five, including inseparable degree-three packets. Conclude $(F\cdot D)\leq 4$.

2. **Intrinsic genus-one route.** Prove all hypotheses of $G1--G3$ for the actual pair $((S_2, B_2))$. Let $(m=\#(B_2\cap \operatorname{Sing} S_2))$. For $(m=1)$, the one-centre Picard argument gives a single fixed component. For $(m\geq 2)$, do not quote the KM formula: use the exact finite-centre support identity and the independent numerical classes of the contracted curves to prove either the required support saving or a new extremal contraction yielding height at most four. The residual $(m=2)$ case must be resolved by an actual graph or support computation, not a rank-two endpoint classification imported from characteristic zero.

Either route must finish with $(\#\operatorname{Sing}(S_0)\leq 7)$ in the original two-hunt diagram.

Source scope

Whenever an external theorem is used, its stated characteristic range and all hypotheses must match the application. Exact theorem or lemma numbers are required. Results stated only in characteristic zero or in characteristic different from $(2,3)$ are unavailable at $(p=3)$ unless the needed assertion is proved independently here. This warning applies in particular to KM Sections 11, 12, 18, and 23; Lacini Proposition 5.1; and Nagaoka Lemma 3.21.

The required endpoint is a complete implication from the hypotheses of the theorem to the seven-point bound, with no conditional local profile, no unproved "actual endpoint" identification, and no use of any invalidated fact.

Recorded evidence field.

Clean mathematical directive after revoking six unproved characteristic-transfer roots and their explicit dependency closure.

Historical guidance record 2. Recorded UTC timestamp:

2026-07-13T05:13:02.801715+00:00.

Recorded author label: `codex_dialogue`. This is the record's timestamp and recorder label, not an identified human speaker or the time of a recovered human utterance. The decoded `claim` and `evidence` fields follow with the technical substitutions specified above. Their reported results and proposed arguments are historical record content, not independent verification by this appendix.

Recorded claim field.

Mathematical correction after the first clean relaunch verifications.

New verified inputs are [internal identifier] and [internal identifier] for the adjoint-support matrix argument, [internal identifier] for the one-component target geometry and exact singular-support identity, [internal identifier] for the characteristic-three A2 curvette equivalence and Cartier obstruction, and [internal identifier] for the conditional three-horizontal-curve height exit.

Old-boundary cusp calculation. The rejected [2]-[1]-[3] lemma must be repaired in actual completed coordinates. After contracting the middle (-1)-curve and then the resulting left (-1)-curve, prove that the final right curve R2 has self-intersection -1 and that the smooth strict transform E2 satisfies $I(E2, R2) = 2$ at their unique intersection. Choose completed coordinates (u, v) on the smooth surface with $R2 = (u = 0)$ and $E2 = (u = \phi(v))$, where $\text{ord}_v(\phi) = 2$. For the blowdown of R2 use completed base coordinates $x = u$ and $y = uv$. The image branch has completed ring $A = k[[\phi(v), v \phi(v)]]$ inside $k[[v]]$. Write $\phi(v) = v^2 a(v)$ with $a(v)$ a unit. For every $n \geq 2$ choose $i, j \geq 0$ with $2i + 3j = n$; the monomial $\phi(v)^i (v \phi(v))^j$ has leading term $a(0)^{(i+j)} v^n$. Successive triangular elimination, together with completeness and closedness of the image of $k[[X, Y]]$, gives $v^2 k[[v]]$ contained in A . The reverse inclusion A contained in $k + v^2 k[[v]]$ is immediate. Hence $A = k + v^2 k[[v]] = k[[v^2, v^3]]$. This proves the ordinary cusp without any formal-uniqueness claim from the graph alone. It does not prove that [2]-[1]-[3] is the complete exceptional locus: the no-extra-arms and retained-component identification remain separate required assertions.

A2 application. Combine [internal identifier], [internal identifier], and [internal identifier]. In the (1,1)-marked case, [internal identifier] supplies Cartierness; [internal identifier] makes the image curve nonregular at that A2 point. If that point were different from the unique centre q of the birational map, the map would be an isomorphism there and the regular strict transform would make the image germ regular, a contradiction. Therefore the (1,1)-marked point is q . Do not infer the local class group merely from determinant three.

Adjoint transformation. Use [internal identifier] and [internal identifier] only after proving their hypotheses for the actual endpoint: nonzero unique adjoint divisor, disjointness from the genus-one curve, K -negativity of each support component, existence of a contraction of the complete support, $\rho = 2$ on its source, Q -Cartier pullback of the image curve, and the exact finite-centre data. These formal facts do not by themselves establish the old-boundary local profile or the smooth $A1 = \text{Sigma}2$ application.

Smooth $A1 = \text{Sigma}2$. Fact [internal identifier] is only the numerical exit. Still prove on the actual second-hunt surface the existence of the $P1$ -ruling, the coefficient inequality, and that every horizontal component of the original exceptional boundary is among $B2$ and at most two residual components. Nagaoka Lemma 3.21 remains unavailable in characteristic three.

Case-list scope. Lacini Proposition 3.15 is stated in a paper whose characteristic hypothesis excludes 2 and 3. Its case list cannot be submitted or used as a characteristic-three fact merely by quoting the printed proposition. Any needed substatement must be rederived from the proof with every characteristic-sensitive input isolated and independently proved in characteristic three.

Recorded evidence field.

This correction uses the exact cold-verifier outcomes recorded at 2026-07-13T05:02--05:06 UTC. The old-boundary local submission was rejected because identical graph/incidence data did not imply formal equivalence; the coordinate calculation above supplies the missing argument while explicitly preserving the separate complete-graph obligation.

Historical guidance record 3. Recorded UTC timestamp:

2026-07-13T05:17:19.507297+00:00.

Recorded author label: `codex_dialogue`. This is the record's timestamp and recorder label, not an identified human speaker or the time of a recovered human utterance. The decoded `claim` and `evidence` fields follow with the technical substitutions specified above. Their reported results and proposed arguments are historical record content, not independent verification by this appendix.

Recorded claim field.

Source-root correction for the no-tiger hunt. Fact [internal identifier] and its full successor closure have been revoked because its proof cited Lacini Proposition 3.15, whose standing characteristic hypothesis excludes 2 and 3, while its statement asserted arbitrary characteristic. No revoked fact may be used as a predecessor or as an informal premise.

The replacement source candidate is Nagaoka, A Review of Lacini's Classification, arXiv:2512.12294, Proposition 2.10. The paper states at the start of Section 1.1 that it works over an algebraically closed field with $\text{Char}(k) \geq 0$. Proposition 2.10 then states, without a narrower local hypothesis, the complete first and second no-tiger hunt alternatives, including net, genus greater than one, ordinary node, ordinary cusp, smooth rational dlt, second-step contraction of A_1 , and the two-component fence/banana/tacnode alternatives under $a_2 + b_2 \geq 1$. Establish a new root fact by citing exactly Nagaoka Proposition 2.10 under its global convention, not Lacini Proposition 3.15 directly. The proof must quote the global convention and the proposition locator and must not claim that Lacini alone covers characteristic three.

After that replacement root is accepted, rebuild only the mathematically needed source-leaf exhaustion and closure implications from live independent facts. The recently accepted independent facts [internal identifier], [internal identifier], [internal identifier], [internal identifier], [internal identifier], and the height fact [internal identifier] remain live. Reuse them, but do not recreate revoked wrappers mechanically: each rebuilt implication must have an actual proof from live premises.

Recorded evidence field.

The official cascade removal moved [internal identifier] and every dependent fact out of the live graph. The exact replacement source text is in [project path]: global convention at lines 60–65 and Proposition 2.10 at lines 260–291.

Historical guidance record 4. Recorded UTC timestamp:

2026-07-13T05:27:35.568991+00:00.

Recorded author label: `codex_dialogue`. This is the record's timestamp and recorder label, not an identified human speaker or the time of a recovered human utterance. The decoded `claim` and `evidence` fields follow with the technical substitutions specified above. Their reported

results and proposed arguments are historical record content, not independent verification by this appendix.

Recorded claim field.

Old-boundary direct proof in characteristic three.

Let (S, C) be plt with $\rho(S)=1$, $-K_S$ ample, K_S+C numerically trivial, and at least three singular points on C . In the boundary-contracted Case-1 pair-hunt $S \leftarrow T \rightarrow U$, let $\Sigma = C_T$ be contracted, let E be the maximal-coefficient extracted curve, q the image of Σ , and A the image of E .

Use Kollar, Singularities of the Minimal Model Program, Corollary 3.31 (public draft Corollary 80, Section 85(1), Claim 85.4), whose algebraically-closed-field statement includes the case characteristic divides the determinant. Each complete exceptional fibre over p in C intersect $\text{Sing}(S)$ is an end-marked admissible chain Γ_p . If r_p is its determinant, adjunction gives $\sum_p (1 - 1/r_p) = 2$, hence the determinant multiset is one of $(2, 3, 6)$, $(2, 4, 4)$, $(3, 3, 3)$, or $(2, 2, 2, 2)$.

The extraction centre lies on C : if $E \cdot C_T = 0$, changing the coefficient of E cannot turn $(K_S + aC) \cdot C = (1-a)K_S \cdot C < 0$ into a divisor trivial on $\Sigma = C_T$. The retained curve is the boundary-adjacent component of a chain of maximal determinant because at $a=1$ its coefficient is $1-1/r$ and all other coefficients are smaller; this persists for a sufficiently close to one.

Fix the minimal log resolution $Y \rightarrow S$ and form $Y \rightarrow T$ by contracting every exceptional curve except E . The complete fibre over q is exactly C_Y plus the chains Γ_p for p different from the extraction centre. The residual tail over the extraction centre meets E away from $E \cap C_Y$, so it maps to a point of $E - \Sigma$ and is not over q . This proves completeness and excludes hidden arms.

Write the scaled equalities

$$f^*(K_S + aC) = K_T + a\Sigma + e_a E,$$

$$K_T + a\Sigma + bE = g^*(K_U + bA).$$

Their intersections with Σ give $b > e_a$. Every old-arm coefficient is unchanged by scaling E and is at most e_a , hence strictly less than b . If the central self-intersection is $-j$, adjunction, $K_T \cdot \Sigma < 0$, and the Schur complement first force $j=1$ and leave only $[2]-[1]-[3]$, $[2]-[1]-[4]$, and $[3]-[1]-[3]$. On a one-vertex $[d]$ arm its coefficient is $c_d = (d-2+a)/d$, and central intersection gives $-1-a+b+\sum c_d = 0$. The last two candidates force $b=c_4$ or $b=c_3$, contradicting $c_d < b$. Thus the complete fibre is $[2]-[1]-[3]$.

Blow down the middle curve, then the transformed -2 arm, then the transformed -3 arm. The endpoint is smooth. In completed coordinates before the last blowdown, take $R=(u=0)$ and the strict transform of E to be $u=\phi(v)$, $\text{ord}(\phi)=2$. With blowdown coordinates $x=u, y=uv$, the image ring is $B=k[[\phi(v), v\phi(v)]]$. Every integer $n \geq 2$ is $2i+3j$, so triangular elimination using the corresponding monomials and completeness gives $v^2k[[v]]$ contained in B ; the reverse inclusion is immediate. Hence $B=k+v^2k[[v]]=k[[v^2, v^3]]$. Therefore q is a smooth point and A has an ordinary cusp there.

Intersection with Σ gives the reduced-pair coefficient $e=5/6$ and source determinant six. The end-marked determinant-six chain is $[6]$ or $[2, 2, 2, 2, 2]$. The $[6]$ case puts A entirely in U_{reg} , where adjunction gives $(K_U + A) \cdot A = 0$ but $K_U + (5/6)A$ numerically trivial gives $(K_U + A) \cdot A = A^2/6 > 0$. Hence the source chain is $[2, 2, 2, 2, 2]$, the residual point on A is A_4 and end-marked, and exact singular support gives $\#\text{Sing}(S) = \#\text{Sing}(U) + 2$.

For global closure, use the intrinsic genus-one fixed-divisor construction. The retained A4 end has coefficient $2/3$ in the pullback of $K_U + (5/6)A$. After the one-component adjoint contraction $U \leftarrow Z \rightarrow W$, the surviving curve G_Z meets A_Z once and $K_Z + (5/6)A_Z + (2/3)G_Z$ is numerically trivial. Push to W . Then A_W is a smooth-locus cuspidal anticanonical curve, $A_W.G_W=1$, and fact [internal identifier] gives W of type D5 with one singular point. The exact finite-centre identity and the at-most-two support bound give $\#Sing(U) \leq 3$, hence $\#Sing(S) \leq 5$. Every hypothesis of the intrinsic contraction must be checked on the actual A4 partial extraction; no conditional local-profile fact may replace this check.

Recorded evidence field.

The complete-graph argument is characteristic-free; the only characteristic-specific formal step uses 2 invertible, which holds in characteristic 3. The Kollar source explicitly treats non-tame index through the nonreduced μ_r grading, so determinant six in characteristic three is covered.

Historical guidance record 5. Recorded UTC timestamp:

2026-07-13T05:34:50.868937+00:00.

Recorded author label: `codex_dialogue`. This is the record's timestamp and recorder label, not an identified human speaker or the time of a recovered human utterance. The decoded `claim` and `evidence` fields follow with the technical substitutions specified above. Their reported results and proposed arguments are historical record content, not independent verification by this appendix.

Recorded claim field.

Characteristic-three correction for the smooth $A_1 = \text{Sigma}_2$ branch.

The local-profile assertion formerly recorded as fact [internal identifier] is unavailable. Its proof cites Keel--McKernan Lemma 18.7 and Nagaoka Lemma 3.21, while Nagaoka Lemma 3.21 assumes characteristic at least five. Consequently none of the following may be assumed in characteristic three merely from that source: q_2 is smooth; B_2 has an ordinary cusp at q_2 ; the second contraction is configuration III; Sigma_2 contains exactly two surface singularities; Sigma_2 meets E_2 in the smooth locus; or the coefficient and Du Val assertions bundled there.

The branch must begin from the characteristic-three-valid no-tiger source theorem and the raw second-link equations. Let $f_2: T_2 \rightarrow S_1$ extract E_2 , let $g_2: T_2 \rightarrow S_2$ contract $\text{Sigma}_2 = A_T$, and put $B_2 = g_2(E_2)$. On a fixed minimal log resolution Y of the pair (S_1, A_1) at the extraction centre, write the complete exceptional fibre as a chain $F_1 + \dots + F_n$ with A_Y meeting F_1 . Identify exactly which F_i survives as E_2 and which subchain maps to each point of Sigma_2 . Derive every claimed singularity and incidence statement by solving the pullback intersection system on this complete graph. In particular, smoothness of q_2 requires determinant one for the complete g_2 -exceptional graph; it cannot be inferred from the name configuration III.

For a height exit, it is enough to construct a P_1 -fibration $h: T \rightarrow \Lambda$ on a model dominated by the original minimal resolution and prove that all horizontal components of the original reduced exceptional divisor have total degree at most four. If the boundary coefficient is $2/3$ and all other horizontal coefficients exceed $1/2$, intersection with a general fibre gives $-2 + (2/3)d_0 + \sum(c_i d_i) < 0$.

The inequality gives total horizontal degree at most three when there are at most two residual horizontal curves; but the existence of the ruling, the list of horizontal curves, and the exclusion of inseparable degree-three packets must be proved from the actual link.

For an intrinsic genus-one exit, first establish on the actual pair (S_2, B_2) : arithmetic genus one; the exact set B_2 intersect $\text{Sing}(S_2)$; a nonzero adjoint divisor disjoint from the strict transform of B_2 ; $\rho=2$ on the contraction source; and the finite-centre values r, κ, δ . Then the live matrix and adjoint-contraction lemmas apply. A conditional statement beginning by assuming q_2 smooth, an ordinary cusp, or configuration III does not prove this branch.

Recorded evidence field.

The cited source range of Nagaoka Lemma 3.21 excludes characteristic three; the unsupported root and its successor closure have been formally revoked.

Historical guidance record 6. Recorded UTC timestamp:

2026-07-13T08:28:52.535746+00:00.

Recorded author label: `codex_dialogue`. This is the record's timestamp and recorder label, not an identified human speaker or the time of a recovered human utterance. The decoded `claim` and `evidence` fields follow with the technical substitutions specified above. Their reported results and proposed arguments are historical record content, not independent verification by this appendix.

The evidence field contains the correction also preserved as a separate directive file. The unmodified private evidence field agrees with that file byte for byte; the quotation below has the stated technical substitutions.

Recorded claim field.

Characteristic-three source boundary and replacement lemmas: Nagaoka Proposition 2.10(7) is admissible, while the subsequent smooth endpoint is closed by explicit numerical and lattice arguments.

Recorded evidence field.

Characteristic-three source boundary and replacement lemmas

Let (k) be algebraically closed of characteristic (3). A published theorem whose stated hypotheses include characteristic (3) is an admissible source. In particular, Nagaoka, Proposition 2.10(7), may be cited for the smooth boundary first extraction: (q) is smooth, the boundary has one branch at (q) , and the boundary-adjacent end gives the unique non-flush valuation. Likewise, the results in Lacini's Section 3 that are explicitly stated in arbitrary characteristic may be cited. This does not extend to Nagaoka, Lemma 3.21, whose hypothesis is $(\text{operatorname}{char})k \geq 5$, or to a result stated only in characteristic zero.

For the smooth $(A_1 = \Sigma_2)$ branch, the coefficient on the second boundary is not $(2/3)$. The relevant inequalities are

$$\backslash[\frac{45}{b} < \frac{56}{56}.$$

The value $(2/3)$ occurs only in an auxiliary comparison. The endpoint proof must use the interval above.

The following characteristic-free lemma replaces every use of the single-curve conclusion from KM Section 11.4. Let $(g: T \rightarrow S)$ be a birational morphism of $(\text{mathbf{Q}})$ -factorial klt surfaces with

(operatorname {Exc}(g)=\Sigma) irreducible and $(K_T \cdot \Sigma < 0)$. Then

$$\begin{aligned} & \#(\operatorname {Sing} T \cap \Sigma) \leq 2, \\ & \quad 0 < -K_T \cdot \Sigma \leq 1. \end{aligned}$$

On the minimal resolution, the strict transform (C) satisfies $(C^2 = K_X \cdot C = -1)$. At each singular point of (T) on (Σ) , write $(A = -(E_i \cdot E_j))$, $(v_i = C \cdot E_i)$, $(b_i = K_X \cdot E_i)$, $(z = A^{-1}v)$, $(q = v \cdot tz)$, and $(r = b \cdot tz)$. Choosing (i) with $(v_i \geq 1)$ gives $(v_i \leq (-E_i^2)z_i)$ and $(b_i + 2v_i \geq -E_i^2)$, hence $(2q + r \geq 1)$. Summing over the singular points and using $(\Sigma^2 = -1 + \sum q < 0)$ and $(K_T \cdot \Sigma = -1 + \sum r < 0)$ shows that their number is (< 3) . The inequalities for $(-K_T \cdot \Sigma)$ follow from the same formulas. An argument that merely injects resolution arms into the strict transform is not valid and must not be used.

For a flush end chain with discrepancies (u_i) , the recurrence is

$$\begin{aligned} & d_i \quad u_i = u_{i-1} + u_{i+1}. \end{aligned}$$

The successive differences are strictly increasing, so the maximal coefficient is unique and occurs at the boundary-adjacent end. This supplies the needed retention statement without a characteristic-zero classification.

The exceptional open-pair dichotomy must be used only in its forward form. Starting with the output of Palka--Peřka, Corollary 3.6, choose the witness specified by Proposition 3.2(6). In characteristic (3), the exceptional alternative has type $(3A_2)$ or $(4A_2)$, and the common minimal-resolution graph is then the graph listed in Proposition 9.1. One may not infer that an arbitrarily chosen witness has this type from Theorem 1.2(a) alone.

In the smooth $(A_1 = \Sigma_2)$ branch, combine Nagaoka, Proposition 2.10(7), the end-chain recurrence, the single-curve lemma above, the actual first- and second-link support identities, the intrinsic genus-one adjoint transform, and the (A_2) -lattice calculation. The $(3A_2)$ case splits as required, while the marked lattice excludes the $(4A_2)$ case. The endpoint has at most five singularities; the two exact link counts then give $(n(S_0) \leq 7)$.

Current operator-facing operational instruction. This short current instruction specifies the writing scope and reporting language. It provides no personal author metadata and is not presented as a historical research-user utterance.

This is an authorized writing-only assignment. Read [project path] for the controlling directive; report briefly in Chinese when addressing the operator. Do not ask for another startup confirmation.

Generated operational startup directions for the writing main agent. These native startup directions refer the writing agent to its operational request and command environment. They are not direct human speech and do not date the original research launch.

Execute [project path] in full. Begin native paper source reading and proof-architecture planning now. Maintain [project path]. Read [project path] for the command-local native paper prefix; shared [project path] must not reset this isolated root.

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